

Supplementary Information for *When dendritic structure helps local credit assignment*

Houman Safaai, Maceo Richards and Bernardo L. Sabatini

Scope of the Supplementary Information

This document records the derivations, implementation conventions, analysis contracts, sensitivity analyses and numerical boundaries underlying the main article. We distinguish the evidential levels throughout. First, exact path-product identities and regular-tree learning concern the directed feedforward conductance network. Second, the structural MICrONS analysis uses an ancestry-based route-sensitivity proxy on measured morphology. Third, focal perturbations use an exact adjoint of a reciprocal passive cable model on the same morphology. Neither MICrONS model measures learning in the animal. Fourth, the active-conductance ensemble tests the exact local Jacobian around nonlinear steady states; channel-density draws are nested sensitivity samples, not animals or kinetic simulations. Fifth, measured-response analyses use all eligible scans and independently observed visual responses to test task alignment, not to infer an endogenous teaching signal. Sixth, the alignment-controlled quadratic experiment imposes exact credit and tests a conditional optimization prediction; it is not independent biological evidence. Seventh, the 50-seed credit-phase experiment tests spectral alignment, hierarchy-depth matching, projection denoising and oracle branch reliability in a fixed-state quadratic family; it is not a trained positive-conductance dendrite. Eighth, a separate 50-seed experiment trains positive excitatory conductances while fixed nonnegative shunts realize oracle branch-reliability factors under an exact state clamp; it does not learn the shunts or show a dendrite-exclusive operation. Ninth, the inhibitory census is a synapse-resolved structural association in one mouse. Tenth, the task-interference calculation is an exact static consequence. Eleventh, the six-animal brain-computer-interface analysis is an external retrospective consistency test of signed neuron identity. Twelfth, the nonlinear physical-depth experiment trains the production positive-rate population model on a selected hierarchical task after transparently reported accessibility pilots; it is not a full $D_r \times H \times K \times \rho$ phase grid or an impossibility result for grouped point models. Exact transported errors and fitted projection coefficients are information oracles wherever they appear.

The primary analysis unit is the independent training seed for artificial networks, the reconstructed cell for anatomy and perturbation analyses, the postsynaptic target cell for measured-response and inhibitory-census analyses, and the animal for the signed-credit reanalysis. Sites, channel draws, scans, splits, synapses, stimulus trials, epochs and Monte Carlo streams are nested observations. All reconstructed cells come from one animal and are not independent biological replicates at the animal level.

S1 Exact gradients in a directed conductance tree

1.1 Steady-state compartment voltage

Let a rooted dendritic tree have soma node 0. A non-somatic compartment n receives synaptic conductances g_i^{syn} with presynaptic activity x_i and reversal potential E_i , child-to-parent coupling conductances $g_{c \rightarrow n}^{\text{den}}$, and leak g_n^L with reversal E_L . A child transmits $a_c = f_c(V_c)$, where f_c is the branch output nonlinearity (named “reactivation” in the

implementation) and is the identity when that mechanism is disabled. The steady-state voltage is

$$g_n^{\text{tot}} = g_n^L + \sum_{i \in \mathcal{I}_n} g_i^{\text{syn}} x_i + \sum_{c \in \mathcal{C}_n} g_{c \rightarrow n}^{\text{den}}, \quad (1)$$

$$V_n = R_n^{\text{tot}} \left(g_n^L E_L + \sum_{i \in \mathcal{I}_n} g_i^{\text{syn}} x_i E_i + \sum_{c \in \mathcal{C}_n} g_{c \rightarrow n}^{\text{den}} a_c \right), \quad R_n^{\text{tot}} = (g_n^{\text{tot}})^{-1}. \quad (2)$$

For nonnegative conductances and activities, Eq. 2 is a conductance-weighted average of leak, synaptic reversal potentials and child activities. This boundedness is useful for numerical stability, but is not required for the chain-rule factorization below.

1.2 Local sensitivities and synaptic eligibility

Differentiating the numerator and denominator of Eq. 2 gives

$$\frac{\partial V_n}{\partial g_i^{\text{syn}}} = x_i R_n^{\text{tot}} (E_i - V_n), \quad (3)$$

$$\frac{\partial V_n}{\partial V_c} = f'_c(V_c) g_{c \rightarrow n}^{\text{den}} R_n^{\text{tot}}, \quad (4)$$

$$\frac{\partial V_n}{\partial g_{c \rightarrow n}^{\text{den}}} = R_n^{\text{tot}} (a_c - V_n). \quad (5)$$

Thus the exact gradient of a synaptic conductance is

$$\frac{\partial \mathcal{L}}{\partial g_i^{\text{syn}}} = \underbrace{x_i R_n^{\text{tot}} (E_i - V_n)}_{\text{local eligibility}} \underbrace{\frac{\partial \mathcal{L}}{\partial V_n}}_{\text{compartment error}}. \quad (6)$$

The eligibility contains quantities available at the synapse or its host compartment. All remaining task dependence lies in the compartment error.

1.3 Transport of the somatic error

Let $p(n)$ be the unique parent of non-somatic node n , and let $\delta_0^V = \partial \mathcal{L} / \partial V_0$. The chain rule gives

$$\frac{\partial \mathcal{L}}{\partial V_n} = \frac{\partial \mathcal{L}}{\partial V_{p(n)}} f'_n(V_n) R_{p(n)}^{\text{tot}} g_{n \rightarrow p(n)}^{\text{den}}. \quad (7)$$

Because the graph is a tree, iterating Eq. 7 along the unique path from n to the soma yields

$$\tilde{\alpha}_n = \prod_{(i \rightarrow k) \in \text{path}(n \rightarrow 0)} f'_i(V_i) R_k^{\text{tot}} g_{i \rightarrow k}^{\text{den}}, \quad (8)$$

$$\frac{\partial \mathcal{L}}{\partial V_n} = \tilde{\alpha}_n \delta_0^V. \quad (9)$$

The effective path gain $\tilde{\alpha}_n$ includes every local transfer derivative. With identity reactivation, it reduces to the conductance-stage gain α_n^{cond} obtained by removing the f'_i factors. Equation 9 is an exact derivative, not a separately fitted model parameter.

1.4 Raw-parameter transforms

Trainable raw parameters θ are transformed to nonnegative conductances by a monotone map $g = \psi(\theta)$. The optimizer receives

$$\frac{\partial \mathcal{L}}{\partial \theta} = \frac{\partial \mathcal{L}}{\partial g} \psi'(\theta). \quad (10)$$

This positive transform preserves the sign relation between the conductance gradient and the local rule. Active synapse masks are applied after forming the conductance-space gradient. The exact-gradient diagnostics compare gradients in the same parameter space and include the transform derivative.

1.5 General implicit-adjoint form

The preceding path product is a corollary of a more general steady-state identity. Suppose $\mathbf{F}(\mathbf{V}, \mathbf{g}, \mathbf{x}) = \mathbf{0}$ and the compartment Jacobian $J_V = \partial \mathbf{F} / \partial \mathbf{V}$ is nonsingular. Define the adjoint $\mathbf{q}$ by

$$J_V^\top \mathbf{q} = \nabla_{\mathbf{V}} \mathcal{L}. \quad (11)$$

The implicit-function theorem gives

$$\frac{\partial \mathcal{L}}{\partial g_i} = -\mathbf{q}^\top \frac{\partial \mathbf{F}}{\partial g_i}. \quad (12)$$

For the residual convention $\mathbf{F} = G\mathbf{V} - \mathbf{b}$, a conductance synapse at compartment n has $\partial \mathbf{F} / \partial g_i = x_i(V_n - E_i)\mathbf{e}_n$, and therefore

$$\frac{\partial \mathcal{L}}{\partial g_i} = x_i(E_i - V_n)q_n. \quad (13)$$

Equations 11–13 recover the directed path product when J_V is triangular. For the reciprocal passive cable, $J_V = G$ is symmetric positive definite, so $G^{-\top} = G^{-1}$. A general nonsymmetric or recurrent operator instead requires the inverse transpose. Active steady states enter through their Jacobian whenever the required differentiability and nonsingularity conditions hold; no passive symmetry is assumed by the general identity.

1.6 Conductance-specific control of path gain

Holding coupling conductances and the forward state fixed, add inhibitory conductance $\Delta G_k^I \geq 0$ at compartment k . For a compartment n whose path crosses k ,

$$\frac{\alpha_n^{\text{cond}}(\Delta G^I)}{\alpha_n^{\text{cond}}(0)} = \prod_{k \in \mathcal{A}(n)} \frac{g_k^{\text{tot}}}{g_k^{\text{tot}} + \Delta G_k^I}, \quad \frac{\partial \log \alpha_n^{\text{cond}}}{\partial G_k^I} = \begin{cases} -R_k^{\text{tot}}, & k \in \mathcal{A}(n), \\ 0, & k \notin \mathcal{A}(n), \end{cases} \quad (14)$$

where $\mathcal{A}(n)$ contains the compartments through which credit from n passes to the soma. Equation 14 proves path-selective attenuation. It does not prove that attenuation improves optimization.

For $z_n = \log \alpha_n^{\text{cond}}$ and path-dependent attenuation β_n , the perturbed log gain is $z'_n = z_n - \beta_n$, so

$$\text{Var}(z') = \text{Var}(z) + \text{Var}(\beta) - 2\text{Cov}(z, \beta). \quad (15)$$

Shunting narrows this distribution only if $\text{Cov}(z, \beta) > \text{Var}(\beta)/2$. Path-gain compression is therefore an operating-point-dependent empirical outcome rather than a theorem.

66 S2 Local rules and feedback objects

67 2.1 Three-factor update

68 The three-factor rule retains the exact local eligibility and substitutes an available voltage-space teaching field $\hat{\delta}_n$
69 for the exact compartment error:

$$\hat{\nabla}_{g_i^{\text{syn}}} \mathcal{L} = \left\langle x_i R_n^{\text{tot}} (E_i - V_n) \hat{\delta}_n \right\rangle_B, \quad \hat{\nabla}_{g_{c \rightarrow n}^{\text{den}}} \mathcal{L} = \left\langle R_n^{\text{tot}} (a_c - V_n) \hat{\delta}_n \right\rangle_B, \quad (16)$$

70 where $\langle \cdot \rangle_B$ denotes a mini-batch average. Excitatory and inhibitory contacts use the same expression; their reversal
71 potentials set the sign and magnitude of the driving force.

The four- and five-factor rules multiply Eq. 16 by slowly varying, bounded stage-level scalar reliability estimates. For stage ℓ , mini-batch t , and example b , the implementation averages the recorded layer voltage and corresponding somatic population activity over their feature axes, giving $\bar{V}_{\ell,b}^{(t)}$ and $\bar{V}_{0,b}^{(t)}$. The archived runs used the uncentered dot-product mode,

$$s_{\ell,t} = \frac{1}{B} \sum_{b=1}^B \bar{V}_{\ell,b}^{(t)} \bar{V}_{0,b}^{(t)}, \quad (17)$$

$$m_{\ell,1} = s_{\ell,1}, \quad m_{\ell,t} = 0.9m_{\ell,t-1} + 0.1s_{\ell,t} \quad (t > 1), \quad (18)$$

$$r_{\ell,t}^{\text{4F}} = \text{clip}_{[0.1,2.0]}(m_{\ell,t}). \quad (19)$$

72 Thus $r_{\ell,t}^{\text{4F}}$ is one scalar for a recorded stage, not a separate branch-wise signal, and is not a centered covariance or
73 a Pearson correlation. Recorded voltages are detached from automatic differentiation, and the same mini-batch
74 supplies $s_{n,t}$ and the local update. The positive clamp prevents a negative or very small dot product from reversing
75 or eliminating the update.

The five-factor rule retains this multiplier and adds a conditional variance-ratio factor. Let $Y_\ell^{(t)}$ be the recorded stage voltage and $X_\ell^{(t)}$ its source-resolved parent proxy, each centered across mini-batch t . With $\mathcal{E}_{0.1}[\cdot]$ denoting an EMA with update rate 0.1, define

$$v_{Y,\ell,t} = \mathcal{E}_{0.1}[\langle (Y_\ell^{(t)})^2 \rangle], \quad v_{X,\ell,t} = \mathcal{E}_{0.1}[\langle (X_\ell^{(t)})^2 \rangle], \quad c_{YX,\ell,t} = \mathcal{E}_{0.1}[\langle Y_\ell^{(t)} X_\ell^{(t)} \rangle], \quad (20)$$

$$\beta_{\ell,t} = \frac{c_{YX,\ell,t}}{v_{X,\ell,t} + 10^{-3}}, \quad v_{\text{res},\ell,t} = \max\{v_{Y,\ell,t} - \beta_{\ell,t} c_{YX,\ell,t}, 10^{-12}\}, \quad (21)$$

$$\phi_{\ell,t} = \text{clip}_{[0.25,4.0]} \left(\frac{v_{Y,\ell,t}}{v_{\text{res},\ell,t}} \right), \quad \hat{\nabla}^{\text{5F}} = r_{\ell,t}^{\text{4F}} \phi_{\ell,t} \hat{\nabla}^{\text{3F}}. \quad (22)$$

76 If $v_{X,\ell,t} \leq 10^{-12}$, the implementation sets $\phi_{\ell,t} = 1$. These factors are empirical preconditioners. They are neither
77 needed for the exact factorization nor extra task-error channels. Under exact transported error, the three-factor
78 rule recovers the exact dendritic parameter gradient in the stated steady-state model. This separates feedback
79 approximation from the validity of the local eligibility term; agreement of trained endpoints with backpropagation
80 is tested empirically rather than assumed from the identity.

81 2.2 Feedback fields

82 Table S13 defines the feedback objects without conflating bandwidth with routing. The matched-width/scalar-
83 fallback field reuses a somatic vector only at a stage of the same width; at wider stages it uses the same robust
84 scalar reduction as the global scalar condition. For a two-dimensional somatic error vector δ_b in example b , the
85 code first computes $\mu_b = N^{-1} \sum_u \delta_{b,u}$. With $\varepsilon = 10^{-6}$,

$$s_b = \begin{cases} \mu_b, & |\mu_b| > \varepsilon, \\ \delta_{b,u^*}, & u^* = \arg \max_u |\delta_{b,u}|, \quad |\mu_b| \leq \varepsilon. \end{cases} \quad (23)$$

For a one-dimensional input the coordinate is retained. The replacement prevents cancellation from silently producing a zero teaching signal. The historical jobs did not record the replacement count, so its cohort-specific activation rate is an explicit output of the clean rerun rather than inferred post hoc. The matched-width field does not repeat each neuron’s coordinate over that neuron’s descendants. The ancestry-shared field does exactly that, using one communicated coordinate per neuron and the tree as a fixed ownership map. Exact transport supplies the entire path derivative and is an information upper bound.

2.3 Conditional gradient alignment

For one tree and one example, suppose the correct somatic error is available but path gain is omitted. Write local eligibility as e_i and let $n(i)$ be the host compartment. Then

$$g_i^{\text{exact}} = e_i \tilde{\alpha}_{n(i)} \delta_0^V, \quad g_i^{\text{shared}} = e_i \delta_0^V. \quad (24)$$

For nonnegative path gains and nonzero gradients,

$$\cos(g^{\text{shared}}, g^{\text{exact}}) = \frac{1}{\sqrt{1 + \text{CV}_w(\tilde{\alpha})^2}}, \quad w_i = (e_i \delta_0^V)^2. \quad (25)$$

This identity explains why path-gain dispersion can matter after the correct somatic coordinate has been supplied. It does not solve the separate problem of supplying the correct coordinate across network layers.

S3 Topology-matched credit capacity

3.1 Ancestry dictionary

Let A_T be the synapse-by-route ancestry matrix of tree T , and let $\boldsymbol{\beta}$ contain state-dependent route gains. A low-bandwidth teaching vector $\mathbf{z} \in \mathbb{R}^K$ generates a synaptic field

$$\hat{\mathbf{g}} = \Phi_{T,K} \mathbf{z}, \quad \Phi_{T,K} = W_e A_T \text{diag}(\boldsymbol{\beta}) S_K, \quad (26)$$

where W_e holds local eligibility or anatomical weights and S_K assigns the K communicated channels to route sites. Topology fixes support. Conductance changes gain and conditioning but does not create an ancestry relation that is absent from A_T .

3.2 Address dimension and the point-neuron resource comparison

Separate the synapse-local eligibility from the task-dependent coefficient field and let $\mathbf{c}_u \in \mathbb{R}^{N_u}$ denote the latter for neuron u . For a prespecified nested family of dictionaries $\Phi_{u,K}$, define

$$d_{\varepsilon,u} = \min \left\{ K : \frac{\mathbb{E} \|P_{\Phi_{u,K}} \mathbf{c}_u\|_W^2}{\mathbb{E} \|\mathbf{c}_u\|_W^2} \geq 1 - \varepsilon \right\}. \quad (27)$$

This definition concerns the task-dependent coefficient after eligibility has been factored out; heterogeneous presynaptic activity alone is therefore not counted as an extra teaching coordinate.

For any fixed $\Phi \in \mathbb{R}^{N_u \times K}$, all fields generated as $\Phi \mathbf{z}$ lie in $\text{col}(\Phi)$ and hence occupy at most $\text{rank}(\Phi) \leq K$ dimensions. Conversely, the least-squares field in that space is $P_\Phi \mathbf{c}_u$, and its expected weighted residual energy is

$$\mathbb{E} \|(I - P_\Phi) \mathbf{c}_u\|_W^2 = \text{tr}[(I - \bar{P}_\Phi) \bar{C}_c], \quad \bar{C}_c = \mathbb{E}[W^{1/2} \mathbf{c}_u \mathbf{c}_u^T W^{1/2}]. \quad (28)$$

Equations 27–28 prove the address-rank statement used in the main text. The ordinary point-neuron local rule has $\Phi = \mathbf{1}$ and therefore one task-dependent coefficient per neuron. A K -route dendritic dictionary can use up to K structured fields inside that neuron. A point-neuron network is not mathematically forbidden from matching them, but doing so requires K separately routed teaching coefficients, K gated parameter groups, or an equivalent expansion into additional units. Accordingly, the comparison is about where address resources are represented, not about an absolute expressive separation between all possible dendritic and point-neuron networks.

3.3 Field capture

Let W be a positive diagonal matrix that defines the coordinate weighting. Set $\bar{\mathbf{g}} = W^{1/2}\mathbf{g}$, $\bar{\Phi}_{T,K} = W^{1/2}\Phi_{T,K}$, and let $\bar{P}_\Phi$ be the ordinary Euclidean orthogonal projector onto the columns of $\bar{\Phi}_{T,K}$. Field capture in this coordinate system is

$$\mathcal{C}_K(\mathbf{g}; \Phi, W) = \frac{\|\bar{P}_\Phi \bar{\mathbf{g}}\|_2^2}{\|\bar{\mathbf{g}}\|_2^2} = 1 - \frac{\|\bar{\mathbf{g}} - \bar{P}_\Phi \bar{\mathbf{g}}\|_2^2}{\|\bar{\mathbf{g}}\|_2^2}. \quad (29)$$

The unweighted case has $W = I$. The MICrONS structural analysis uses summed mapped excitatory synaptic area on the diagonal of W . For transformed gradient covariance $\bar{C}_g = \mathbb{E}[\bar{\mathbf{g}}\bar{\mathbf{g}}^T]$, expected capture for a fixed projector is

$$\mathcal{C}_K(\bar{C}_g; \Phi, W) = \frac{\text{tr}(\bar{P}_\Phi \bar{C}_g)}{\text{tr}(\bar{C}_g)}. \quad (30)$$

Equation 30 states the topology–task matching condition: a sparse anatomical basis is useful only to the extent that task credit lies in its span.

Proposition 1 (Ky–Fan credit-capacity bound). *Let the eigenvalues of $\bar{C}_g \succeq 0$ satisfy $\lambda_1 \geq \dots \geq \lambda_p \geq 0$. For every rank- K orthogonal projector P ,*

$$\frac{\text{tr}(P\bar{C}_g)}{\text{tr}(\bar{C}_g)} \leq \frac{\sum_{j=1}^K \lambda_j}{\sum_{j=1}^p \lambda_j}. \quad (31)$$

Equality is attained by the projector onto the leading K eigendirections. The difference between the right side and anatomical capture is the rank-matched spectral regret.

Proof. Ky–Fan’s maximum principle gives $\max_{\text{rank}(P)=K} \text{tr}(P\bar{C}_g) = \sum_{j=1}^K \lambda_j$. Division by the fixed total spectral energy proves Eq. 31. $\square$

For the spectral screen, $C(\rho) = (1 - \rho)C_0 + \rho C_1$ and $\text{tr}C_0 = \text{tr}C_1$. Hence for two fixed projectors P and Q , the normalized capture difference has the sign of

$$\Delta(\rho) = \text{tr}[(P - Q)C_0] + \rho \text{tr}[(P - Q)(C_1 - C_0)], \quad (32)$$

so an interior crossover is determined exactly by its two endpoints. This affine result distinguishes alignment from rank: increasing K raises the oracle bound, whereas increasing ρ determines whether the anatomical subspace approaches it.

3.4 One-step descent bound

The weighted statement is a Euclidean statement in explicitly transformed parameter coordinates. Define

$$\bar{\mathbf{w}} = W^{-1/2}\mathbf{w}, \quad \mathbf{w} = W^{1/2}\bar{\mathbf{w}}, \quad (33)$$

and define $\bar{\mathcal{L}}(\bar{\mathbf{w}}) = \mathcal{L}(W^{1/2}\bar{\mathbf{w}})$. Then $\nabla_{\bar{\mathbf{w}}} \bar{\mathcal{L}} = W^{1/2}\mathbf{g} = \bar{\mathbf{g}}$. If $\bar{\mathcal{L}}$ has L_W -Lipschitz gradients, the descent lemma gives

$$\bar{\mathcal{L}}(\bar{\mathbf{w}} - \eta \bar{P}_\Phi \bar{\mathbf{g}}) \leq \bar{\mathcal{L}}(\bar{\mathbf{w}}) - \eta \bar{\mathbf{g}}^T \bar{P}_\Phi \bar{\mathbf{g}} + \frac{L_W \eta^2}{2} \|\bar{P}_\Phi \bar{\mathbf{g}}\|_2^2. \quad (34)$$

Orthogonal projection implies $\bar{\mathbf{g}}^T \bar{P}_\Phi \bar{\mathbf{g}} = \|\bar{P}_\Phi \bar{\mathbf{g}}\|_2^2$, hence

$$\bar{\mathcal{L}}(\bar{\mathbf{w}} - \eta \bar{P}_\Phi \bar{\mathbf{g}}) \leq \bar{\mathcal{L}}(\bar{\mathbf{w}}) - \eta \left(1 - \frac{L_W \eta}{2}\right) \|\bar{P}_\Phi \bar{\mathbf{g}}\|_2^2. \quad (35)$$

At $\eta = 1/L_W$, the ratio between the projected- and full-gradient guarantees is Eq. 29. In the original coordinates, the projected step is

$$\mathbf{w}^+ = \mathbf{w} - \eta W^{1/2} \bar{P}_\Phi W^{1/2} \mathbf{g}, \quad (36)$$

which is a particular preconditioned update rather than an ordinary Euclidean gradient step in $\mathbf{w}$. The implementation was checked on 10,000 random positive-definite quadratic losses: there were no bound violations, minimum numerical slack was 2.67×10^{-7} , and the capture identity agreed to 1.89×10^{-15} . That audit checks the Euclidean projector algebra, which also applies after an explicit coordinate transformation; it does not validate synaptic-area weighting as a physical parameter metric. The result is a one-step guarantee only when the tested field is an exact gradient in the stated coordinates. It is not a convergence or generalization theorem and does not show that a biological circuit can infer the oracle coefficients.

S4 Stochastic credit-operator phase theory

4.1 Expected descent with signal, gain and noise

Let $\mathcal{L} : \mathbb{R}^p \rightarrow \mathbb{R}$ have L -Lipschitz gradient at the state under study. Write $\mathbf{g} = \nabla \mathcal{L}(\mathbf{w})$ and let $\hat{\mathbf{g}} = \mathbf{g} + \boldsymbol{\xi}$, where $\mathbb{E}\boldsymbol{\xi} = 0$ and $\mathbb{E}\boldsymbol{\xi}\boldsymbol{\xi}^\top = \Sigma$. A fixed matrix M collects every restriction imposed between the population gradient and the parameter update: coordinate ownership, route span, coefficient gain and any linearized local decoder. The update is

$$\mathbf{w}^+ = \mathbf{w} - \eta M \hat{\mathbf{g}}. \quad (37)$$

Proposition 2 (Stochastic credit-operator bound). *For Eq. 37,*

$$\mathbb{E}\mathcal{L}(\mathbf{w}^+) \leq \mathcal{L}(\mathbf{w}) - \eta \mathbf{g}^\top M \mathbf{g} + \frac{L\eta^2}{2} \left[\|M\mathbf{g}\|_2^2 + \text{tr}(M\Sigma M^\top) \right]. \quad (38)$$

If $A = \mathbf{g}^\top M \mathbf{g} > 0$ and $B = \|M\mathbf{g}\|_2^2 + \text{tr}(M\Sigma M^\top) > 0$, the right side is minimized by $\eta^* = A/(LB)$ and guarantees decrease

$$U(M) = \frac{A^2}{2LB}. \quad (39)$$

If $A \leq 0$, the bound does not certify a positive step.

Proof. The descent lemma gives $\mathcal{L}(\mathbf{w} + \mathbf{d}) \leq \mathcal{L}(\mathbf{w}) + \mathbf{g}^\top \mathbf{d} + (L/2)\|\mathbf{d}\|_2^2$. Substitute $\mathbf{d} = -\eta M(\mathbf{g} + \boldsymbol{\xi})$, take expectations, use $\mathbb{E}\boldsymbol{\xi} = 0$ and $\mathbb{E}\|M\boldsymbol{\xi}\|_2^2 = \text{tr}(M\Sigma M^\top)$, then minimize the resulting scalar quadratic in η . $\square$

At fixed infinitesimal update norm, $-\mathbf{g}$ is the steepest Euclidean descent direction by Cauchy–Schwarz. Consequently, no routed local operator provides a uniform first-order improvement over exact full-batch backpropagation on the same parameters and objective. Equation 39 instead compares restricted or noisy estimators and exposes when rejecting stochastic energy compensates for discarded signal.

Corollary 1 (Projected stochastic crossover). *Consider the identity-curvature quadratic $\mathcal{L}(\mathbf{w}) = \frac{1}{2}\|\mathbf{w} - \mathbf{w}^*\|_2^2$, initialize at $\mathbf{w} = 0$, set $\mathbf{g} = -\mathbf{w}^*$, use unit step size, and let P be an orthogonal projector. The expected post-update losses of the unprojected and projected unbiased gradients are*

$$\mathbb{E}\mathcal{L}[-(\mathbf{g} + \boldsymbol{\xi})] = \frac{1}{2} \text{tr}(\Sigma), \quad (40)$$

$$\mathbb{E}\mathcal{L}[-P(\mathbf{g} + \boldsymbol{\xi})] = \frac{1}{2} [\|(I - P)\mathbf{g}\|_2^2 + \text{tr}(P\Sigma P)]. \quad (41)$$

The projected estimator is better precisely when

$$\text{tr}[(I - P)\Sigma] > \|(I - P)\mathbf{g}\|_2^2, \quad (42)$$

that is, rejected noise exceeds discarded signal. A local estimator using the same P but additional coefficient noise cannot exceed backpropagation plus P in expectation.

For completeness, at a common step $0 < \eta < 2$ the unprojected-minus-projected expected loss is

$$\frac{\eta}{2} \left\{ \eta \text{tr}[(I - P)\Sigma] - (2 - \eta)\|(I - P)\mathbf{g}\|_2^2 \right\}. \quad (43)$$

Hence projection is better exactly when $\text{tr}[(I - P)\Sigma] > [(2 - \eta)/\eta]\|(I - P)\mathbf{g}\|_2^2$; the unit-step result above is the special case $\eta = 1$. Smaller common steps therefore require proportionally more rejected noise to compensate for the discarded signal.

4.2 Interior optimum along the nested route ladder

Every feedback field studied in this work is, up to coefficient noise, a projector on one nested ladder, $P_{\text{scalar}} \subset P_{\text{neuron}} \subset P_{\text{ancestry}, K} \subset I$. Along such a ladder the capture of Eq. 29 is monotone by construction, but the guaranteed utility is not. For an orthogonal projector with retained signal energy $S = \mathbf{g}^\top P \mathbf{g} = \|P \mathbf{g}\|_2^2$ and admitted noise $N = \text{tr}(P \Sigma P)$, the optimal-step utility of the credit-operator bound is

$$U(S, N) = \frac{S^2}{2L(S + N)}. \quad (44)$$

Let one rung up the ladder add marginal signal $\Delta S \geq 0$ and marginal noise $\Delta N \geq 0$. Substituting into Eq. 44 and clearing denominators gives the exact marginal condition

$$U(S + \Delta S, N + \Delta N) > U(S, N) \iff \Delta S [(S + N)(2S + \Delta S) - S^2] > S^2 \Delta N. \quad (45)$$

The bracket is positive, so for small marginal additions the condition reduces to a threshold on the marginal signal-to-noise ratio, $\Delta S / \Delta N > S / (S + 2N)$: the utility keeps rising while each added route scale contributes proportionally more signal than the running threshold, and falls once the added scales admit mostly noise. When the signal spectrum decays faster than the admitted noise across ladder scales, U therefore attains an interior maximum at a finite budget rather than at full rank. Equations 44 and 45, together with the interior-maximum construction, are verified numerically in the released test suite (`tests/test_laminar_optimality.py`).

Two frozen experiments locate this optimum empirically (Supplementary Fig. S16; analysis code is included in the release). In the hierarchy-depth phase experiment the initialization utility of the aligned tree, computed per seed and task depth, placed its argmax at the trained loss minimum in 135 of 200 seed–task pairs, and the group-mean argmax matched the trained optimum exactly at $H = 1, 2, 3$; at the full-rank boundary $H = 4$ the one-step bound was conservative and favoured $D_r = 2$, reflecting its single-step horizon. In the trained subtree factorial the same initialization utility reproduced the exact ties at $K = 8$, where every span coincides, but did not predict the trained $K = 4$ ancestry advantage over the best matched control (6 of 20 seeds; the mean utility contrast at $K = 4$ is negative against both the published four-control set and its three implementable, non-oracle members). The trained $K = 4$ advantage is therefore a training-dynamics effect that accrues along the trajectory rather than a property of one-step initialization geometry, consistent with the main-text observation that initial capture alone does not determine learning.

4.3 Laminar route families and hierarchical task covariance

The distinction between a point neuron and a dendritic tree can be stated as a constraint on the feasible credit operators at fixed rank. A point neuron addresses its synapses with the single shared column **1**: a rank-one operator with uniform support. An ancestry dictionary is a rank- K operator whose columns form a *laminar* family: any two supports are nested or disjoint. An unconstrained rank- K feedback matrix is the comparison class. For the balanced binary tree used in the factorial, the span of the K -route ancestry dictionary equals the span of the K coarsest vectors of the orthonormal tree-Haar basis $W = [\mathbf{h}_1, \dots, \mathbf{h}_n]$ (constant, then sibling-subtree differences, coarse to fine) for every admissible budget $K \in \{1, 2, 4, \dots, n\}$.

Proposition 3 (Laminar capture of hierarchical covariance). *Let the task-credit covariance be $C = W \Lambda W^\top$ with $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$ ordered so that $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n \geq 0$ along the coarse-to-fine Haar order. Then for every admissible budget K the ancestry dictionary attains the Ky–Fan bound,*

$$\frac{\text{tr}(P_{A,K} C)}{\text{tr}(C)} = \frac{\sum_{i \leq K} \lambda_i}{\text{tr}(C)} = \max_{\text{rank}(P)=K} \frac{\text{tr}(P C)}{\text{tr}(C)}. \quad (46)$$

For a general covariance C with eigenpairs $(\lambda_i, \mathbf{v}_i)$ ordered by decreasing eigenvalue, the shortfall from the Ky–Fan bound decomposes exactly as

$$\sum_{i \leq K} \lambda_i - \text{tr}(P_{A,K} C) = \sum_{i \leq K} \lambda_i \|(I - P_{A,K}) \mathbf{v}_i\|_2^2 - \sum_{j > K} \lambda_j \|P_{A,K} \mathbf{v}_j\|_2^2, \quad (47)$$

the leading-eigenspace energy outside the laminar span minus the trailing energy that the span retains. It is therefore bounded above by the former term, and is zero exactly when the span of the top- K eigenvectors of C is contained in the ancestry span.

Proof. The ancestry span equals the span of the K coarsest Haar vectors, which under the stated ordering are the top- K eigenvectors of C ; a rank- K orthogonal projector maximizes $\text{tr}(PC)$ exactly on a top- K eigenspace (Ky–Fan). For general C , expanding $\text{tr}(P_{A,K}C) = \sum_j \lambda_j \|P_{A,K} \mathbf{v}_j\|_2^2$ and $\|P_{A,K} \mathbf{v}_i\|_2^2 = 1 - \|(I - P_{A,K}) \mathbf{v}_i\|_2^2$ for $i \leq K$ gives the stated decomposition; it vanishes iff $P_{A,K}$ projects onto a top- K eigenspace. $\square$

Thus anatomy is an optimal rank- K feedback basis precisely for hierarchically organized credit: a node offers one coefficient per neuron, a tree offers a wavelet hierarchy, and the hierarchy is the right basis exactly when task-gradient covariance is organized on the tree. The spectral phase experiment realizes both regimes: at full alignment and $K = 4$ ancestry attained the rank-matched Ky–Fan bound to numerical precision, whereas at random alignment its advantage over sampled random routes vanished (main text). The attainment, the shortfall identity and the failure case with fine-scale-dominant covariance are verified numerically in `tests/test_laminar_optimality.py`.

4.4 Point-neuron partition residual and route-coefficient error

Let c_i be the exact task-dependent coefficient remaining after local eligibility has been factored out, let $w_i > 0$ be coordinate weights, and let $\pi(i)$ assign synapse i to a group that must share one teaching coefficient. The best allowed field minimizes $\sum_i w_i [c_i - z_{\pi(i)}]^2$.

Proposition 4 (Shared-coefficient residual). *For every group G , the optimum is its weighted mean*

$$\bar{c}_G = \frac{\sum_{i \in G} w_i c_i}{\sum_{i \in G} w_i}, \quad (48)$$

and the irreducible residual is

$$\mathcal{R}(\pi) = \sum_G \sum_{i \in G} w_i (c_i - \bar{c}_G)^2. \quad (49)$$

Refining a partition cannot increase $\mathcal{R}$; it decreases the residual only when an old group contains coefficient variation resolved by the new groups.

Proof. Differentiate the separable weighted least-squares objective with respect to each z_G . The refinement statement follows because the old solution remains feasible when two or more refined coefficients are constrained to be equal. $\square$

The ordinary point-neuron rule has one group per neuron. Dendritic ancestry offers a particular nested sequence of refinements, but is helpful only if the task coefficient varies across those subtrees. An explicitly grouped point model can implement the same partition, which is why the resource-matched point controls in the main article coincide with the dendritic implementation.

For a general dictionary Φ and positive diagonal W , let $\mathbf{z}^* = \arg \min_{\mathbf{z}} \|\mathbf{c} - \Phi \mathbf{z}\|_W^2$ and let $\hat{\mathbf{z}}$ be the coefficients actually supplied by a local circuit. Weighted projection orthogonality yields the exact decomposition

$$\|\mathbf{c} - \Phi \hat{\mathbf{z}}\|_W^2 = \underbrace{\|\mathbf{c} - \Phi \mathbf{z}^*\|_W^2}_{\text{address / representation error}} + \underbrace{\|\Phi(\hat{\mathbf{z}} - \mathbf{z}^*)\|_W^2}_{\text{coefficient-estimation error}}. \quad (50)$$

If the coefficient error is unbiased with covariance Ω_z , the expected second term is $\text{tr}(\Phi^T W \Phi \Omega_z)$. The Gram matrix $G_\Phi = \Phi^T W \Phi$ therefore controls how coefficient error is expressed. Small positive eigenvalues amplify the inverse problem used to estimate $\mathbf{z}^*$, while column coherence makes individual route coefficients poorly identifiable even when their combined field is stable. Address rank alone is not a complete learning metric.

4.5 Static route gains preserve span but can change conditioning

Proposition 5 (Route-wise static-gain invariance). *Let B be an invertible diagonal matrix of fixed nonzero route gains. Then*

$$\text{col}(\Phi B) = \text{col}(\Phi), \quad P_{\Phi B, W} = P_{\Phi, W}. \quad (51)$$

Thus route-wise static scalar gains cannot change address rank, projection residual or spectral capture. They can change the Gram matrix from $\Phi^\top W \Phi$ to $B^\top \Phi^\top W \Phi B$ and hence alter conditioning and coefficient dynamics.

This proposition concerns nonzero scalar rescaling of existing route columns. Spatially varying row gains, state-dependent gating and gains that reach zero can change the effective span or active support; those cases are not covered by Eq. 51.

4.6 Finite-time coefficient learning has a bias–variance crossover

Equal span does not imply equal coefficient dynamics. Let the realized field be $\mathbf{f}_t = \Phi \mathbf{z}_t$, normalize vanilla coefficient descent by $\lambda_{\max}(\Phi \Phi^\top)$, and define $A = \Phi \Phi^\top / \lambda_{\max}$. For noisy rate-based target estimates $\mathbf{g} + \boldsymbol{\xi}_t$, with $\mathbb{E} \boldsymbol{\xi}_t = 0$ and $\text{Cov}(\boldsymbol{\xi}_t) = \sigma^2 I/n$, the field iteration is

$$\mathbf{f}_{t+1} = \mathbf{f}_t - cA(\mathbf{f}_t - \mathbf{g} - \boldsymbol{\xi}_t). \quad (52)$$

Proposition 6 (Exact finite-time coefficient risk). *Assume $\mathbf{f}_0 = 0$, $\mathbf{g} \in \text{col}(\Phi)$ and let $(\mu_j, \mathbf{u}_j)$ be the positive eigenpairs of A , with $\theta_j = \mathbf{u}_j^\top \mathbf{g}$. For independent isotropic noise and a stable step, the expected population loss after T updates is*

$$\mathbb{E} \frac{\|\mathbf{f}_T - \mathbf{g}\|_2^2}{2} = \frac{1}{2} \sum_j \left[(1 - c\mu_j)^{2T} \theta_j^2 + \frac{\sigma^2 (c\mu_j)^2 [1 - (1 - c\mu_j)^{2T}]}{1 - (1 - c\mu_j)^2} \right]. \quad (53)$$

For Gram-preconditioned descent, $A = P_\Phi$ and all positive $\mu_j = 1$; therefore every dictionary with the same projector has the same field trajectory under common observations.

Proof. Project Eq. 52 onto each eigenvector. The deterministic error contracts by $1 - c\mu_j$ per step. Independent innovations have variance $(c\mu_j)^2 \sigma^2/n$ and accumulate as a finite geometric series. Summing half the modal second moments gives Eq. 53. Gram preconditioning replaces $\Phi^\top \Phi$ by its pseudoinverse on the positive subspace, so the induced field operator is P_Φ . $\square$

Equation 53 makes the low-data boundary explicit. Small μ_j retain more finite-time bias but admit less estimator variance. For two same-span operators, write their risk difference as $\Delta B + (\sigma^2/n)\Delta V$. If $\Delta B > 0$ and $\Delta V < 0$, the ordering reverses at $n^* = -\sigma^2 \Delta V / \Delta B$: slow coordinates can help below n^* by implicit noise filtering, whereas well-conditioned coordinates help above it by reducing bias. This is a finite-horizon regularization effect, not new address capacity and not an unconditional advantage of ill-conditioning.

4.7 Branch reliability shrinkage

Partition parameter coordinates into disjoint branch blocks b . Let $S_b = \|\mathbf{g}_b\|_2^2$ and $N_b = \text{tr}(\Sigma_b)$. For $M = \text{blockdiag}(a_b I_b)$ and fixed step $\eta = c/L$, Eq. 38 guarantees a loss decrease of

$$\mathcal{G}_c(\mathbf{a}) = \frac{c}{L} \sum_b \left[a_b S_b - \frac{c a_b^2}{2} (S_b + N_b) \right]. \quad (54)$$

Proposition 7 (Fixed-step reliability-optimal attenuation). *Equation 54 is maximized independently at*

$$a_b^* = \min \left\{ 1, \frac{S_b}{c(S_b + N_b)} \right\}, \quad (55)$$

for a physical attenuator constrained to $0 \leq a_b \leq 1$. The best common gain is $a_g^* = \min\{1, \sum_b S_b / [c \sum_b (S_b + N_b)]\}$, and the branch-specific guarantee is no smaller because its feasible set contains every common gain. At $c = 1$, Eq. 55 reduces to the reliability factor $S_b / (S_b + N_b)$; in that case $\mathcal{G}_{\text{branch}} = (2L)^{-1} \sum_b S_b^2 / (S_b + N_b)$ and $\mathcal{G}_{\text{global}} = (2L)^{-1} (\sum_b S_b)^2 / \sum_b (S_b + N_b)$, with equality when reliability is constant across blocks.

Proof. Differentiate each concave quadratic in Eq. 54 and project its unconstrained maximizer onto $[0, 1]$. Restricting all coordinates to a common value gives a_g^* . The branchwise optimum cannot be worse than this restricted solution. For $c = 1$, the closed-form comparison also follows from Cauchy–Schwarz applied to $S_b / \sqrt{S_b + N_b}$ and $\sqrt{S_b + N_b}$. $\square$

The result is an oracle prescription, not a claim that a neuron knows S_b or N_b . It also has no dendrite-exclusive content: any point implementation provided with the same branch partition and multiplicative factors is identical. Its biological relevance depends on whether local conductance can estimate or realize those factors.

4.8 Passive focal shunts attenuate a positive adjoint

Proposition 8 (No sign reversal in a passive grounded tree). *Let G be a symmetric positive-definite nonsingular M -matrix, let $\mathbf{s} \geq 0$, and define the passive adjoint $\mathbf{q} = G^{-1}\mathbf{s}$. Adding a nonnegative shunt κ at compartment k gives $G' = G + \kappa \mathbf{e}_k \mathbf{e}_k^T$ and*

$$\mathbf{q}' = \mathbf{q} - \frac{\kappa q_k}{1 + \kappa (G^{-1})_{kk}} G^{-1} \mathbf{e}_k. \quad (56)$$

Both G^{-1} and $(G')^{-1}$ are elementwise nonnegative, hence $0 \leq \mathbf{q}' \leq \mathbf{q}$ componentwise. A focal passive shunt attenuates but cannot reverse a positive adjoint field.

Proof. Equation 56 is the Sherman–Morrison identity. A nonsingular M -matrix has a nonnegative inverse; adding a nonnegative diagonal term preserves the M -matrix property. The subtracted vector is nonnegative, and $\mathbf{q}' = (G')^{-1}\mathbf{s} \geq 0$. $\square$

This ordering does not by itself guarantee descendant selectivity relative to a matched current injection; that spatial comparison depends on the Green’s function, focal site and electrotonic regime and is evaluated numerically in the focal experiments.

Corollary 2 (Spectral size of a focal passive perturbation). *Under the preceding assumptions, let $H = G^{-1}$ and $H' = (G + \kappa \mathbf{e}_k \mathbf{e}_k^T)^{-1}$. The inverse-operator change is rank one and*

$$\|H' - H\|_2 = \frac{\kappa \|\mathbf{H} \mathbf{e}_k\|_2^2}{1 + \kappa H_{kk}} \leq \frac{\kappa \|H\|_2^2}{1 + \kappa / \lambda_{\max}(G)}. \quad (57)$$

Consequently, for any adjoint source $\mathbf{s}$, $\|(H' - H)\mathbf{s}\|_2 \leq \|H' - H\|_2 \|\mathbf{s}\|_2$.

Proof. Sherman–Morrison gives $H' - H = -\kappa (\mathbf{H} \mathbf{e}_k)(\mathbf{H} \mathbf{e}_k)^T / (1 + \kappa H_{kk})$. The norm of this symmetric rank-one matrix is its scalar coefficient times $\|\mathbf{H} \mathbf{e}_k\|_2^2$. Use $\|\mathbf{H} \mathbf{e}_k\|_2 \leq \|H\|_2$ and $H_{kk} \geq \lambda_{\min}(H) = 1 / \lambda_{\max}(G)$ for the bound. $\square$

Equation 57 makes the electrotonic boundary explicit: a well-conditioned, strongly grounded cable has small $\|G^{-1}\|_2$ and therefore cannot show a large global adjoint change at fixed shunt dose. The bound is global and conservative; the descendant-enriched spatial contrast remains a Green’s-function question.

4.9 Why depth can help or hurt

Let $P_1 \preceq P_2 \preceq \dots \preceq P_{D_r}$ be nested tree-route projectors. Increasing depth reduces the address bias $\|(I - P_{D_r})\mathbf{g}\|^2$ but increases admitted noise $\text{tr}(P_{D_r} \Sigma P_{D_r})$ whenever the new modes carry stochastic energy. Equation 42 therefore predicts an interior optimum when task signal is concentrated above one hierarchy level and noise continues into finer levels. This is the controlled role isolated by the hierarchy-depth phase experiment. It is distinct from nonlinear

forward composition, which can make deeper dendritic subunits representationally useful even when the backward address span is unchanged.

Depth can also impose a transport cost. In an idealized ensemble with equal local eligibility, let the route gain be $A = \prod_{\ell=1}^{D_p} G_\ell$ and assume independent $\log G_\ell$ with variances σ_ℓ^2 . The cosine between a shared coefficient and the exact gain-weighted field in the large-route limit is

$$\frac{\mathbb{E}A}{\sqrt{\mathbb{E}A^2}} = \exp\left(-\frac{1}{2} \sum_{\ell=1}^{D_p} \sigma_\ell^2\right). \quad (58)$$

Additional levels can therefore add address resolution while simultaneously making uncompensated shared feedback less aligned. The balance depends on task hierarchy, gain variability, noise and the available coefficient estimator; “more depth” has no unconditional sign.

4.10 Controlled phase experiment

The experiment used the 16-column orthonormal Haar basis of a balanced binary tree. The spectral screen mixed a tree-aligned covariance with an independent rotation having the same eigenvalues and compared ancestry, random rank and dense principal-eigenspace projectors. The depth screen sampled targets from tree levels no finer than H and assigned increasing noise to levels finer than H ; aligned and leaf-rewired projectors were updated with identical noise draws for 80 steps. The projection screen directly crossed signal retention and noise retention. The reliability screen crossed eight branch blocks with five degrees of oppositely ordered signal and noise energy. Full parameters, seeds, gates and row counts are frozen in `configs/credit_phase_theory/confirmatory.json`; seed-level tables, summaries and contrasts are in `source_data/credit_phase_theory/`.

The two canary seeds tested numerical artifacts only. Fifty untouched confirmatory seeds supplied the inferential unit; intervals used 20,000 paired seed-bootstrap draws and central paired contrasts used two-sided Wilcoxon tests. All full-rank projectors agreed to 4.44×10^{-16} , route-wise static gain rescaling preserved the projector to 1.33×10^{-15} , and the explicit point gate matched reliability-aligned attenuation exactly. This fixed-state quadratic experiment validates the phase logic but does not join oracle reliability to learned positive conductances.

4.11 State-matched positive-conductance reliability experiment

We next tested the physical realization of Eq. 55 in a separate trained branch model. Every input rate x_{bid} was nonnegative and every trainable excitatory conductance was $g_{bd} = \text{softplus}(q_{bd})$. For branch b , define total unshunted conductance and voltage

$$G_b = g^L + \sum_d x_{bd} g_{bd}, \quad V_b = \frac{\sum_d x_{bd} g_{bd}}{G_b}. \quad (59)$$

A fixed shunt $\kappa_b \geq 0$ had reversal zero. To isolate its effect on input resistance from its effect on the forward feature, an oracle current $\kappa_b V_b$ was added on each example. The perturbed state is therefore

$$V'_b = \frac{\sum_d x_{bd} g_{bd} + \kappa_b V_b}{G_b + \kappa_b} = V_b, \quad \frac{\partial V'_b}{\partial q_{bd}} = \frac{x_{bd}(1 - V_b)}{G_b + \kappa_b} \text{sigmoid}(q_{bd}). \quad (60)$$

Relative to the unshunted eligibility, the physical input-resistance factor is $G_b/(G_b + \kappa_b)$. The compensating current was recomputed from the pre-shunt state and treated as a clamp, not differentiated as an independently learned pathway. Consequently this manipulation deliberately removes forward feature suppression while retaining conductance attenuation in the local update.

The binary task used eight branches, four inputs per branch, 512 training examples and 2,048 held-out examples. Input rates were generated from a positive baseline plus a class-dependent branch pattern and clipped at zero. Initial full-data gradients supplied $S_b = \|\mathbf{g}_b\|_2^2$. Coefficient-noise variance was then scaled so that

$$N_b = S_b \exp(-2hx_b), \quad x_b \in [-1, 1], \quad (61)$$

for eight evenly spaced branch indices and $h \in \{0, 0.5, 1, 1.5, 2\}$. The statistical reliability was $r_b = S_b / (S_b + N_b)$, lower-bounded at 0.03 for numerical stability. Because the common step was $\eta = c/L$ with $c = 0.5$, the corrected fixed-step target was $a_b^* = \min\{1, r_b/c\}$ and the initial shunt was $\kappa_b = \overline{G}_b[(a_b^*)^{-1} - 1]$. Shunts remained fixed during the 40 updates; the excitatory conductance parameters were trained, and the same coefficient-noise realizations were used across matched methods within seed.

Nine conditions separated the relevant alternatives: exact clean backpropagation, noisy credit without a shunt, a state-matched additive control, the step-consistent best single global shunt, reliability-aligned, seed-shuffled and anti-aligned shunts, an explicit point gate receiving the identical branch factors, and aligned shunting with clean credit. The point gate is a resource control, not a biological assertion: because it receives the same branch partition and factors, its update must equal Eq. 60. The clean-backpropagation condition retains the unattenuated exact gradient and is the optimization reference.

The preceding pilot is retained as a historical artifact because it used r_b despite taking $c = 0.5$. After identifying that inconsistency, the corrected configuration and predictions were frozen before outcomes for two new artifact-only canary seeds. Confirmatory execution then used 50 fresh paired seeds 8100–8149 with unchanged configuration and runner hashes. All 2,250 expected seed-condition rows completed. The minimum input rate was zero, maximum state mismatch was 2.22×10^{-16} , and independently computed point-gate and additive-control trajectory differences were 4.00×10^{-15} and 1.31×10^{-14} . A new finite-difference gate held the compensating current fixed while perturbing an excitatory parameter; its maximum relative error was 3.60×10^{-8} . All prespecified gates passed.

At $h = 2$, reliability-aligned minus best-global one-step test-loss reduction was 0.001817 (95% paired seed-bootstrap interval 0.001618–0.002020), positive in 49/50 seeds (two-sided Wilcoxon $P = 3.55 \times 10^{-15}$). Aligned minus unshunted reduction averaged 0.001339 (0.000594–0.002096), positive in 33/50 seeds ($P = 0.0030$). After 40 updates, aligned-minus-global test loss was -0.020404 (-0.022220 to -0.018705), favoring aligned in 50/50 seeds ($P = 1.78 \times 10^{-15}$). Aligned shunting also had lower final loss than shuffled and anti-aligned shunts by 0.009204 and 0.013781, respectively. However, aligned-minus-unshunted final loss was 0.000425 (-0.002456 –0.003071; $P = 0.154$), so the immediate improvement did not establish a sustained loss difference from no shunt. Exact clean backpropagation retained the lowest mean final loss. These results validate a positive-conductance approximation and ordering of fixed-step oracle reliability factors under a state clamp; they do not establish autonomous shunt learning, dendrite-exclusive computation or general superiority to unshunted learning.

4.12 Same-span noisy coefficient-learning experiment

We tested the distinction between address and coefficient errors with a new rate-based factorial. The 16-leaf balanced-tree Haar basis supplied the common rank-eight span through depth three. The tested dictionaries were the eight orthonormal Haar modes, all 15 nested indicators through depth three, and the same nested indicators scaled by positive gains evenly spaced on the log scale from $\exp(-3)$ to $\exp(3)$. The dictionaries' projectors agreed to 9.44×10^{-16} and their positive-spectrum Gram condition numbers were 1, 15 and 388.52.

Each independent seed drew a unit-norm target in the common span. At every update the estimator observed the target plus isotropic Gaussian noise with standard deviation $2/\sqrt{n_{\text{eff}}}$, where $n_{\text{eff}} \in \{4, 16, 64, 256\}$. The same 80 observation vectors were used for every matched parameterization. Vanilla coefficient descent used $0.2/\lambda_{\text{max}}(\Phi^T \Phi)$; the oracle coordinate control used $0.2(\Phi^T \Phi)^+$. Population field loss was recorded after 1, 5, 20 and 80 updates. The target address residual was below 1.45×10^{-30} in every condition, so all reported loss is within-span coefficient-estimation error.

The predictions were frozen before two artifact-only canary seeds and 50 paired seeds 9100–9149. The prediction that Haar coordinates would always reduce finite-time error was false. At $n_{\text{eff}} = 4$, raw-minus-Haar final loss was -0.2774 (95% paired interval -0.3260 to -0.2313) and scaled-minus-Haar was -0.1906 (-0.2500 to -0.1335): slower modes filtered high estimator noise. At $n_{\text{eff}} = 256$, the effects reversed to 0.02555 (0.02171–0.02952) and 0.17091 (0.14576–0.19686), positive in 50/50 seeds for both contrasts. Equation 53, added to explain this falsification, predicted median crossovers at effective sample sizes 38.53 and 8.25. The Gram-preconditioned field trajectories agreed across dictionaries to 1.16×10^{-15} . Thus static parameterization can alter finite-data

learning without altering the attainable credit field.

4.13 Nonlinear physical depth in the production rate model

The hierarchy-depth screen above changes the resolution of a linear route projector. We therefore added a separate bridge to the original nonlinear population implementation. The registered `hierarchical_gain_load` generator emits 64 strictly positive excitatory rates followed by 64 strictly positive inhibitory rates. Three independent zero-mean Gaussian node fields define fine, coarse and global lognormal gain factors. Their product multiplies signal-bearing excitatory coordinates in the distal block; the three inhibitory blocks separately measure the corresponding factors. Lognormal means are corrected by $-\sigma^2/2$ at every factor, so changing hierarchy or sensor alignment does not silently change first moments.

The matched condition uses the same latent node in excitatory signal and its inhibitory sensor. In the zero-alignment condition, each sensor instead uses an independent node field with the same scale and grouping. In the shuffled condition, a nonzero cyclic trial shift is applied to the complete inhibitory field after class balancing. The tree-placement condition leaves every data tensor unchanged and reverses the fine and global feature ranges across the fine/coarse/proximal inventory tiers. These controls remove task-tree alignment in different ways: latent correspondence, example-wise access and physical placement, respectively. The final placement control reverses the fine and global ranges and preserves their 21/22/21 widths. An initial three-cycle was invalidated by the prospective resource gate because it changed candidate mask slots; that complete execution remains in the audit archive and does not enter inference.

Branch factors are ordered soma-to-distal by the production layout planner. For architecture $[b_1, \dots, b_{D_p}]$, the number of nonsomatic units is $\sum_{j=1}^{D_p} \prod_{r=1}^j b_r$. Thus D1 [8] has eight distal units per soma; D2 [2, 3] has two proximal and six distal units; and D3 [2, 1, 2] has two proximal, two middle and four distal units. The unary middle expansion is deliberate: all depths have eight nonsomatic units. The production structured-mask builder assigns all 4/2/2 tiers to the single D1 stage, fine+coarse tiers to the D2 distal stage and the global tier to its proximal stage, and one fine, coarse or global tier to each D3 stage. Construction-time resource records additionally verified equal trainable parameters (66,178), active synapses (14,336), candidate slots (21,760) and persistent state scalars per sample (2,944). The physical branch law was the same positive-conductance division used elsewhere in the paper, with identity reactivation and learned softplus-positive synaptic weights.

The original canary used signal contrast 0.24, training gain SD 0.25 and an unseen test gain SD 1.2. Deeper networks fit training better, but BP and both LocalCA transports remained at chance on the shifted test distribution (Supplementary Fig. S15A). This failed the accessibility gate and is retained as a negative result. A two-seed test gain ladder showed monotone erosion under increasing shift. A coupling ladder at matched train/test gain identified a seed-consistent interaction: from child conductance 1 to 64, the D3-minus-D1 accuracy contrast increased by 0.1236 on average, while D1 changed minimally. A signal ladder at child conductance 16 then located the non-ceiling signal-contrast boundary. Contrast 0.72 narrowly failed the frozen D1 accessibility threshold (mean 0.597), and the separately run 0.80 boundary passed with D1, D2 and D3 means 0.6086, 0.6737 and 0.9227. These exploratory seeds 10100–10141 were excluded from inference.

Before observing any seed 10200 outcome, we froze a 270-fit replication at signal contrast 0.80, per-factor train/test log-gain SD 0.25 and child conductance 16. The seven cohorts were aligned shunting BP; zero-alignment, shuffled-sensor and reversed-placement shunting BP; aligned raw-additive BP; and aligned or reversed-placement shunting LocalCA with shared-soma and path transport. Every cohort crossed D1, D2 and D3 over the same ten paired seeds 10200–10209. The primary contrasts, resource/accessibility gates, bootstrap procedure and claim boundaries were fixed in the dated experiment contract. In particular, aligned D3–D1 alone tests forward capacity. Task-tree specificity requires an aligned-minus-control depth interaction, and a local-credit claim requires the corresponding LocalCA contrast. Raw additive is a mechanism control; a point model with sufficient flexible capacity can in principle emulate the hierarchy, a boundary tested directly in the same-task controls below.

All 270 frozen fits passed the completeness, finite-metric, no-fallback and exact-resource gates. With aligned shunting BP, mean accuracy was 0.6096, 0.6752 and 0.9182 at D1–D3, giving a paired D3-minus-D1 effect of 30.86 percentage points (95% seed-bootstrap interval 30.45–31.30; 10/10 positive; exact sign-flip $P = 0.001953$).

The raw depth effects were 0.02 points for independent sensors, -0.01 for shuffled trials and -0.43 for reversed placement. Their aligned-minus-control depth interactions were 30.84, 30.87 and 31.28 points, respectively, all positive in 10/10 paired seeds. Aligned raw-additive BP instead fell from 0.5733 to 0.5190. Under LocalCA, aligned D3-minus-D1 was 16.95 points (16.37–17.44) for shared-soma feedback and 27.90 points (27.24–28.53) for path transport. The corresponding aligned-minus-reversed interactions were 17.72 points (17.27–18.18) and 28.68 points (28.07–29.27), both positive in 10/10 seeds. These interactions, rather than the aligned depth effects alone, support task–tree specificity.

As a separate post-training mechanism diagnostic, we replayed one fixed held-out batch through each of the 30 aligned-shunting BP checkpoints. Exact path LocalCA reproduced the selected autograd conductance-gradient direction at every checkpoint (minimum cosine 0.9999999999994; no missing gradient tensors). Mean shared-soma cosine was 0.9457 at D1, 0.7990 at D2 and 0.7674 at D3. Its paired D3-minus-D1 change was -0.1784 (95% seed-bootstrap interval -0.2082 to -0.1490 ; 0/10 positive pairs). This replay validates the implemented path recursion and shows that a common neuronal error becomes a poorer directional proxy as additional physical stages differentiate branch states; it was not a separately preregistered performance endpoint.

4.14 Same-task point, nonserial and credit-coordinate controls

We added three controls without changing the calibrated task, data, seeds, initialization policy or stopping rule. First, an all-active grouped-star model used exactly the same branch layers, structured masks and parameters as the serial tree, but each nonsomatic layer read the external input independently; the retained child aggregators were summed only once at the soma. State-dictionary values and resource counts were identical at initialization. D1 logits were numerically identical, as required. At D1 the serial-minus-star contrast was -0.005 percentage points (-0.067 to 0.054 ; 6/10 positive; $P = 0.844$). At D2 and D3 it was 6.54 (6.20–6.88) and 30.81 points (30.41–31.26), respectively, both positive in 10/10 seeds ($P = 0.001953$). Under reversed placement the D3 serial-minus-star contrast was -0.43 points (-0.68 to -0.16 ; 1/10 positive; $P = 0.0137$), and the paired alignment interaction was 31.24 points (30.65–31.77; 10/10 positive; $P = 0.001953$). Serial divisive composition, rather than the inventory of branch units, is therefore required for the aligned effect in this implementation.

Second, point MLP widths were chosen before fitting to match either 15,055 active parameters or, to the nearest realizable integer width, 66,200 total parameters (the tree has 66,178). Their mean test accuracies were 0.9861 and 0.9902. They exceeded serial D3 by 6.79 points (6.19–7.34) and 7.20 points (6.75–7.68), respectively, in 10/10 paired seeds ($P = 0.001953$ for each). Neither point model was given the hierarchy. This rejects an absolute expressivity advantage for the tree while leaving a matched structured-resource interpretation intact.

Third, soma-broadcast autograd retained the standard synaptic derivatives but replaced each nonsomatic loss derivative by the loss derivative at its owning soma. Forward values and logits were identical to full BP before training. With the standard BP optimizer, full BP minus soma broadcast at D3 was 0.64 points (0.15–1.17; 8/10 positive; $P = 0.0332$). Only after this 140-fit primary extension completed did a configuration audit reveal that the frozen LocalCA reference used a different, split optimizer. A dated amendment therefore froze 60 additional broadcast fits using the LocalCA optimizer before their outcomes were viewed. This diagnostic broadcast reached 0.7771 at aligned D3, compared with 0.7778 for shared LocalCA: broadcast minus LocalCA was -0.06 points (-0.23 to 0.09 ; 5/10 positive; $P = 0.492$). Changing only the broadcast optimizer cost 13.47 points (12.89–14.05; 10/10 positive). Within LocalCA, path transport exceeded the shared coordinate by 10.96 points (10.34–11.58; 10/10 positive), while full BP remained 3.09 points above path LocalCA (2.48–3.79; 10/10 positive). Thus, for this task, the tested eligibility expression adds no detectable penalty after matching teaching coordinate and optimizer; path specificity and optimizer regime account for the larger separations. The amendment is diagnostic rather than prospectively confirmatory, and no comparison turns autograd into a biological mechanism.

Across the primary and amended controls, all 200 new fits were finite, contained no fallback warning and used exactly seeds 10200–10209. Serial and star models had 66,178 trainable parameters, 14,336 active contacts, 21,760 candidate slots and 2,944 persistent state scalars. Seed-level outcomes, paired contrasts, configuration manifests and a machine-readable audit are distributed in `source_data/point_dendrite_credit_controls/`.

4.15 Alignment-dose interpolation of the physical-depth effect

After the $\alpha = 0$ and 1 endpoint results were available, we froze an interpolation at $\alpha \in \{0.25, 0.50, 0.75\}$ without changing any other generator, architecture, optimizer, stopping or seed setting. The 90 new fits crossed three physical depths with ten paired seeds. Combined with the frozen endpoints, paired D3-minus-D1 effects were 0.019, 0.207, 0.820, 3.126 and 30.855 percentage points at $\alpha = 0, 0.25, 0.50, 0.75, 1$, respectively. The corresponding positive-pair counts were 6, 8, 10, 10 and 10 of ten. Exact two-sided sign-flip tests gave $P = 0.844$ at $\alpha = 0$, 0.0703 at 0.25 and 0.00195 at every higher level.

The frozen primary summary fit an ordinary least-squares line to the five paired depth effects within each seed. Its mean slope was 25.84 percentage points per unit α (95% paired-seed bootstrap interval 25.43–26.26; 10/10 positive; exact sign-flip $P = 0.001953$). This positive summary does not make the response linear. The change from $\alpha = 0.25$ to 0.75 was only 2.92 points (2.65–3.17), whereas the final 0.75 to 1 step contributed 27.73 points (27.35–28.11). The interpolation therefore supports a steep boundary near complete alignment in this calibrated task. It does not supply a second task family, hierarchy depth or biological replication.

A frozen-source guard stopped 26 scheduled jobs when an unrelated source edit occurred during the first array. The completed pre-edit jobs were retained and the stopped jobs rerun because the exact source diff only introduced an optional rule override that none of these resolved configurations enabled. Both manifest hashes, the condition-level deduplication and the equivalence argument are recorded in the dated mid-array source-equivalence report. All 90 unique intermediate fits passed the final finite-metric, no-fallback, seed and equal-resource gates. Machine-readable outcomes and paired contrasts are in `source_data/physical_alignment_dose/`.

4.16 Literal grouped-point control and independent hierarchy depth

The all-active grouped star above retains serial-shaped child aggregators as independent arms. We therefore implemented a stricter grouped-point control in which each nonsomatic level receives the raw external excitatory and inhibitory streams independently and projects directly to its owning soma. Every local branch module, structured mask and shunting equation is retained, but no level inherits another branch state. Each direct projection has the same scalar count as the serial child aggregator it replaces and copies its initialized raw conductances. Tests required copied projection parameters, finite gradients and exact D1 logits against the serial hierarchy.

After the preceding $H = 3$ endpoints were known, a dated contract froze 60 literal grouped-point fits crossing aligned or reversed placement, D1–D3 and the same paired seeds 10200–10209. The frozen serial outcomes were retained unchanged as the reference. Under alignment, grouped-point means were 0.6096, 0.6100 and 0.6095 at D1–D3. Serial-minus-grouped-point was 30.87 percentage points at D3 (95% paired-seed bootstrap interval 30.48–31.31; 10/10 positive; exact sign-flip $P = 0.001953$), but -0.44 points after placement reversal (-0.69 to -0.16). The resulting alignment interaction was 31.31 points (30.73–31.84; 10/10 positive). The earlier grouped star minus the literal grouped point was 0.06 points at aligned D3 (0.01–0.13; 7/10 positive; $P = 0.0840$), which did not pass the frozen directional claim gate.

We separately changed hierarchy depth before viewing any new outcome. The $H = 2$ generator used two 32-feature sensor tiers, fixed inventory counts 4/4 and exact-resource D1 [8] and D2 [4, 1] morphologies. All remaining task, signal, gain, axial-coupling, optimizer and stopping settings were inherited without retuning. Aligned and reversed placement were crossed with serial BP, literal grouped-point BP and serial LocalCA with shared-soma or exact-path transport over ten fresh paired seeds 10300–10309, yielding 160 fits.

Aligned serial BP rose from 0.6588 at D1 to 0.9672 at D2. The 30.84-point depth effect (30.46–31.23; 10/10 positive; $P = 0.001953$) became -1.56 points under reversed placement, giving a 32.40-point interaction (32.10–32.70). Grouped-point depth effects were 0.02 points under alignment (-0.02 to 0.06) and -0.02 under reversal (-0.06 to 0.02); their alignment interaction was 0.04 points (-0.03 to 0.11). Serial-minus-grouped at aligned D2 was 30.82 points (30.41–31.23), and its architecture-by- placement interaction was 32.36 points (32.02–32.71). Shared-soma LocalCA retained a 21.20-point aligned depth effect (19.34–23.00) and a 22.93-point placement interaction (21.02–24.77). Exact-path LocalCA retained 31.38 points (30.95–31.83) and a 33.36-point interaction (33.00–33.75). Every positive interaction was ordered in 10/10 seeds with exact sign-flip $P = 0.001953$.

537 All 220 new fits were finite and stage complete, with no fallback or nonfinite alert, the expected seeds and exact
 538 within-hierarchy resource equality. The second hierarchy is a replication within the same synthetic mechanism-
 539 matched task family, not an independent dataset. Configurations, frozen manifests, run-level metrics, source hashes
 540 and every paired contrast are in `source_data/remaining_physical_experiments/`.

541 S5 Artificial-tree experiments

542 5.1 Architecture and input sparsity

543 MNIST images were converted to tensors, flattened to 784 inputs and not normalized or augmented. The 60,000-
 544 image training set was split by seed into 48,000 training and 12,000 validation images; the official 10,000-image
 545 test set was evaluated only after training. The core contained 128 modeled excitatory neurons and no separate
 546 inhibitory-neuron population. Each neuron had morphology [3, 3]: three proximal compartments, three distal
 547 children per proximal compartment and one soma, for 13 compartments per neuron. There were no somatic input
 548 synapses.

549 At each of the 12 dendritic compartments, separate TopK parameter arrays selected 40 excitatory and 20
 550 inhibitory contacts from the same 784-pixel input. The model therefore contained 92,160 active input contacts
 551 among 2,408,448 candidate synaptic parameters. Inactive local weights were not updated. Child-to-parent couplings
 552 and all active conductances were learned after a softplus transform. A linear 128×10 decoder produced the class
 553 scores. The archived model-resource record contained 2,414,602 trainable raw parameters in total, including
 554 candidate synaptic arrays, couplings, reactivation parameters and decoder.

555 5.2 Raw additive control

The configuration alias `dendritic_additive` set shunting and additive normalization to false and used the raw
 additive mode. For compartment n , its source implementation can be written

$$E_n^{\text{add}} = \sum_{i \in \mathcal{S}_n^E} g_i^E x_i, \quad I_n^{\text{add}} = \sum_{i \in \mathcal{S}_n^I} g_i^I x_i, \quad (62)$$

$$C_n^{\text{add}} = \sum_{c \in \mathcal{C}_n} g_{c \rightarrow n}^{\text{den}} a_c, \quad V_n^{\text{add}} = E_n^{\text{add}} - I_n^{\text{add}} + C_n^{\text{add}}, \quad a_n = f_n(V_n^{\text{add}}). \quad (63)$$

556 Consequently,

$$\frac{\partial V_n^{\text{add}}}{\partial g_i^E} = x_i, \quad \frac{\partial V_n^{\text{add}}}{\partial g_i^I} = -x_i, \quad \frac{\partial V_n^{\text{add}}}{\partial g_{c \rightarrow n}^{\text{den}}} = a_c, \quad \delta_c^V = \delta_n^V g_{c \rightarrow n}^{\text{den}} f'_c(V_c). \quad (64)$$

557 Thus this control retained subtractive E/I input, the same branching graph and reactivation, but omitted the
 558 conductance denominator, reversal-potential driving force and parent-compartment resistance. Raw-parameter
 559 gradients in both architectures also included the softplus derivative.

560 Every compartment used the learned reactivation

$$f_n(V) = \frac{\tanh[m_n(V - b_n)] + 1}{2}, \quad f'_n(V) = \frac{m_n}{2} \{1 - \tanh^2[m_n(V - b_n)]\}. \quad (65)$$

561 The shunting architecture used its analytical initialization policy. Because the raw additive voltage scale differs, the
 562 additive architecture used its predefined occupancy-quantile policy: the empirical 0.1 and 0.9 voltage quantiles were
 563 mapped to reactivation outputs 0.1 and 0.9, with minimum quantile width 10^{-3} , maximum slope 50 and reversion
 564 on an invalid fit. These were architecture-specific, prespecified gate policies rather than identical initial gate values.

565 5.3 Training protocols and archive boundary

566 The primary feedback-definition experiment used the three-factor rule, a locally updated decoder, mini-batches of
 567 256 and 180 epochs. Adam used $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\varepsilon = 10^{-8}$ and no weight decay. Raw TopK parameters, child

couplings, reactivation parameters and decoder used learning rates 1.5×10^{-3} , 7×10^{-4} , 5×10^{-4} and 1.5×10^{-3} , respectively. Local gradients were clipped elementwise at 5. The best validation checkpoint was restored; early stopping and mixed precision were disabled. Runs used deterministic PyTorch settings, disabled TF32 and cuDNN benchmarking, and used the highest float32 matrix-multiplication precision.

The reported feedback comparison contains 15 seeds (42–56) per architecture and feedback condition. It is a clean current-code cohort run after freezing the design and replacement policy. Complete resolved configurations, checkpoints, logs, source and output hashes, and job records are retained for all 60 runs. The clean cohort replaced the earlier mixed archive irrespective of the observed effect; the superseded rows remain in the audit archive but are not used in the article.

The Fashion-MNIST replication retained this architecture, contact pattern, three-factor local rule, local decoder, optimizer, 180-epoch limit and architecture-specific initialization without retuning. Images were flattened to 784 inputs without normalization or augmentation. For each fresh seed 10200–10209, 48,000 of the 60,000 training images were used for fitting and 12,000 for validation; the official 10,000-image test set was used only after checkpoint selection. Sixty fits crossed shunting and additive trees with scalar fallback, neuron-indexed feedback and exact path transport.

Mean accuracies for scalar, neuron-indexed and exact path fields were 0.8393, 0.8839 and 0.8829 in shunting trees and 0.8442, 0.8820 and 0.8833 in additive trees. Neuron identity improved over scalar fallback by 4.46 percentage points (95% paired-seed interval 4.09–4.88) in shunting trees and 3.78 points (3.43–4.14) in additive trees; every seed difference was positive in both architectures (exact two-sided sign-flip $P = 0.001953$). Exact path minus neuron identity was -0.11 points (-0.21 to -0.01 ; 2/10 positive; $P = 0.0957$) and 0.13 points (-0.08 to 0.35 ; 8/10 positive; $P = 0.314$), respectively. Thus the dominant scalar-to-identity bottleneck replicated, while within-tree transport added no reliable benefit on this dataset.

The preregistered replication gate required at least 8/10 positive identity differences and an interval above zero in each architecture. The artifact gate required all 60 finite endpoints, complete checkpoints, exact registered seeds and no fallback warning; all gates passed. A frozen-source guard split each architecture across two source manifests after a concurrent edit. The exact diff added an optional rule override that no resolved configuration enabled; the audit retains both manifests. Seed-level outcomes, contrasts and source identities are in `source_data/fashion_feedback_ladder/`.

The exact-transport factorial was a separate five-seed protocol (seeds 42–46): it used 200 epochs, a ReLU core output, and crossed three- versus five-factor eligibility with local versus backpropagated decoder updates. TopK, coupling, reactivation and decoder learning rates were 1.5×10^{-3} , 7×10^{-4} , 5×10^{-4} and 1.5×10^{-3} . Its same-architecture backpropagation reference used the same seeds, 200-epoch budget, ReLU output, and validation-based checkpoint selection, with method-specific learning rates of 10^{-3} for TopK and decoder and 5×10^{-4} for coupling and reactivation. Test data were not used for checkpoint selection. Tables S9–S11 provide the frozen specifications and archive status; Table S14 reports the principal outcomes.

The result is not that shunting universally improves learning. Preserving neuron identity recovers most of the scalar-feedback deficit in both tested architectures. The separate exact-transport experiment establishes that the local three-factor eligibility is sufficient when the remaining compartment error is supplied accurately.

Exact reconstruction of the analytic gradient against automatic differentiation used 100 diagnostic runs. Median relative error was 7.43×10^{-8} and the maximum was 1.88×10^{-7} . Parameter tensors, rather than scalar tensor elements, were the diagnostic units.

5.4 Prospective depth-by-feedback experiment and checkpoint diagnostic

The prospective experiment crossed two tasks, two neuron models, one to four dendritic stages, and three local teaching fields at ten paired seeds. A matched backpropagation condition was included for each task–model–depth–seed combination, giving 480 local-learning and 160 backpropagation runs. The local conditions all used the theorem-derived three-factor eligibility. One modeled population of 128 neurons was used so that equal neuron indices in different feedforward populations could not share a teaching coordinate. The design, configurations, seeds, audit criteria and analysis code were frozen before confirmatory outcomes were inspected.

Every run was required to contain a resolved configuration, checkpoint, complete learning curves, resource record and finite held-out metrics. The audit also required every expected dendritic stage, nondegenerate learned couplings and the absence of transport-shape fallbacks or critical log errors. All 640 runs passed those artifact checks. A later outcome-independent input- validity audit excluded the 160 synthetic-noise shunting runs because their signed transfer output is not a valid positive-conductance drive. The 480 retained runs contain both neuron models on MNIST and the additive model on the noise task. Nominal depth changed compartment and parameter count and is therefore interpreted as a depth stress rather than a capacity-matched comparison. Table S15 summarizes the retained feedback contrasts.

To separate feedback dimension from assignment, an additional frozen cohort compared correct neuron-indexed sharing with a fixed derangement at depths two and four. The derangement preserves the number and per-example distribution of teaching coordinates, the forward network, and the training path; only the coordinate-to-tree address changes. Applying the same validity rule removed the 40 synthetic-noise shunting runs. Table S17 reports the six retained paired contrasts from 120 runs.

A second frozen cohort tested inhibitory dose on the noise-resilience task. It crossed two neuron models, inhibitory-contact counts of 0, 5, 10, 20 and 40 per branch, backpropagation or three local feedback fields, and ten paired seeds, for 400 runs. Its primary endpoint was the change from dose 0 to dose 40 in the shunting-minus-additive accuracy difference. Because all 200 shunting cells used the invalid signed-input convention, the full family is excluded from figures and inference; its executions remain in the public validity ledger.

5.5 Detached clean exact-transport/backpropagation audit

The replacement implementation audit crossed exact path transport or standard backpropagation, MNIST or the synthetic noise-resilience task, additive or positive-conductance shunting cores, depths one through four and seeds 42–51. It therefore contained 320 training runs and 160 method pairs. All accepted runs executed from detached tracked-clean source commit 74792ca91724066148f36c758191744a315d62b3; the nested synthetic-data repository was fixed at commit 4f3612a52756c3691e5bd269c6465c02f240c1e8. The additive synthetic model retained its signed transfer output. The positive-conductance shunting model used an explicit ReLU output transfer for both learning methods before conductances entered the denominator. This is a corrected input-valid audit, not a byte-identical replication of the historical signed-shunting cells.

Acceptance required a resolved configuration, seed record, learning summary, finite final metrics, finite stage-complete checkpoint, checkpoint and configuration hashes, and the newest scheduler and training logs without a fatal error, non-finite training event or transport-shape fallback. If device memory was insufficient only for full-dataset metric materialization, the runtime logged the event and recomputed the same final metric in streaming batches; the collector recorded this path separately and still required a finite endpoint. Initial attempts that lacked the ignored nested data repository, requested insufficient host or GPU memory, or reached the signed- shunting input guard produced no accepted endpoint. Missing configurations were rerun unchanged on H200 accelerators. Inference paired exact transport and backpropagation by seed within task, core and depth. The four plotted task–core effects first averaged the four depths within seed and then used a 20,000-draw paired-seed bootstrap interval; two-sided Wilcoxon tests are supplied as descriptive implementation checks rather than equivalence tests. All 320 runs passed these gates. Forty-two used the explicitly logged streaming recomputation for at least one terminal metric; every accepted final metric was finite. Table S16 reports the four depth-averaged comparisons. Across all 160 depth-specific method pairs, mean exact-minus-backpropagation accuracy was -0.0545 percentage points and the largest absolute individual pair difference was 5.90 points. The latter precludes interpreting the audit as pointwise equality even though each depth-averaged task–core interval included zero.

A third cohort compared fixed spatial and random sparse input maps at depth four. The 320 runs crossed two tasks, two neuron models, backpropagation or three local feedback fields, two maps, and ten paired seeds. Each distal branch had 21 contacts. Spatial branches recursively partitioned the 28×28 feature plane into 16 disjoint regions per neuron, whereas random branches sampled independently from all 784 coordinates. Supplementary Fig. S7 reports the 240 validity-qualified learning contrasts together with a configuration-level coverage audit. MNIST retains both cores; the noise task retains the additive core. Because the effect persists under backpropagation

and on the randomly projected task, it is treated as a forward sparse-connectivity prior rather than credit-routing evidence.

A fourth frozen cohort varied depth while holding 16 terminal branches and approximately fixed input contact budget. The four depth conditions contained 960, 960, 968 and 960 active contacts per soma, respectively. It crossed two neuron models, backpropagation or three local feedback fields, four depths and ten paired seeds, giving 320 runs. The input-valid synthetic comparison retains the 160 additive runs. This control matches terminal branches and contacts, not trainable parameters: deeper networks still contain more compartments and more coupling and reactivation parameters. Supplementary Fig. S8 reports their complete depth profiles and the prespecified depth-four-minus-depth-one contrasts.

We then evaluated scalar, neuron-indexed and exact teaching fields at the 120 input-valid backpropagation checkpoints on one frozen test batch. Only non-somatic stages and their trainable branch parameters entered the field and gradient vectors. For each candidate field we computed scaled error-field capture, eligibility-weighted gradient capture and cosine, and the loss change after matching its update norm to the exact gradient. The primary relative step was 10^{-5} ; 10^{-6} , 10^{-4} and 10^{-3} are retained as sensitivity values. The association interval resampled checkpoints while keeping paired feedback fields together. The diagnostic did not update or select checkpoints. Table S20 reports the primary operating point. Supplementary Fig. S9 displays the complete diagnostic.

5.6 Frozen phase-1 trained subtree-address experiment

This experiment implements the primary within-neuron support contrast from the trained-address contract on a deliberately minimal two-sibling model. It does not implement the complete architecture-by- K factorial. The phase-1 JSON configuration and executable were hashed before canary seed 4099 was run. The canary checked artifact integrity and nondegeneracy only: neuron-shared capture was 0.506, correct-route capture was 1.000, correct-route held-out accuracy was 0.814, and analytic exact transport agreed identically with analytic backpropagation. Because all frozen gates passed, the same unchanged files were used for confirmatory seeds 4100–4109.

For seed-specific unit vectors $\mathbf{u}_A$ and $\mathbf{u}_B$, binary label sign $y \in \{-1, 1\}$ and context $c \in \{0, 1\}$, the two input means were

$$\mathbb{E}[\mathbf{x}_A | y, c] = y [\mathbb{I}(c = 0) - 0.75\mathbb{I}(c = 1)]\mathbf{u}_A, \quad (66)$$

$$\mathbb{E}[\mathbf{x}_B | y, c] = y [\mathbb{I}(c = 1) - 0.75\mathbb{I}(c = 0)]\mathbf{u}_B. \quad (67)$$

Each stream had 12 coordinates and independent Gaussian noise with standard deviation 1.15. The 2,048-example train and test splits contained exactly 512 examples in each context–label cell. The somatic logit selected branch A when $c = 0$ and branch B when $c = 1$. Both inputs nevertheless remained active, preventing the one-coordinate rule from acquiring branch specificity through a zero eligibility on the distractor.

Let $\delta = \sigma(z) - \mathbb{I}(y = 1)$ and $\mathbf{r}(c) = (\mathbb{I}[c = 0], \mathbb{I}[c = 1])$. Exact and correct-subtree coefficients were $\delta\mathbf{r}(c)$; within-neuron derangement used $\delta\mathbf{r}(c)_{:-1}$; the shared neuronal and same-depth controls used $\delta(1, 1)$; and the random rank-two control used $\delta\mathbf{r}(c)R_s^T$ for a seed-fixed orthogonal rotation R_s . The gated point emulation used one somatic coefficient together with the context-gated features $(\mathbb{I}[c = 0]\mathbf{x}_A, \mathbb{I}[c = 1]\mathbf{x}_B)$, making its alternative address resource explicit. All conditions had 24 trainable weights and the same paired initialization and batch order. The random dense control had four routing nonzeros; the sparse two-route conditions had two.

Training used mini-batch SGD, learning rate 0.3, batch size 64 and 60 epochs. The switch endpoint applied another 12 epochs to 1,024 context-1 examples with balanced labels and measured the change in context-0 accuracy. For gradient geometry, exact and candidate per-example gradients were concatenated over both branches. Scaled capture was squared cosine after optimal scalar rescaling. The one-step candidate was normalized to the exact batch-gradient norm and evaluated at step 0.25. Table S1 reports every confirmatory condition; no failed condition was removed.

The very poor deranged loss and the variable random-map loss were retained as prespecified failures. The equality of correct subtree routing, exact transport, backpropagation and gated point emulation is equally important: in this shallow model the exact path coefficient is only the binary branch gate. The result therefore isolates address

Table S1: Frozen phase-1 trained within-neuron address results. Values are means across ten independent paired confirmatory seeds; intervals for the principal differences are reported in the main text and Source Data.

Condition	Channels	Test accuracy	Initial capture	Switch forgetting
Neuron shared	1	0.7624	0.5095	0.5566
Same-depth shared	1	0.7624	0.5095	0.5566
Correct sibling subtrees	2	0.8063	1.0000	0.0000
Within-neuron derangement	2	0.1963	0.0000	0.0006
Random dense rank two	2	0.5847	0.2791	−0.0014
Exact transport	2	0.8063	1.0000	0.0000
Analytic backpropagation	2	0.8063	1.0000	0.0000
Gated point emulation	1 + gated eligibility	0.8063	1.0000	0.0000

alignment but does not establish a dendrite-specific advantage over a point system supplied with equivalent gated parameter groups.

5.7 Confirmatory subtree-address bandwidth and architecture factorial

The complete factorial replaced the two-sibling task with eight context- selected streams arranged in a balanced binary hierarchy. Context and binary label were exactly balanced in train and test sets, every stream remained active on every example, and the sign of the task signal reversed outside the selected context. This prevents a shared coefficient from recovering the required address through silent eligibility. The prespecified feedback budgets were $K \in \{1, 2, 4, 8\}$ nested routes. Twenty paired confirmatory seeds (4200–4219) crossed five representations and 27 method–budget combinations, giving 135 fits per seed and 2,700 fits in total.

The dendritic tree, explicitly gated point model, flat-compartment model and grouped-point model were implementation-equivalent controls: each contained 64 trainable weights, 64 active input contacts and one output nonlinearity, and each received the same routed coefficient field. The fifth representation was a degree- and depth-matched rewired tree. Its resource counts were unchanged, but its ancestry groups were permuted relative to the task hierarchy. Feedback families comprised correct ancestry, fixed within-neuron derangement, depth-interleaved groups, random sparse groups, a seed-fixed random rank- K map, and a task-fitted rank- K upper bound. Neuron-shared, exact-transport and analytic-backpropagation references completed the design. Route fields were unit-normalized per example. Data, initialization, mini-batch order and every stochastic control map were keyed to seed, method family and K , so matched representations saw identical stochastic realizations.

A preconfirmatory canary required exact context balance, zero context–label correlation, nonzero activity in all streams, neuron-shared gradient capture below 0.2, exact capture of one, numerical equality of analytic exact transport and backpropagation, zero fallback counts, and exact agreement among the four implementation-equivalent representations. The first large execution was invalidated after a post-run audit detected that stochastic control maps were not paired across representation labels. Those rows were quarantined before scientific interpretation. The random streams were corrected, the implementation-equivalence check was promoted to a canary gate, and the entire 2,700-fit cohort was rerun. Only this corrected cohort is reported.

Table S2 gives the primary bandwidth contrasts. Correct ancestry strongly exceeded route derangement at $K = 2$ and $K = 4$, but its comparison with the best matched non-anatomical control was deliberately more stringent: ancestry lost at $K = 1$ and $K = 2$, won by 0.0127 accuracy at $K = 4$ (95% paired seed-bootstrap interval 0.0059–0.0197; 15/20 seeds; two-sided Wilcoxon $P = 0.0048$), and tied at full rank. Correct routing on the task-matched tree exceeded the rewired tree at $K = 2$ and $K = 4$ in all 20 seeds. The four matched representations were exactly equal in every condition (maximum held-out-accuracy difference zero). The experiment therefore supports an intermediate-bandwidth advantage for a task-aligned nested address, while showing that an explicitly routed point implementation can reproduce it.

5.8 Trained partition-residual reconstruction

We quantified irreducible address error in the complete factorial as a post-hoc mechanism diagnostic. The unchanged simulation script and frozen configuration matched their archived SHA256 hashes. For every example b and leaf coordinate i , we defined the exact coefficient field $c_{bi} = \delta_b e_{c_b, i}$, coordinate weight $w_{bi} = \|\mathbf{x}_{bi}\|_2^2$, and active route field ϕ_{bi} . Allowing the best single coefficient for that route on each example gives

$$z_b^* = \frac{\sum_i w_{bi} \phi_{bi} c_{bi}}{\sum_i w_{bi} \phi_{bi}^2}, \quad \mathcal{C}_{\text{part}} = 1 - \frac{\sum_{b,i} w_{bi} (c_{bi} - z_b^* \phi_{bi})^2}{\sum_{b,i} w_{bi} c_{bi}^2}. \quad (68)$$

This is the normalized complement of the shared-coefficient residual in Eq. 49. It grants an optimal route coefficient and therefore isolates address/support error from the coefficient scale actually used during training.

All 20 seeds and 135 conditions per seed were reconstructed at initialization and after 80 training epochs. Reconstructed archived losses agreed within 4.97×10^{-9} and accuracies within 5.01×10^{-11} , the precision of the archived decimal tables; the address-plus-coefficient Pythagorean decomposition closed within 7.28×10^{-12} . On the dendritic-tree representation, trained correct-ancestry capture was 0.162, 0.315, 0.547 and 1.000 at $K = 1, 2, 4, 8$. Fixed ownership derangement had the common scalar value at $K = 1$ and zero capture at $K = 2, 4, 8$, because its active support excluded the target leaf (Supplementary Fig. S23a).

For each seed we averaged implementation-equivalent representations within feedback family and budget, then correlated trained capture with held-out accuracy across conditions. The mean within-seed Spearman correlation was 0.824 (95% seed-bootstrap interval 0.810–0.838; 20/20 positive; two-sided Wilcoxon $P = 1.91 \times 10^{-6}$). Restricting the analysis to the six route families and $K < 8$ removed the exact full-rank boundary and gave 0.639 (0.610–0.665; 20/20 positive; $P = 1.91 \times 10^{-6}$; Supplementary Fig. S23b,c). This association is descriptive and post hoc. In particular, ancestry, depth-interleaved and random sparse partitions had similar address capture at matched K , so the residual does not explain the small trained ancestry advantage at $K = 4$.

5.9 Inherited regular-tree figures and expanded regime archive

The three figures below are reproduced without redrawing from the final NeurIPS/arXiv revision of the regular-tree study. Their PDF assets, panel definitions, and frozen generator scripts are preserved byte for byte in the journal repository. We retain them here because they show the original mechanistic and learning controls in exactly the form in which they were first reported. The following expanded archive reorganizes the same validated regular-tree evidence alongside additional task and stress controls; it is an extension, not a replacement for these inherited displays.

The journal synthesis retains the validated task, inhibition-dose, depth, broadcast-noise, rule-family, error-source, feedback-rank and flattened CIFAR-10 controls from the original study. These analyses use the historical frozen configurations rather than a newly tuned joint sweep. Most conditions contain five independent seeds; the rule-family and error-source comparisons contain three and are descriptive. Broadcast noise is zero-mean Gaussian noise added to the available teaching signal, with σ measured relative to the unperturbed signal scale. The depth sweep changes the number of dendritic stages at fixed somatic population width. CIFAR-10 images are flattened and use no convolutional front-end or competitive vision recipe. These experiments define operating regimes and feedback limits; cross-architecture differences do not isolate a backward shunting mechanism. Selected endpoints are reported in Supplementary Figs. S1, S2, S3, and S4, and Table S21.

S6 MICrONS morphology reconstruction

6.1 Cohort and data sources

The main analysis used eight layer 2–5 intratelencephalic cells from the functionally co-registered MICrONS cohort [1]. Stable nucleus identifiers were resolved to root IDs in materialization 1822 of minnie65_public.

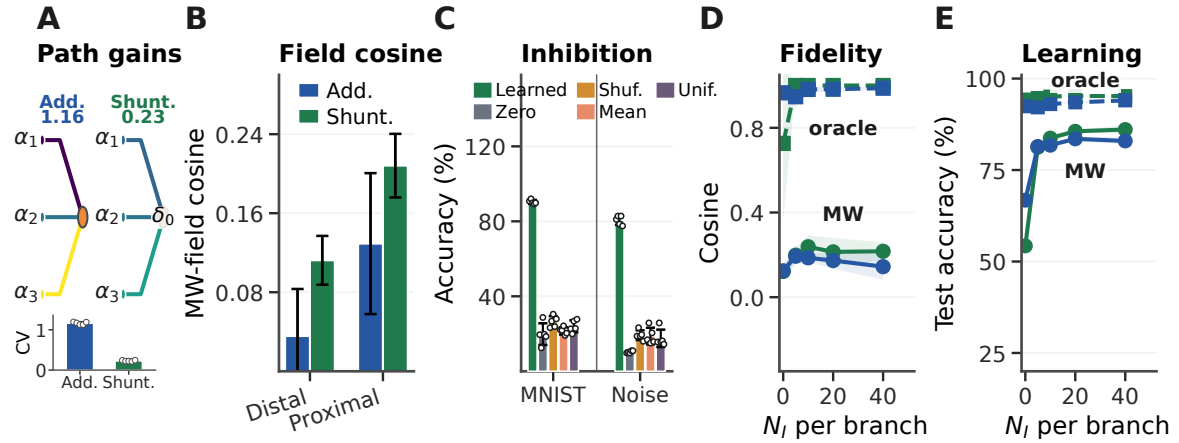


Figure S1: **Mechanistic chain from path gains to learning, reproduced from the original regular-tree study.**

A, Conductance-stage path-gain field at five inhibitory synapses per branch; the inset shows paired-seed coefficients of variation. **B**, Stage-resolved cosine between the matched-width/scalar-fallback field and the exact MNIST error on distal and proximal compartments. The width-matched somatic stage is excluded because its cosine is one by construction. **C**, Post-training intervention in the learned shunting model: inhibition is zeroed, sample shuffled, replaced by a per-branch batch mean, or replaced by one uniform matched mean. **D,E**, Feedback fidelity and learning on the noise-resilience task; solid lines denote matched-width/scalar-fallback feedback and dashed lines exact transported error. Error bars are ± 1 s.d. over checkpoints. This figure is the unchanged NeurIPS/arXiv asset generated by the archived original script. Its cross-architecture comparisons define operating regimes and do not establish a general backward shunting advantage. Noise-task shunting cells in panels **D,E** use the signed-input convention that the journal input-validity rule excludes for fresh cohorts; the corrected non-negative-transfer audit in the main text supersedes them for inference.

Dendritic skeletons were downloaded through the MICrONS skeleton service and incoming synapses came from synapses_pni_2. Table S22 reports the frozen cell-level inputs to the structural analysis.

Degree-two chains were collapsed into branch-to-branch cable segments. A synapse was assigned to the nearest dendritic cable node and retained if its distance was at most $5\mu\text{m}$. Across the eight cells, 28,669 classified incoming synapses passed the mapping and E/I classification criteria. The mean mapping pass fraction was 0.980. Presynaptic coarse cell type supplied E/I identity when available. Otherwise, high-confidence target structure was used as a secondary proxy from the in-development synapse_target_predictions_ssa_v2 table: spine targets were classified as excitatory and shaft or soma targets as inhibitory. This table does not currently have a stable public release identifier. Proxy and direct presynaptic labels agreed for 73.0% of 2,012 jointly labeled contacts. The principal capacity ordering was therefore repeated on the 2,077 directly typed contacts only.

6.2 Normalized electrical proxy

Raw axial coupling was proportional to radius squared divided by cable length; it was normalized by the median positive value within cell and clipped to 0.05–20. Raw leak was proportional to radius times the larger of cable length and radius; it was normalized by its median positive value and clipped to 0.1–10. Excitatory and inhibitory synaptic areas were normalized by the median nonzero total synaptic area. The resulting quantities are passive sensitivity weights, not fitted membrane conductances. The soma was the root of each compressed tree. Parent traversal verified acyclicity and one path from every retained segment to the soma. The compressed cable model used reciprocal axial coupling and a symmetric passive conductance matrix. It is therefore distinct from the directed child-to-parent artificial network of Supplementary Note 1. The two models share a local synaptic sensitivity

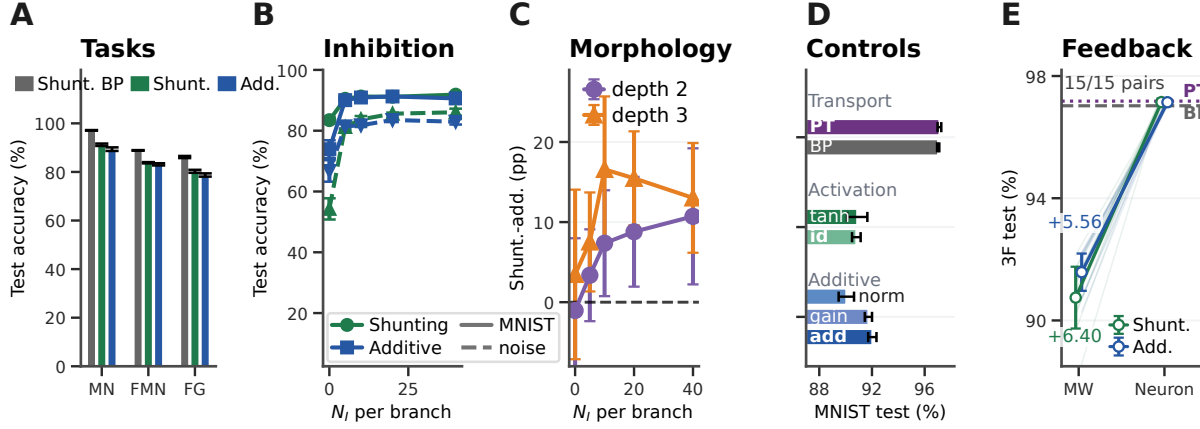


Figure S2: **Matched-capacity performance, regime dependence, and feedback definition, reproduced from the original regular-tree study.** **A**, Matched shunting backpropagation and five-factor local learning on MNIST, Fashion-MNIST, and figure-ground MNIST. **B**, Inhibitory dose-response. **C**, Descriptive morphology-by-inhibition performance. **D**, Exact-transport, reactivation, normalized-eligibility, gain-matched-additive, and additive-core controls on MNIST. **E**, Fifteen-seed comparison showing the gain from preserving one soma teaching coordinate for every compartment of the corresponding modeled neuron. Error bars are ± 1 s.d. This is the unchanged NeurIPS/arXiv figure asset; the prospective journal experiments extend its feedback result across depth, input-valid task-core conditions, and routing controls.

multiplied by a soma-derived compartment adjoint, but only the directed model has the path product in Eq. 9.

For excitatory segment i and candidate inhibitory route k , the direct model route matrix was

$$\Gamma_{ik} = A_{ik}\beta_k, \quad (69)$$

where A_{ik} indicates that k lies on the path from i to the soma and β_k is its normalized inhibitory-conductance fraction. This is a normalized ancestry-based route-sensitivity proxy. It is not the exact Green's-function path derivative of the reciprocal passive model used for the focal intervention. The participation rank of the excitatory-area-weighted kernel was $(\sum_j s_j^2)^2 / \sum_j s_j^4$, where s_j are singular values.

S7 Structural credit-capacity analysis

Perturbation fields were generated by multiplying independent route amplitudes by the kernel in Eq. 69. For each cell and channel count, the tested dictionaries were: a dense principal-component oracle; selected real ancestry paths; random nonempty paths from the same tree; bins of cable depth; and a column-wise ancestry shuffle that preserved dictionary dimensions and sparsity. Ancestry routes were ranked by β_k times the fraction of total mapped excitatory synaptic area among their descendants.

Both fields and dictionaries were multiplied by the square root of summed mapped excitatory synaptic area before ordinary least-squares projection, as defined in Eq. 29. Twenty independent Monte Carlo streams were used, each containing 384 perturbation fields and 64 random or shuffled dictionaries per cell and channel count. Streams measure numerical sensitivity and were averaged within cell before inference. The eight reconstructed cells were the units of analysis; they came from one animal.

At eight channels, morphology-selected paths captured 0.487 of the weighted energy in the model-derived route-perturbation fields (hierarchical 95% interval 0.400–0.581) at 6.92% dense-feedback wiring. This was 0.850 of the dense oracle's capture. Table S23 reports all matched controls. On the directly typed subset, weighted capture was 0.782 for the dense oracle, 0.696 for morphology, 0.347 for random paths, 0.447 for depth bins and 0.308

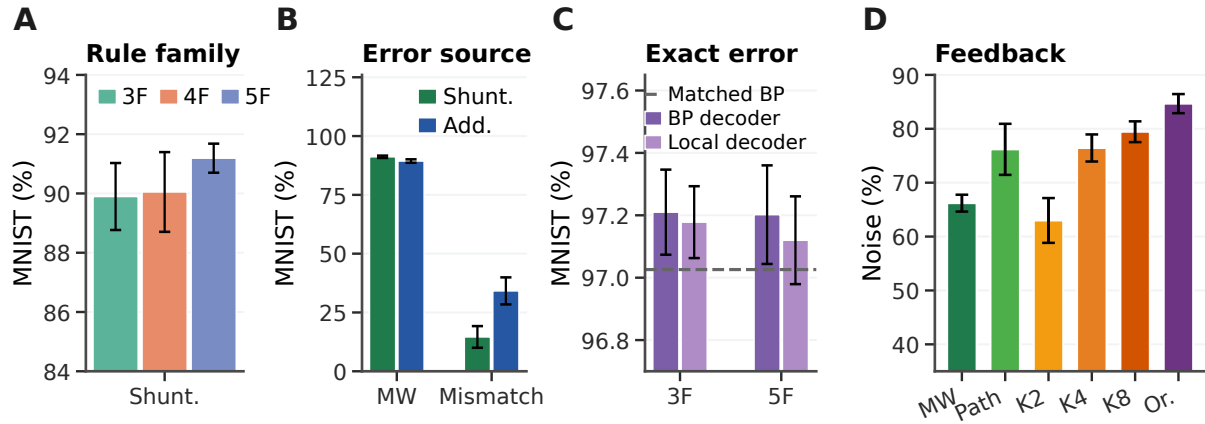


Figure S3: **Rule and feedback controls identify the principal learning bottleneck, reproduced from the original regular-tree study.** **A**, Within-shunting comparison of the three-, four-, and five-factor rules in an archived three-seed cohort. **B**, Somatic teaching error versus an unmatched compartment-local quantity. **C**, Five-seed exact-transport factorial showing that the theorem-derived three-factor rule approaches matched backpropagation when the required compartment error is supplied. **D**, Noise-resilience feedback ladder spanning scalar, path-propagated, random low-rank, and transported-oracle fields. Error bars are ± 1 s.d. This is the unchanged NeurIPS/arXiv figure asset and generator; exact transport is an information oracle. Noise-task shunting cells in panel **D** use the signed-input convention that the journal input-validity rule excludes for fresh cohorts; the corrected non-negative-transfer audit in the main text supersedes them for inference.

for ancestry shuffling. Ancestry-route capture exceeded every structural control in all eight cells. The direct-type analysis addresses sensitivity to E/I labels, but its smaller input set is not a population estimate. These fields are not observed task gradients. Their descent interpretation is conditional on treating them as exact gradients in the transformed coordinate system; the alignment-controlled analysis supplies an explicit optimization test.

We also froze a 55-cell sensitivity universe before computing outcomes, mapped it to the public MICrONS v661 static release, and excluded the original eight cells by stable nucleus identifier. Twenty independently generated perturbation streams were averaged within cell. At eight channels, morphology paths in the 47 remaining cells captured 0.756 of modeled field energy with 8.08% of dense wiring, compared with 0.839 for the dense oracle, 0.302 for random paths, 0.441 for depth bins and 0.330 for ancestry shuffles (Supplementary Fig. S10b–d). The morphology advantage over every structural control was positive in 47/47 cells. The cohort is from the same mouse, is layer-imbalanced, descends from a convenience export, and uses an earlier reconstruction release. Direct presynaptic E/I labels were available for 11,024 incoming synapses before mapping (4.71%); 10,516 mapped contacts entered the analysis. It is therefore a disjoint-cell robustness analysis, not an independent-animal or population replication. The frozen candidate and endpoint-exclusion manifests accompany the source data.

To remove the direct generator–dictionary correspondence, we formed an independent response operator on the original eight trees. A focal shunt of 10^{-3} times local leak-plus-synaptic conductance was added at every inhibitory-bearing segment of the reciprocal passive model. After an independently solved state-matching compensatory current restored the baseline somatic voltage, we finite-differenced the log magnitude of the exact excitatory-conductance gradient at every excitatory-bearing segment. The resulting response operator was full rank in all eight cells and placed 17.6% of its weighted energy, on average, outside direct ancestry support. For a dictionary Φ , expected capture was evaluated exactly as the squared Frobenius norm after projection of this operator onto $W^{1/2}\Phi$, divided by its total weighted Frobenius norm.

We added 200 surrogate trees per cell. At each topological depth, child nodes were randomly assigned among the original parent slots, preserving every node’s depth and every parent’s out-degree. All edge and node attributes

remained attached to their original child nodes. The same leverage rule was then applied to the surrogate dictionary. Row-shuffled selected paths preserved each selected column's values and nonzero count. At eight channels, morphology exceeded random real paths and row-shuffled paths in every cell, but did not reliably exceed depth bins or degree- and depth-matched surrogate trees (Table S24). This control supports sparse ancestry structure relative to random assignment while bounding the component specific to fine topology beyond depth and degree.

S8 Focal shunting perturbations

8.1 Adjoint formulation and matched additive control

For the passive conductance matrix G , voltage was $\mathbf{V} = G^{-1}\mathbf{b}$. For a loss depending on somatic voltage, the adjoint error was

$$\mathbf{q} = G^{-1}\mathbf{e}_s \frac{\partial \mathcal{L}}{\partial V_s}, \quad \frac{\partial \mathcal{L}}{\partial w_i} = q_i(E_E - V_i). \quad (70)$$

A focal shunt η_k gives $G' = G + \eta_k \mathbf{e}_k \mathbf{e}_k^T$. The Sherman–Morrison identity yields

$$\mathbf{q}' = \mathbf{q} - \frac{\eta_k q_k}{1 + \eta_k (G^{-1})_{kk}} G^{-1} \mathbf{e}_k. \quad (71)$$

Let $\mathbf{c}_k = G^{-1} \mathbf{e}_k$ be the k th Green's-function column and let $r_k = c_{k,k} = (G^{-1})_{kk} > 0$. Then

$$\Delta \mathbf{q} = -a_k \mathbf{c}_k, \quad a_k = \frac{\eta_k q_k}{1 + \eta_k r_k}, \quad \frac{\partial |a_k|}{\partial \eta_k} = \frac{|q_k|}{(1 + \eta_k r_k)^2} > 0. \quad (72)$$

The perturbation amplitude saturates at $|q_k|/r_k$, while its normalized spatial profile is fixed by $\mathbf{c}_k$ in this linear passive state. For a descendant set $D(k)$ and a prespecified matched comparison set $C(k)$, define

$$S_k(D, C) = \frac{\text{median}_{i \in D(k)} |c_{k,i}|}{\text{median}_{j \in C(k)} |c_{k,j}|}. \quad (73)$$

Because the common amplitude $|a_k|$ cancels, $S_k > 1$ is necessary and sufficient for a positive descendant-minus-comparison contrast in absolute adjoint change at fixed k . It is only a transport-level criterion. Relative or logarithmic changes additionally divide by the baseline q_i , and the full synaptic gradient also contains the post-perturbation driving force. This is why axial coupling, leak, background conductance and local input resistance must be varied explicitly rather than inferred from topology alone. The first-order-current-matched additive control injected $\eta_k(E_I - V_k)$ at the same node without changing G . An independently solved state-matching compensatory current injection restored baseline somatic voltage after both interventions. This is not a continuously adjusted voltage clamp. Therefore the somatic scalar error was fixed, while shunting changed adjoint transport and the additive control did not. Dendritic voltages were not clamped, so both interventions could also change local driving forces.

8.2 Frozen endpoint and validation

Before running the analysis, we fixed the site-selection rule, primary dose, comparison groups, and localization endpoint. Focal sites were selected by a depth-stratified rule without access to gradient outcomes. A site required at least three excitatory descendants and three comparison sites. At most 16 eligible sites were sampled per cell across quartiles of topological depth. The primary model used excitatory and inhibitory synaptic scales of 0.35 and normalized reversal potentials $E_E = 1$ and $E_I = -0.2$. Synapses were classified as descendants, ancestors, sister-subtree sites or unrelated sites. Unrelated synapses within each template were matched to descendant cable depth. For baseline gradient g_i and perturbed gradient g'_i , we defined $m_i = |\log[(|g'_i| + \varepsilon)/(|g_i| + \varepsilon)]|$, where $\varepsilon = 10^{-15} \max(1, \max_i |g_i|)$. For each site, the localization index was the median m_i among descendants minus the median among depth-matched unrelated synapses. Site effects were first averaged within cell.

The somatic loss was one-half the squared difference between somatic voltage and a target one normalized unit below its baseline value. Conductance doses were 0.25, 0.5, 1, and 2 times the local leak-plus-synaptic conductance. The state-matching current was solved after each perturbation and then treated as fixed when evaluating the post-intervention conductance gradient. The topology control sampled 200 relation templates from other focal sites, stratified by focal-site topological-depth quartile when another site was available; otherwise it sampled another site in the same cell.

The primary unit-dose test included 101 sites in eight cells. Mean localization was 0.104 under shunting and 0.035 under matched additive perturbation. The paired cell-level difference was 0.069 log units (95% bootstrap interval 0.053–0.086), positive in all eight cells. A relation control sampled 200 templates with the stratified-with-fallback procedure above. True focal relations exceeded this control by 0.080 (0.059–0.098), again in all eight cells. The dose contrast increased from 0.018 at 0.25 local-conductance units to 0.128 at two units.

We separated the two factors in the exact gradient algebraically. Writing $d_i = E_E - V_i$ and $d'_i = E_E - V'_i$, full shunting uses $q'_i d'_i$, adjoint-only substitution uses $q'_i d_i$, and driving-force-only substitution uses $q_i d'_i$. At unit dose, their mean localization indices were 0.104, 0.130 and 0.026, respectively, compared with 0.035 for the matched additive control. Adjoint-only minus driving-force-only localization was 0.104 (95% interval 0.080–0.125), positive in all eight cells.

More directly, full shunting and the driving-force-only state share the same post-shunt voltage field and differ only in whether the adjoint transport field is updated. Their localization difference was 0.079 and was positive in all eight cells. This is the exact factor-freeze contrast plotted in main focal-perturbation figure, panel h.

Because the gradient factors and the localization statistic are nonlinear, these four payoffs are not an additive decomposition. We therefore allocated the full-shunt-minus-additive contrast with two-factor Shapley values. For v_{00} equal to matched-additive localization, v_{10} adjoint-only, v_{01} driving-force-only and v_{11} full-shunt localization,

$$\phi_q = \frac{1}{2}[(v_{10} - v_{00}) + (v_{11} - v_{01})], \quad (74)$$

$$\phi_d = \frac{1}{2}[(v_{01} - v_{00}) + (v_{11} - v_{10})]. \quad (75)$$

The identity $\phi_q + \phi_d = v_{11} - v_{00}$ held at every site to a maximum absolute residual of 5.6×10^{-17} . Averaging sites within cells, the adjoint allocation was 0.087 (hierarchical 95% interval 0.063–0.111), while the driving-force allocation was -0.017 (-0.029 to -0.009). The former was positive and the latter negative in all eight cells. Within this algebraic allocation, the adjoint change accounted for more than the net positive contrast, while the driving-force change partially cancelled it. The matched additive reference is the relevant control but is not a literal no-factor corner of a physical 2×2 intervention. Using the unperturbed field as that corner assigned 0.104 (0.074–0.136) to the adjoint and approximately zero to the driving force. Both analyses allocate a nonlinear modeled payoff; the factor substitutions are not independent biological interventions.

All conductance matrices in the primary analysis were positive definite. The somatic state-matching residual was numerically zero. Analytic adjoint gradients agreed with finite differences to a maximum relative error of 1.03×10^{-6} . Table S25 reports prespecified conductance-scale and reversal-potential sensitivity runs. The direction of the shunting-minus-additive effect was positive in every condition, with seven or eight positive cells. These results establish a mechanism in the passive model on measured trees. They do not establish endogenous inhibitory teaching signals in vivo.

With proxy-classified contacts removed, the primary shunt-minus-additive contrast was 0.101 (0.075–0.129) across 55 sites and was positive in all eight cells. In the disjoint v661 cohort, 45 cells supplied 235 eligible sites. The same contrast was 0.138 (0.116–0.162), positive in 45/45 cells. Forty cells with at least two eligible sites supplied the reassigned-relation control; the true relation exceeded this control by 0.118 (0.094–0.144), with 39 positive cells and one numerical tie (main v661 robustness figure, panels f,g). Cells, rather than sites or relation draws, were the units of inference.

We also computed axial and leak conductances in physical units from the same compressed geometry. A cylindrical non-root segment had axial conductance $\pi r^2 / (R_a \ell)$ and membrane leak $2\pi r \ell / R_m$; the root was assigned spherical area $4\pi r^2$. All values were converted to nS. Because MICRONS synapse size is not an electrical conductance, synaptic sizes retained the prespecified relative scaling with the median physical leak as reference.

We varied R_a and R_m without changing site selection, relative dose, the state-matching somatic current, gradient definition or additive control. Table S26 shows the resulting electrotonic dependence. The large normalized-proxy contrast was recovered only in low- R_m , high-conductance settings; at $R_a = 150 \Omega \text{cm}$ and $R_m = 15,000 \Omega \text{cm}^2$ it was negligible or small. These are sensitivity settings, not fits to the recorded cells, and the model remains passive and steady-state.

We then froze a phase-1 matrix before its numerical canary. It fixed $R_a = 150 \Omega \text{cm}$ and crossed $R_m \in \{300, 1000, 15,000\} \Omega \text{cm}^2$, distributed leak-like background conductance of 0, 1 or 4 times baseline leak, and three dose definitions. Relative-local doses were 0.25, 1 and 2 times membrane-plus- synaptic conductance; fixed doses were 0.05, 0.5 and 5 nS; input-normalized doses were 0.25, 1 and 4 times $[(G^{-1})_{kk}]^{-1}$. The same 101 sites in eight cells supplied 16,362 site-condition-intervention rows. Every conductance matrix was positive definite (minimum eigenvalue 0.107 nS) and the maximum somatic-state residual was 3.33×10^{-16} .

At fixed absolute dose, the conductance-specific contrast did not follow the raw focal-shunt localization alone. At 0.5 nS, shunt-minus-additive localization was 0.0104 (0.0080–0.0126), 0.0073 (0.0044–0.0104), and -0.0039 (-0.0075 to -0.0004) at $R_m = 300, 1,000$ and $15,000$, respectively. At 5 nS the corresponding values were 0.0974, 0.0741 and 0.0061; the last interval included zero. Distributed background conductance restored the input-normalized effect in the high-resistance condition: at unit input-conductance dose, the contrast changed from -0.0004 with no background to 0.1210 and 0.2587 at background multipliers one and four. Across all 8,181 focal-shunt rows, every descendant gradient magnitude was attenuated, none was enhanced and no sign flip occurred. The passive mechanism is therefore a selective attenuator; signed reversal would require dynamics or active conductances absent from this phase (Supplementary Fig. S11).

8.3 Active steady-state local-Jacobian ensemble

We next tested whether the focal result survives voltage-dependent currents. The physical reconstructed-tree system used $R_a = 150 \Omega \text{cm}$, $R_m = 1,000 \Omega \text{cm}^2$ and distributed background conductance equal to baseline leak. Steady-state Na, K, Ca, HCN and NMDA current families had fixed reversal potentials, half-activation voltages, slopes and gate powers specified in the frozen JSON protocol. Global and segment-level maximal conductances were drawn independently from prespecified log-normal distributions. These draws vary plausible operating regimes; they are not fits to cell-specific channel measurements.

For each draw, a residual-reducing Newton solve found the nonlinear steady state. Acceptance required all voltages to lie in the frozen range, maximum absolute state residual below 10^{-10} , and minimum eigenvalue of the symmetrized local Jacobian above 10^{-8} . The implicit-adjoint calculation then used the exact inverse of this Jacobian. Focal shunting and a first-order-current-matched additive input were evaluated at 0.25, 1 and 4 times local input conductance; an independently solved somatic current restored the baseline somatic voltage. Up to 256 attempts were permitted to obtain 64 accepted draws per cell. All first 64 attempts were accepted in every cell, giving 512 equilibria, 101 focal sites and 38,784 site-draw-condition rows. All values were finite, the largest equilibrium residual was 9.51×10^{-11} and the smallest Jacobian eigenvalue was 1.084.

Channel draws and focal sites were averaged within each of the eight cells before inference. At doses 0.25, 1 and 4, mean shunt localization was 0.159, 0.529 and 1.339, respectively (Table S6). At unit dose, shunting exceeded the additive control by 0.382 (95% cell-bootstrap interval 0.356–0.409), positive in 8/8 cells (two-sided Wilcoxon $P = 0.0078$). Every modeled descendant gradient was attenuated, none was enhanced or changed sign, and descendant gradient energy fell to 0.330 of baseline. This extends the local sensitivity result beyond a passive matrix, but remains a steady-state linearization rather than a kinetic-channel or spiking simulation.

S9 Functional topology and task-derived credit

9.1 Direct anatomy-function join

Functionally imaged excitatory partners were linked to anatomical contacts through MICrONS/CAVE and DANDI mappings. The analysis contained 69 connected presynaptic partners across seven postsynaptic target cells. Au-

964 tomatic matches came from `coregistration_auto_phase3_fwd_apl_vess_combined_v2` in materialization
965 1822. We retained residual at most 10 and score at least zero, then deduplicated each presynaptic root by ascending
966 residual and descending score. All mapped contacts from a retained presynaptic root contributed to its partner-level
967 contact summary. For the one-site-per-partner model, contacts were first grouped by postsynaptic segment. The
968 partner was assigned to the segment with greatest summed contact size, breaking ties by synapse count, and its
969 major branch was the first segment below the soma on that segment's ancestry chain. Thus a partner with contacts
970 on several segments supplied one activity feature and one modeled learning site; its total contact size and synapse
971 count still included all retained contacts.

972 The target universe was the eight-cell morphology pilot. The prospective specification included every target in
973 that universe with a DANDI scan and directly connected, functionally imaged excitatory partners; seven targets
974 qualified. Operational checks required exactly one postsynaptic target in an extract, agreement between partner and
975 contact counts, at least five mapped partners for the topology screen, at least two sites after any requested task filter,
976 and at least one nonempty inhibitory ancestry route. The primary task analysis applied neither a manual-match
977 restriction nor a repeat-reliability threshold. The exact target/session/scan pairs are frozen in Table S27. The
978 archived specification does not record a deterministic rule for choosing among multiple eligible scans of the same
979 target, so these scan choices are treated as fixed pilot assets rather than as the output of a coverage-ranking procedure.
980 The omitted structural target, root 864691136043380566 (nucleus 292648), had one archived trial-manifest row, for
981 session 8/scan 4, but `has_dandi_asset=False` and no DANDI asset identifier or path. No additional exclusion
982 reason is inferred.

983 To remove ambiguity from selecting one scan per target, the extension applied one deterministic rule to the
984 complete extracted scan universe: retain every automatic-conservative target-session-scan cohort with at least
985 five unique connected imaged presynaptic roots. Thirteen scans across the same seven targets qualified. Pairwise
986 statistics were computed within scan, scan values were averaged within target, and the seven targets remained the
987 units of inference. Mean shared-path Spearman correlation was -0.0245 (95% target-bootstrap interval -0.214
988 to 0.147); the partial shared-path correlation after geometry and depth control was -0.0485 (-0.224 to 0.118),
989 positive in three of seven targets. Negative tree-distance correlation and the same-major-branch contrast were also
990 centered near zero (Table S7). Thus the deterministic scan-complete cohort preserves the original null boundary.

991 Each target had 464 trials and 280 unique condition hashes, of which 136 were repeated. Trial families
992 comprised 384 Clip, 40 Monet2 and 40 Trippy presentations. For each trial and region of interest, the response was
993 the mean raw `RoiResponseSeries` fluorescence over samples whose timestamps fell from trial start through trial
994 stop, including both endpoints. Start and stop offsets were zero. No trials were dropped; every response used here
995 was finite. Table S27 gives target-level asset identifiers and sample counts.

996 Functional similarity between two partners was the Pearson correlation of their vectors of mean raw fluorescence
997 over the 136 condition identities with at least two trials. Repeat reliability was computed separately for each region
998 of interest: occurrences of every repeated condition were divided into alternating first and second halves, responses
999 were averaged within each half, and the two vectors of condition means were compared by Spearman correlation.
1000 No reliability correction was applied, and the primary cohort retained all partners regardless of this diagnostic;
1001 thresholds of 0.1 and 0.2 were used only in sensitivity analyses.

1002 The controlled shared-ancestry statistic was computed separately within each target. Functional similarity and
1003 shared-path fraction were rank transformed. The two controls were $\log(1 + d_{\text{Euclidean}})$ and $\log(1 + |d_{\text{soma},i} - d_{\text{soma},j}|)$,
1004 where the second term is the absolute difference in soma path length; each control was also rank transformed. We
1005 regressed each ranked variable on an intercept and the two ranked controls by ordinary least squares, then took the
1006 Pearson correlation between the two residual vectors. The mean target-level partial-rank effect was -0.069 with
1007 a target-bootstrap 95% interval of -0.255 to 0.108 ; three of seven targets had positive effects. Pairwise partner
1008 comparisons were not treated as independent replicates.

1009 The functional cohort is deliberately interpreted narrowly. MICrONS calcium imaging measured excitatory
1010 neurons, observed partners cover only a small fraction of each target's complete inputs, reliability is modest, the
1011 volume contains one animal, and the recordings were not made during learning.

9.2 Task-derived branch credit

A branch model predicted the standardized postsynaptic response from the standardized responses of its connected presynaptic partners. Let $b(i)$ be the dominant major branch receiving partner/site i . The fitted model was

$$d_b = \sum_{i:b(i)=b} x_i w_i, \quad h_b = \tanh(d_b), \quad \hat{y} = \sum_b v_b h_b + c. \quad (76)$$

The objective was

$$\frac{1}{2N} \sum_{t=1}^N (\hat{y}_t - y_t)^2 + \frac{\lambda}{2} (\|\mathbf{w}\|_2^2 + \|\mathbf{v}\|_2^2), \quad \lambda = 10^{-3}, \quad (77)$$

with no penalty on c . The exact voltage-space error at site i was $(\hat{y} - y)v_{b(i)}[1 - \tanh^2(d_{b(i)})]$.

Whole condition-hash stimulus identities, not trials of the same identity, were split: 20% of the 280 identities and all of their trials were held out. Input columns and the target were standardized separately using training-set means and sample standard deviations (ddof = 1); invalid or zero scales were replaced by one and remaining nonfinite standardized values by zero. A stable SeedSequence combined base seed 20260721, target root, stream and replicate. There were ten fixed split and initialization replicates per target.

Input weights were initialized as $\mathcal{N}(0, 0.12^2/n_{\text{site}})$, branch readout weights as $\mathcal{N}(0, 0.25^2/n_{\text{branch}})$, and $c = 0$. Each method used 1,200 full-batch Adam steps with learning rate 0.015, $(\beta_1, \beta_2) = (0.9, 0.999)$ and $\epsilon = 10^{-8}$. The exact model was trained first. Its exact site-error fields on training trials defined the dictionaries and its held-out exact field defined reported capture. All projected-learning methods then restarted from the identical initial state. Only the input-weight error was projected; $\mathbf{v}$ and c always received their exact gradients.

The primary budget requested four channels, reduced only when a target had fewer sites or nonempty candidate routes. Ancestry candidates were columns of ancestry multiplied by local inhibitory fraction g_i/g_{tot} ; routes were ranked by this fraction times their descendant-site fraction. Random paths sampled nonempty candidates uniformly without replacement. Each ancestry-shuffle column was permuted independently over sites. Depth controls were path-length quantile bins, scalar feedback was an all-ones column, and the dense oracle used the leading right singular vectors of the exact training error. Projectors used relative singular-value tolerance 10^{-10} . No held-out response was used to fit a dictionary.

Held-out normalized MSE divided a method's held-out MSE by the MSE of predicting the training mean response. Replicates were averaged within target before inference; the seven target cells were the units. Intervals used 20,000 target bootstrap draws. Paired method comparisons used two-sided Wilcoxon tests, and the capture-utility analysis used method-label permutations stratified within target.

Table S28 reports the four-channel primary analysis. The dense four-dimensional oracle nearly matched exact learning, establishing that feedback dimension alone was not limiting in this surrogate. Ancestry-route did not reliably outperform ancestry shuffling in capture or learning. The within-target association between capture and normalized error was $\rho = -0.150$ with a target-stratified method-label permutation $P = 0.419$. The target-bootstrap 95% interval for ρ was -0.449 to 0.144 . The coefficients are least-squares oracles and therefore upper bounds on the capacity of each feedback family, not biologically implemented encoders.

Sensitivity analyses at one, two and eight channels, with direct manual matches, with repeat-reliability thresholds, and with a linear branch model did not establish a reliable morphology-specific learning benefit. Some subsets improved morphology-specific capture without improving learning. This separation is the empirical reason the article distinguishes anatomical route capacity from task alignment.

9.3 Complete reconstructed-tree learning across all eligible scans

The segment-resolved extension eliminated the major-branch state reduction. Every complete compressed reconstruction was instantiated as a reciprocal passive conductance system. Each imaged partner supplied a trainable non-negative excitatory conductance at its dominant mapped segment. For trial-specific soma prediction error, the exact input gradient factored into the local term $x_i(E_E - V_i)$ and the exact soma-to-segment adjoint from the inverse conductance operator. Readout and bias parameters received exact gradients under every method.

Restricted rules received scalar output error, local eligibility and one fixed four-channel site vector. The vector was calibrated before training from a conductance-weighted topology dictionary, an equal-rank random selection of nonempty anatomical routes, or a column-value-preserving site shuffle. No target response entered this calibration. Ten seed-fixed splits per scan held out complete stimulus identities. Runs were averaged within scan and then within target. The resulting design contained four methods, 13 scans and ten splits, for 520 fits. A run was excluded for missing routes, rank mismatch, nonfinite values, incomplete scan coverage or failed finite differences; no run was excluded. One exact conductance coordinate per run agreed with finite differences to maximum relative error 3.05×10^{-7} .

Mean normalized held-out MSE was 0.788 for exact compartment error, 0.832 for topology, 0.833 for random routes and 0.839 for site shuffling (Table S8). The topology MSE improvement over shuffling was 0.0070 (95% target-bootstrap interval -0.0207 to 0.0321) and over random routes 0.0013 (-0.0090 to 0.0113). At the common exact checkpoint, topology captured 0.232 of the eligibility-weighted update field, compared with 0.358 for random routes and 0.439 for site shuffling. Thus a full segment tree and deterministic scan inclusion do not rescue anatomy alignment for this measured-response objective. This is a defined null result, not evidence against topology–function alignment under other behavioral states, response windows, input coverage or cell-specific biophysical calibration.

1067 **S10 Alignment-controlled optimization**

The alignment-controlled experiment was specified as a constructive positive control for the conditional claim. For each reconstructed cell, we rebuilt the conductance-weighted ancestry kernel and selected eight ancestry routes by inhibitory fraction times descendant excitatory synaptic-area fraction before drawing any gradient fields. Coordinates were weighted by the square root of summed mapped excitatory synaptic area. Let Q be an orthonormal basis for those selected routes. Paired unit vectors $\mathbf{u}_{\parallel} \in \text{span}(Q)$ and $\mathbf{u}_{\perp} \perp \text{span}(Q)$ generated

$$\mathbf{g}(a) = \sqrt{a}\mathbf{u}_{\parallel} + \sqrt{1-a}\mathbf{u}_{\perp}, \quad a \in \{0, 0.2, 0.4, 0.6, 0.8, 1\}. \quad (78)$$

Every field therefore had unit energy and exactly fraction a in the morphology-selected span. The parallel and orthogonal directions were reused across alignment levels.

Controls were eight random paths from the same tree, eight path-distance quantile bins, and an independent row permutation of each selected morphology route. The last control preserved each column’s values and nonzero count while breaking ancestry. Random and shuffled dictionaries were redrawn within each nested stream and held fixed across alignment levels in that stream. Each of the eight cells contributed 40 streams with 256 fields per stream.

Optimization used a positive diagonal quadratic whose curvature spectrum ran from 0.5 to 1.5. The spectrum was independently permuted within each stream and then fixed across methods and alignment levels. One-step comparisons matched update norm at 0.1 times exact-gradient norm. Iterative projected descent used learning rate 0.25 for 20 steps. Progress was normalized by the corresponding full-gradient progress. Supplementary Fig. S5 and Table S33 report the principal values. Across the three control dictionaries and alignment levels, the median within-cell Spearman association between initial capture and 20-step progress was $\rho = 0.990$ (cell range 0.969–0.996). This analysis establishes conditional sufficiency and links capture to loss reduction under a controlled objective. Because alignment is imposed and coefficients are fitted from the exact gradient, it is neither an independent biological result nor a model of how a teaching circuit learns its coefficients.

1088 **S11 Synapse-resolved inhibitory route analysis**

The inhibitory-census endpoints and controls were frozen in a prespecified analysis contract before the data join. The frozen cohort was the complete stable-nucleus-ID intersection between the 47-cell v661 sensitivity cohort and the v795 inhibitory census: 20 targets, with no selection by an outcome. Because the two releases use different coordinate representations, the raw-to-slanted affine map was estimated from paired published inhibitory-synapse locations and required to reproduce all paired locations to less than $10^{-6} \mu\text{m}$ before any target was analyzed. Five

1094 targets had different root IDs across releases and were joined by stable nucleus identity. The source tables contained
1095 116,401 input locations and 2,789 known contacts from 116 inhibitory cells before spatial mapping; 114,775 inputs
1096 and 2,637 known contacts from 114 inhibitory cells mapped within $5\mu\text{m}$. The packaged census table has no
1097 per-target proofreading-status field, so no such status is inferred.

1098 The connection analysis retained one largest synapse per published axonal clump. This avoided counting imme-
1099 diately adjacent contacts from the same presynaptic axonal event as independent control sites. A matched candidate
1100 pool was constructed separately for every retained contact. Matching preserved target cell and compartment and
1101 minimized difference in log distance to the soma. All connection outcomes were averaged within target before
1102 testing. There were 402 inhibitory cell–target connections with at least two retained clumps, nested within 20
1103 targets.

1104 For descendant-domain overlap, let r_k be a binary vector indicating mapped input locations below candidate
1105 site k , multiplied coordinate-wise by the square root of synapse size and normalized to unit length. Two control
1106 sites therefore have overlap $r_k^\top r_l$. This value is one for identical weighted descendant domains and zero for disjoint
1107 domains. It is an anatomical co-control statistic, not a measured correlation of inhibitory activity. Table S29 reports
1108 the frozen target-level endpoints.

1109 Two sensitivity controls followed the primary target-level analysis. First, connection contrasts were averaged
1110 within each of 92 presynaptic inhibitory axons before testing, which accounts for one axon contacting multiple
1111 targets. Second, candidate sites were ranked jointly by proximity in log path distance and three-dimensional
1112 Euclidean distance, and 500 matched assignments were repeated within target. The coarse tree-distance contrast
1113 survived presynaptic-axon aggregation. Shared-path and descendant-domain overlap did not; no endpoint survived
1114 joint 3D and path-distance matching (Table S30). Thus the primary result establishes spatial co-targeting, not
1115 geometry-independent organization over credit domains.

1116 The route-capacity matrix used all actual inhibitory-bearing segments as columns, rather than one column per
1117 presynaptic inhibitory cell. Row weights were square roots of mapped input size and column weights were square
1118 roots of known inhibitory synapse size. Highest column norm defined the selected-site order. Random-site and
1119 shuffled controls were independently generated 200 times per cell and budget. This diagnostic asks whether the set
1120 of observed locations contains an efficient ancestry basis; it does not show that the highest-leverage subset was
1121 selected or used in vivo. Because the target is the same actual-route matrix from which the selected columns are
1122 drawn, this is an internal compressibility analysis rather than an independent routing test.

1123 **S12 External signed-credit analysis and task-interference identity**

1124 The Francioni et al. workbook was downloaded from the source-data link of the published article and frozen by
1125 SHA-256 be1b87481d0f1d1bffc85d8a4cabfd37b7055d640306afab5b3622de0ced8264. The animal-level
1126 contrasts in sheet “Ext 13E” were used directly. The source table’s signed-separation column was required to
1127 equal P+ minus P- to numerical precision. The orthonormal common and signed projections sum exactly to total
1128 two-coordinate energy. Sheet “5E” supplied the four neuron-level distributions in the display but was not used to
1129 increase the sample size of the animal-level test.

1130 For $n = 6$, the one-sided exact random-sign test enumerated all 2^6 sign assignments. Bootstrap intervals sampled
1131 animals with replacement. The directions were fixed by the experimental mapping: P+ positive, P- negative, and
1132 P+ minus P- positive. This is a retrospective test because the prediction was formulated after publication of the
1133 experiment, although it was evaluated from the numerical source table rather than the displayed figure.

1134 **12.1 Route overlap and task interference**

1135 Branch-specific plasticity during motor learning suggests that different tasks can recruit different dendritic regions
1136 [2]. Let $\mathbf{w}_A$ be an optimum of a quadratic Task-A objective with Hessian H_A , and let an approximate Task-B
1137 teaching field produce update $-\eta M_B \mathbf{g}_B$. Expansion about the Task-A optimum removes the first-order term, leaving

1138 the exact loss increase

$$\mathcal{L}_A(\mathbf{w}_A - \eta M_B \mathbf{g}_B) - \mathcal{L}_A(\mathbf{w}_A) = \frac{\eta^2}{2} \mathbf{g}_B^\top M_B^\top H_A M_B \mathbf{g}_B. \quad (79)$$

1139 In the equal-route example, opposing updates on the shared fraction ω have magnitude two, whereas leakage onto
1140 the Task-A-only fraction has magnitude λ . Dividing by route size gives $\eta^2[4\omega + \lambda^2(1 - \omega)]/2$. Numerical vector
1141 updates matched this identity to 4.2×10^{-17} absolute error (Supplementary Fig. S6). Separated routes therefore
1142 protect a learned task only when both anatomical overlap and teaching-field leakage are small. The published SST
1143 perturbation is consistent with the resulting prediction that loss of branch-selective inhibition increases effective
1144 leakage and cross-task interference [2], but it does not uniquely identify the credit-routing mechanism. This static
1145 identity does not model dendritic calcium, spikes, eligibility timing, or the SST manipulation in the source study.

1146 S13 Statistics, reproducibility and claim controls

1147 Central comparisons report paired effect sizes, 95% intervals and the number of positive cell or seed pairs. Tests
1148 were two sided unless a direction had been frozen before analysis. Exact Wilcoxon signed-rank tests were used
1149 for small paired cell cohorts. Reconstructed-cell intervals resampled cells first and, where relevant, one Monte
1150 Carlo stream within cell. Measured-response intervals resampled target cells. Artificial-network summaries used
1151 independently initialized training runs. Inhibitory-census outcomes were first averaged within postsynaptic target.
1152 Signed-credit intervals and exact sign tests used animals; neuron-level source rows were descriptive only. Bootstrap
1153 intervals used 20,000 draws. The four-budget primary subtree scan used Benjamini–Hochberg correction across
1154 $K = 1, 2, 4, 8$; other parameter sweeps and sensitivity analyses received no multiplicity correction, are labeled
1155 secondary, and are not interpreted as independent confirmation.

1156 The following measures are not used to support cross-architecture conclusions: pooled error-field metrics that
1157 mix a somatic stage whose reconstruction is exact by construction with dendritic stages; unweighted path-gain
1158 dispersion as evidence of gradient direction; and post-hoc comparisons between unmatched training cohorts.
1159 Table S32 records the supported interpretation and boundary of each evidence class.

1160 Every main panel is accompanied by a machine-readable source table containing the plotted values, biological
1161 or training unit, nesting variables, seed, condition names, and analysis version. The release audit verifies that figure
1162 scripts read these tables rather than duplicating values in plotting code. Authentication tokens, restricted caches,
1163 and raw data already hosted by MICrONS or DANDI are not redistributed.

1164 The release also includes the three archived regular-tree sweep files, representative fully resolved configurations,
1165 the surviving frozen manifest, the seed-level result tables, and an executable NumPy reference for the raw additive
1166 forward model and derivatives. Exact source copies of the morphology, routing-capacity, focal-perturbation and
1167 projection-bound analyses are provided with SHA-256 provenance. The measured-response directory contains
1168 the CAVE join, DANDI/NWB extraction and task-derived learning scripts together with a machine-readable
1169 four-channel specification and target manifest. The expanded release also contains the stable-ID inhibitory-census
1170 join, coordinate-transform audit, target-level topology tables, the verified Francioni source checksum and extracted
1171 contrasts, and the static interference verification grid. Private filesystem paths in portable configurations were
1172 replaced by paths relative to the article directory; numerical settings were not changed. The provenance manifest
1173 records each origin and every portability edit.

1174 An exact package lock was not archived for the June exact-transport or backpropagation jobs, nor for the
1175 measured-response analysis. These versions are therefore not guessed from the current environment. The complete
1176 July feedback extension froze Python 3.10.13, its Git identity and selected source hashes, but recorded a dirty
1177 tracked worktree. Table S11 and the accompanying reproducibility README distinguish exact, partial and
1178 unverifiable provenance fields.

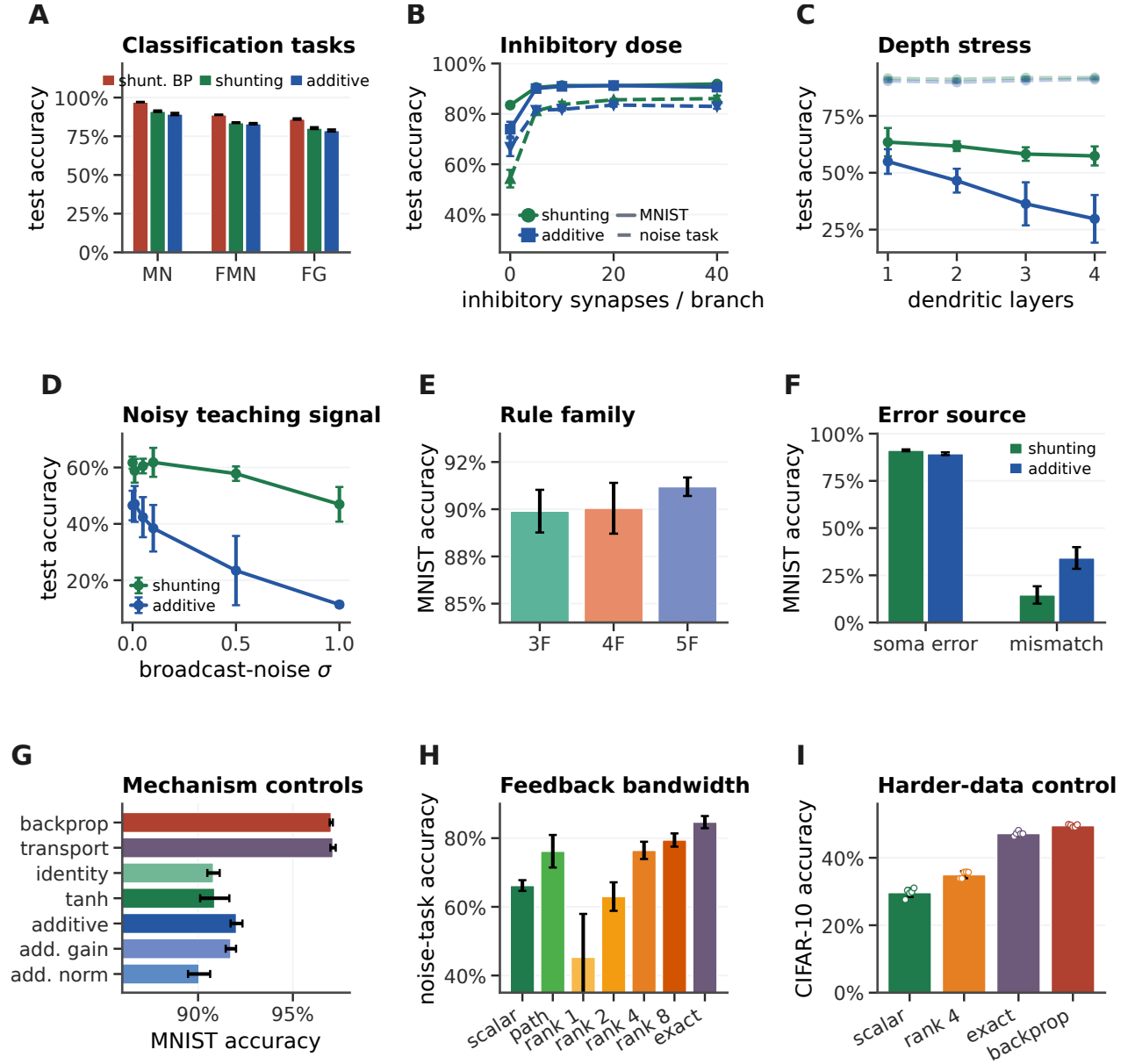


Figure S4: **Regular-tree learning across tasks, operating regimes, and feedback constraints.** **A**, Test accuracy on MNIST (MN), Fashion-MNIST (FMN), and figure-ground MNIST (FG) under shunting backpropagation and low-bandwidth local learning in shunting and additive trees. **B**, Local-learning accuracy as inhibitory synapses per branch vary on MNIST (solid) and the noise-resilience task (dashed). **C**, Accuracy as nominal dendritic depth increases. Solid lines are local learning and faint dashed lines are matched backpropagation; colour denotes shunting or additive architecture. **D**, Accuracy after adding zero-mean Gaussian noise with standard deviation σ to the broadcast teaching signal. **E**, Three-, four-, and five-factor rules within the same shunting, scalar-feedback cohort. **F**, Accuracy when the teaching factor is the somatic error or an unmatched local quantity. **G**, Separate five-seed controls for exact transport, reactivation, and additive gain or normalization, shown with a matched backpropagation reference. **H**, Noise-task accuracy across scalar, path-propagated, random low-rank, and exact-transport feedback. **I**, Flattened CIFAR-10 control in the shunting architecture. Bars and lines show means; error bars show sample standard deviations across independent training seeds ($n = 5$ unless noted). Panels **E,F** use the archived three-seed rule and error-source cohorts.

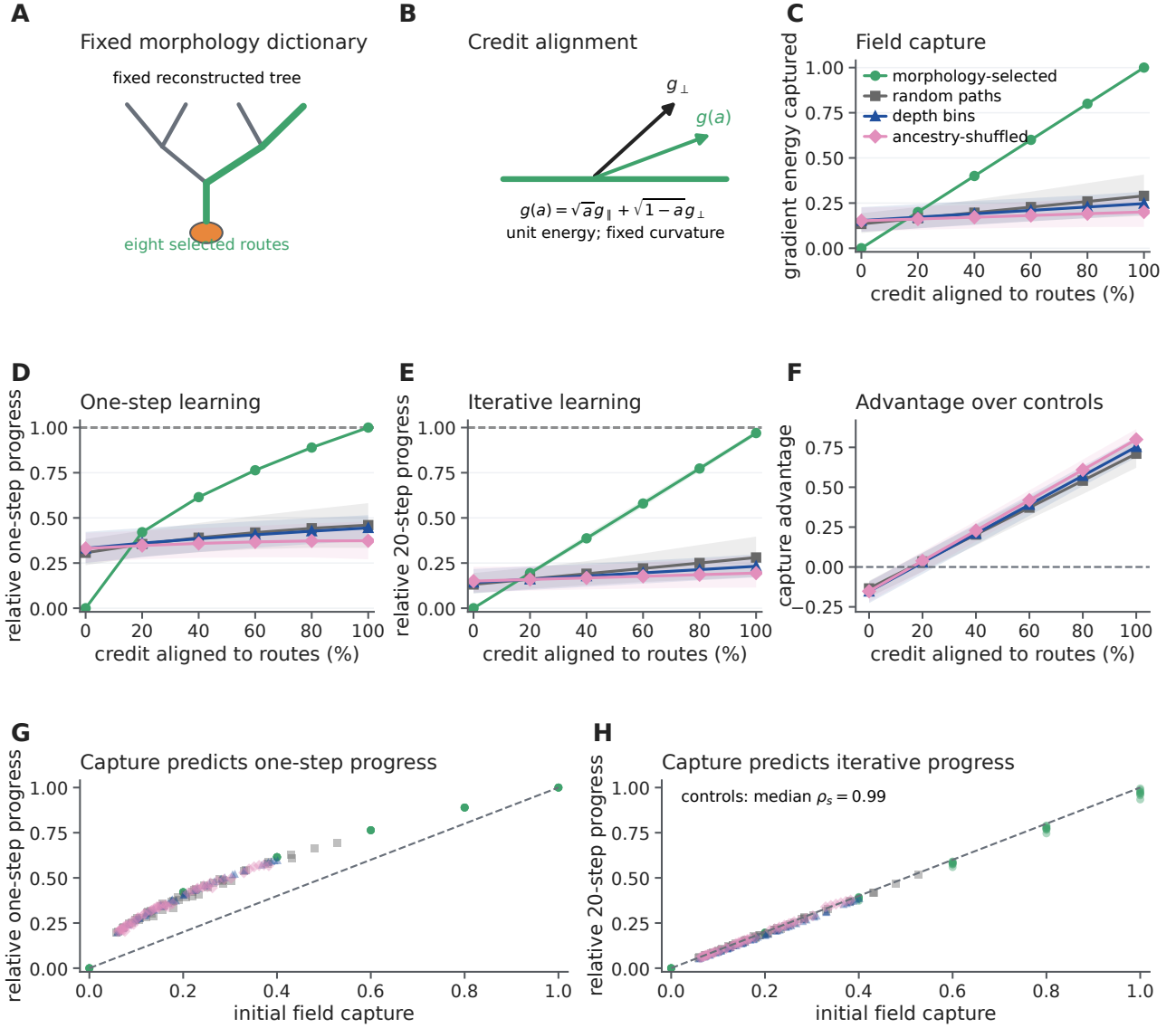


Figure S5: **Constructed route alignment controls projected learning.** **A**, A fixed eight-route dictionary on a reconstructed tree. **B**, Unit-energy credit fields are rotated between its span and the orthogonal complement. **C**, Capture across imposed alignment levels. **D**, Loss reduction from one norm-matched projected-gradient step, relative to the full-gradient step. **E**, Relative loss reduction after 20 projected steps. **F**, Ancestry-route capture relative to random paths, depth bins and ancestry shuffling. **G,H**, Initial capture versus one- and 20-step progress. Means in **C–F** are across eight cells with hierarchical 95% intervals. Ancestry-route capture equals imposed alignment by construction; coefficients are oracle projections. This is a conditional positive control, not independent evidence about the recorded neurons.

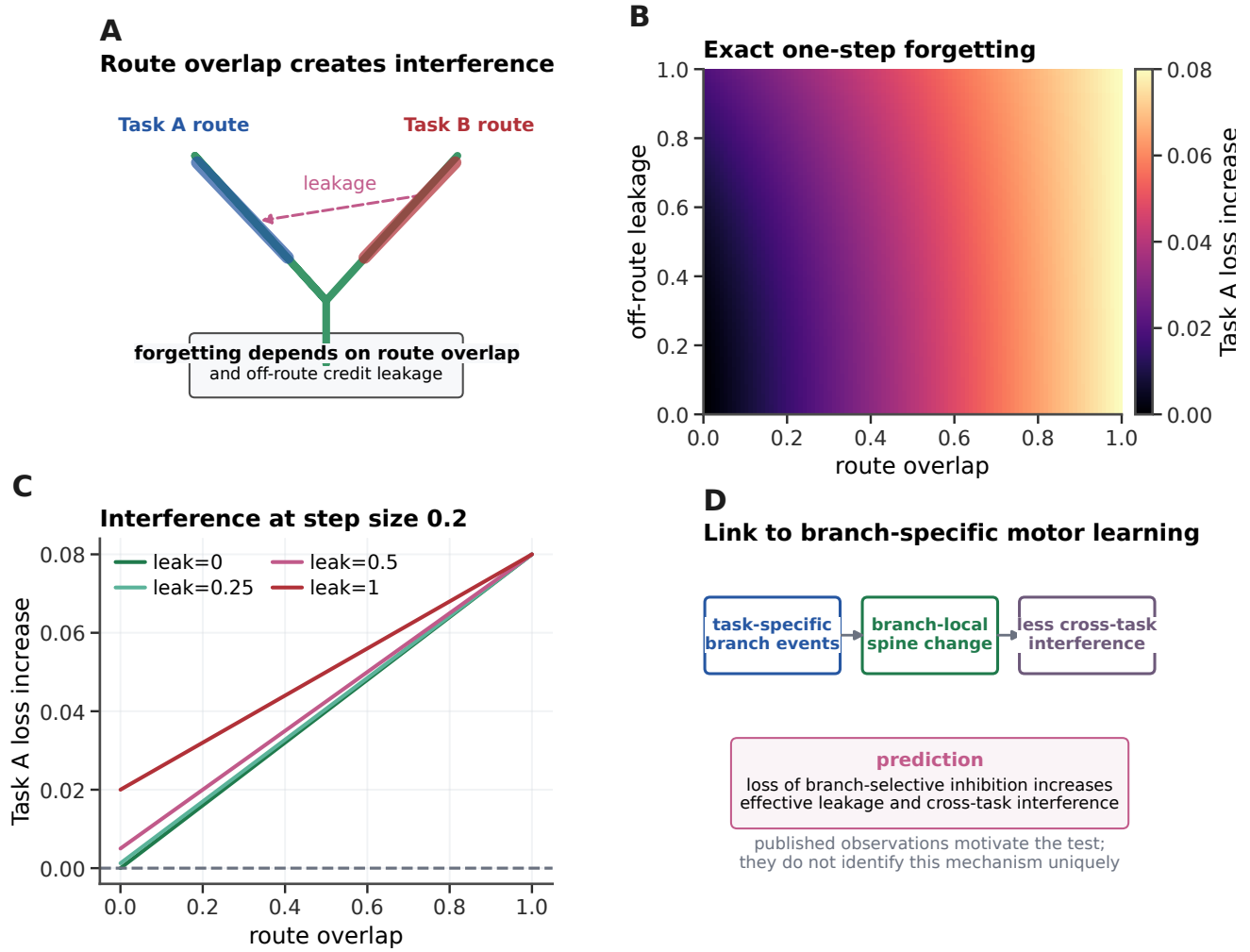


Figure S6: Route overlap and off-route leakage determine one-step task interference. **A**, Two tasks address partly separated dendritic routes; an approximate teaching field can also leak onto the other route. **B**, Exact Task-A loss increase after one Task-B update across route overlap and off-route leakage. **C**, Slices of the exact surface at step size 0.2. **D**, Relation to published branch-specific motor-learning observations and the resulting prediction. Panels **A–C** are an exact static quadratic consequence and numerical verification, not a temporal neural model. Panel **D** is a cross-study interpretation rather than a reanalysis of the original motor-learning data.

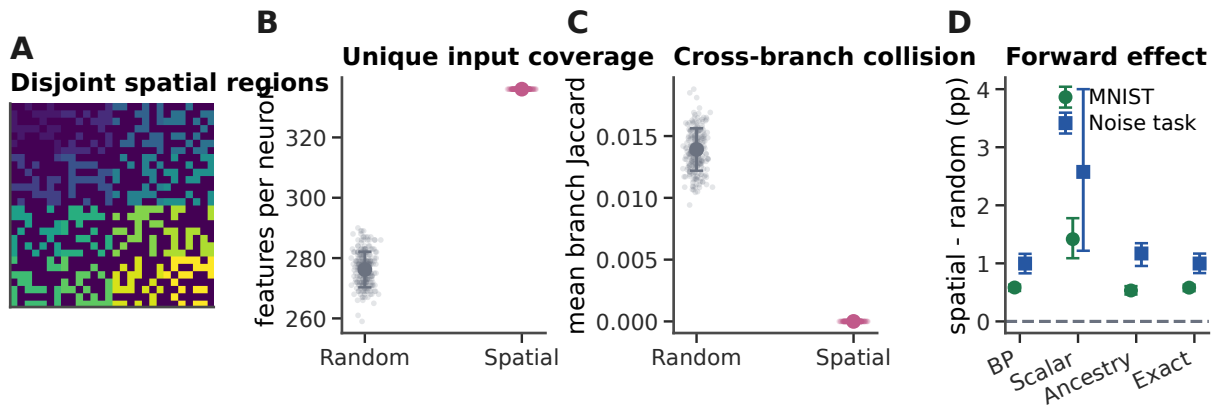


Figure S7: **The fixed spatial map changes forward input coverage.** **A**, Example contacts from one neuron. The 16 distal branches occupy disjoint image regions. **B**, Unique input coordinates reached by each neuron under independently sampled random branches and the spatial map. **C**, Mean pairwise Jaccard overlap between branch input sets. **D**, Spatial-minus-random accuracy under backpropagation (BP), scalar, neuron-indexed ancestry and exact feedback. MNIST averages shunting and additive models within seed; the randomly projected noise task uses the input-valid additive model only. Points and intervals show means and 95% paired-seed bootstrap intervals. Each learning condition contains ten paired seeds. The map has the same number of contacts in both conditions but greater unique coverage and no cross-branch collisions in the spatial condition; its performance effect is therefore not specific to local credit assignment.

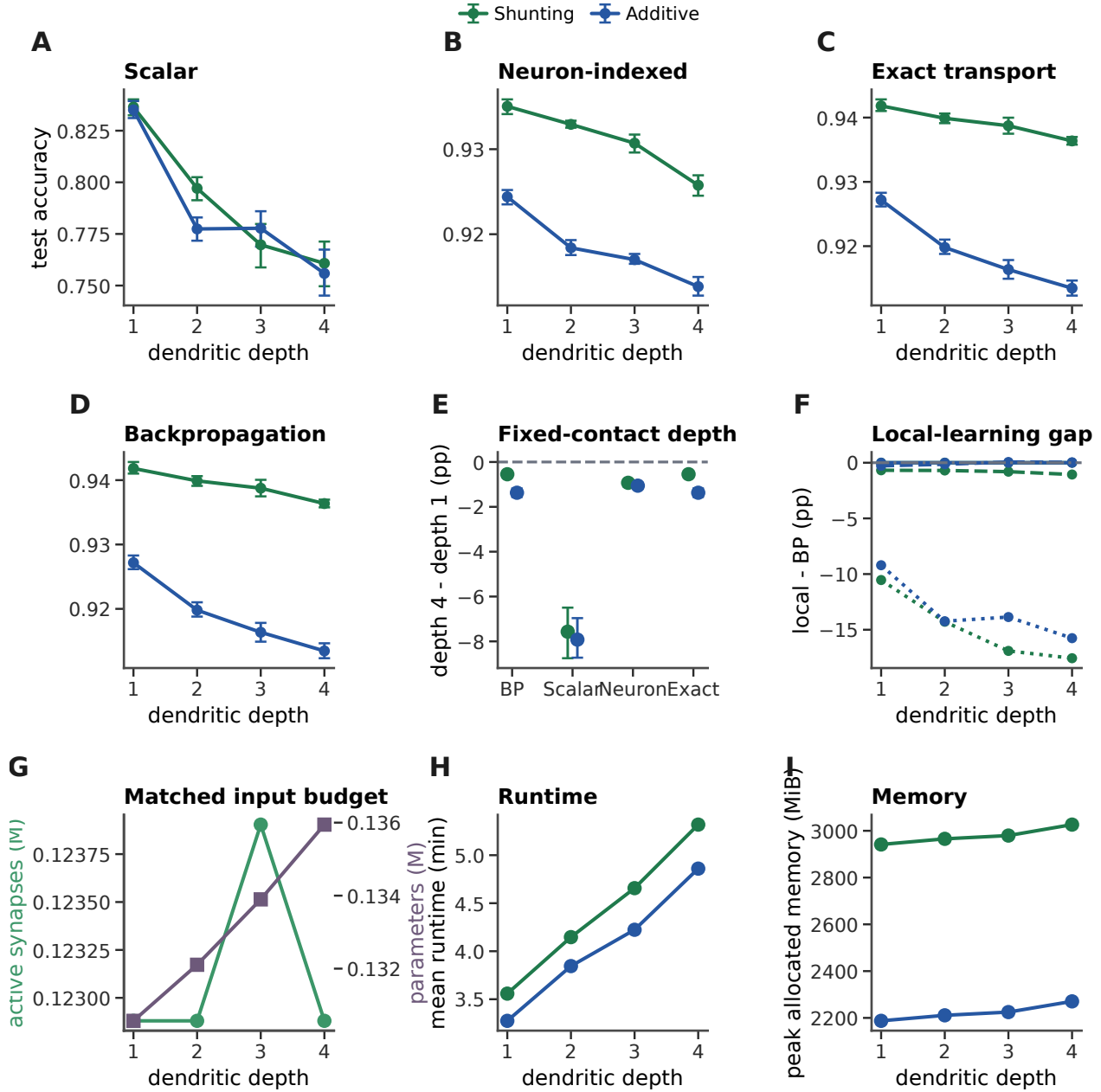


Figure S8: **Fixed terminal-branch and input-contact budgets do not make greater dendritic depth intrinsically beneficial.** A–C, Held-out accuracy across depths one to four under scalar, neuron-indexed ancestry and exact feedback for the input-valid additive network. D, Matched backpropagation. E, Prespecified paired depth-four-minus-depth-one accuracy contrasts. F, The gap between local and matched backpropagated learning. G, Active contacts and trainable-parameter counts. H, I, Runtime and peak memory. Each condition contains ten paired seeds; points and intervals are means and 95% paired-seed bootstrap intervals. All four retained depth-four-minus-depth-one contrasts were negative in 10/10 pairs (Wilcoxon $P = 0.001953$). The design fixes 16 terminal branches and 960–968 active contacts per soma, but does not match compartment or trainable-parameter count.

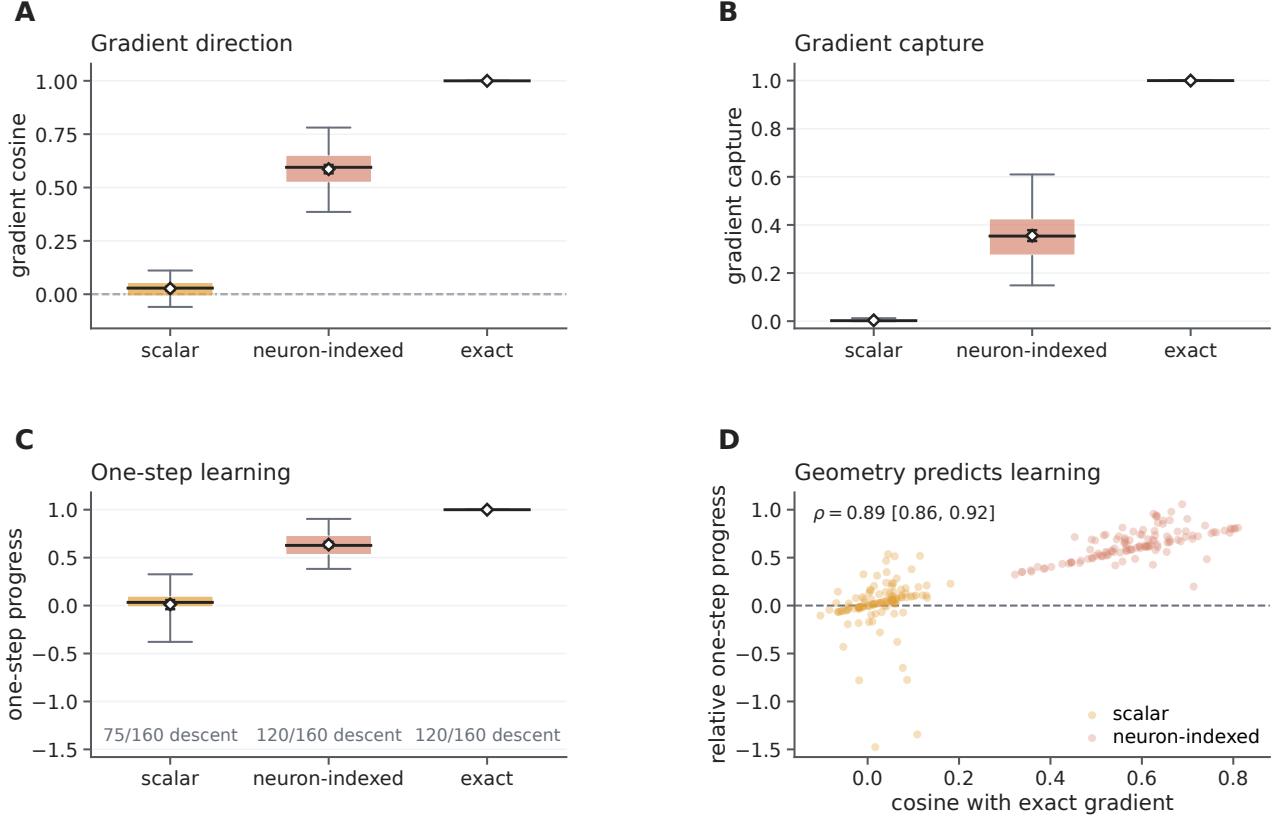


Figure S9: **Credit geometry predicts the loss change available to a local update.** **A**, Cosine between candidate and exact dendritic gradients at 120 input-valid matched backpropagation checkpoints. **B**, Eligibility-weighted gradient energy retained after optimal rescaling. **C**, Loss decrease after matching each candidate update to the exact-gradient norm; labels give the number of checkpoints in which the update decreased loss. **D**, Gradient cosine versus relative one-step progress for scalar and neuron-indexed fields; the interval for Spearman's ρ resamples checkpoints. In **A–C**, boxes show the interquartile range, whiskers the 5th–95th percentiles, and diamonds the mean with a 95% checkpoint-bootstrap interval. All panels use one frozen test batch per checkpoint and relative step 10^{-5} . Exact transport is an information oracle.

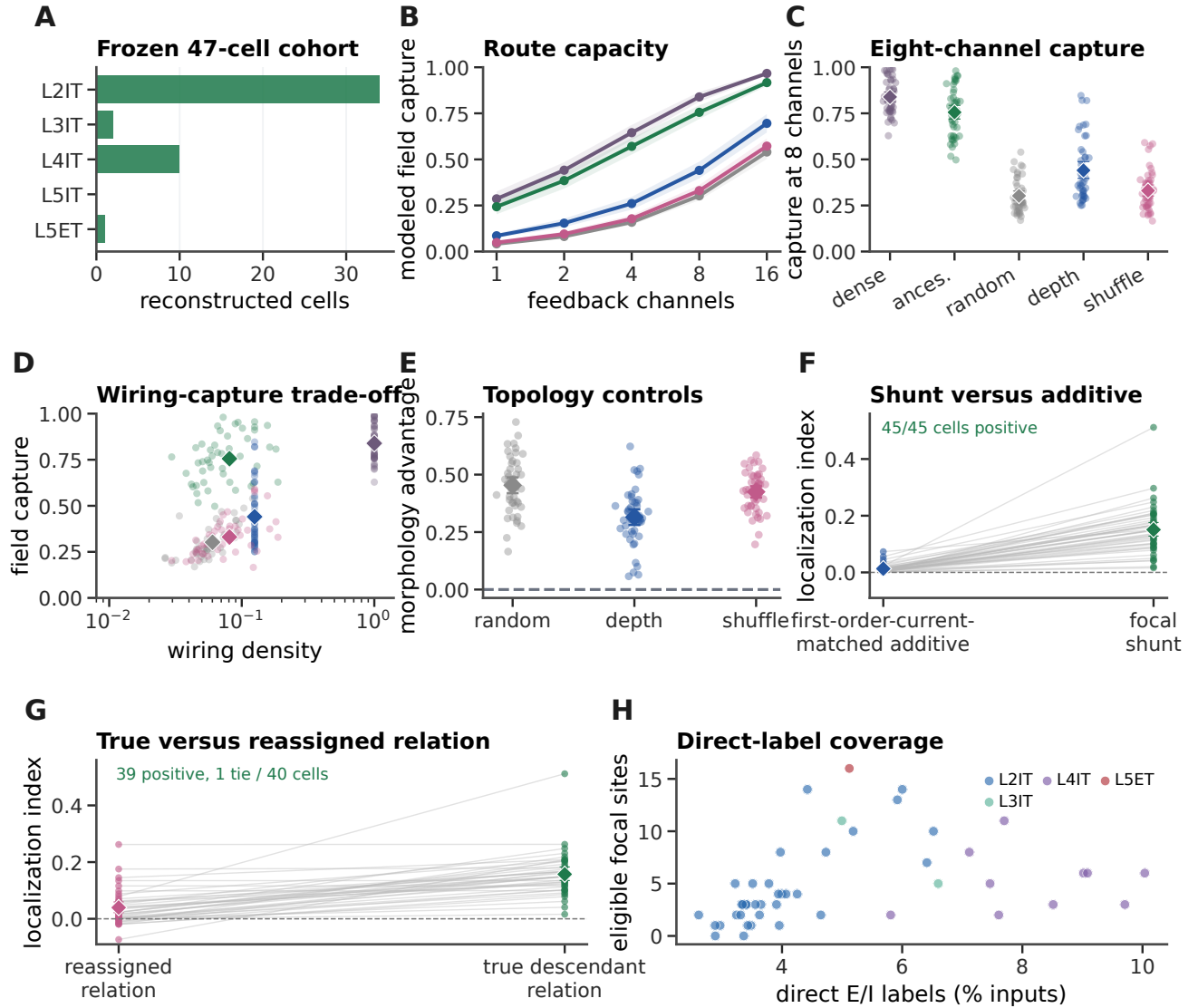


Figure S10: Structural routing and focal shunting remain robust in a disjoint public-v661 cohort. **A**, Composition of the 47-cell frozen analysis cohort by excitatory cell class. **B**, Cell-mean weighted modeled field-energy capture across feedback-channel budgets for a dense PCA oracle, morphology-defined paths, random paths, depth bins, and ancestry shuffles ($n = 47$ cells through eight channels; $n = 46$ at 16 channels). Twenty perturbation streams were averaged within cell; bands are 95% cell-bootstrap intervals. **C**, Cell-level capture at eight channels. **D**, Eight-channel wiring-capture trade-off; morphology paths used 8.1% of dense-feedback wiring. **E**, Within-cell morphology advantage over random paths, depth bins, and ancestry shuffles; all three contrasts were positive in 47/47 cells. **F**, Localization of exact-gradient changes after a unit-dose focal shunt and first-order-current-matched additive perturbation ($n = 45$ cells, 235 sites). **G**, Localization for true descendant relations and relation templates reassigned to other focal sites ($n = 40$ cells, 230 sites). **H**, Direct presynaptic E/I-label coverage and eligible focal sites per cell. Diamonds and intervals in **C**, **E**–**G** show means and 95% cell-bootstrap intervals. Cells, not sites or streams, are the inferential units. The cohort excludes the original eight nuclei but comes from the same mouse. It uses historical v661 reconstructions and direct E/I calls only (4.71% of incoming synapses). These are structural and passive-model tests, not measurements of learning in vivo.

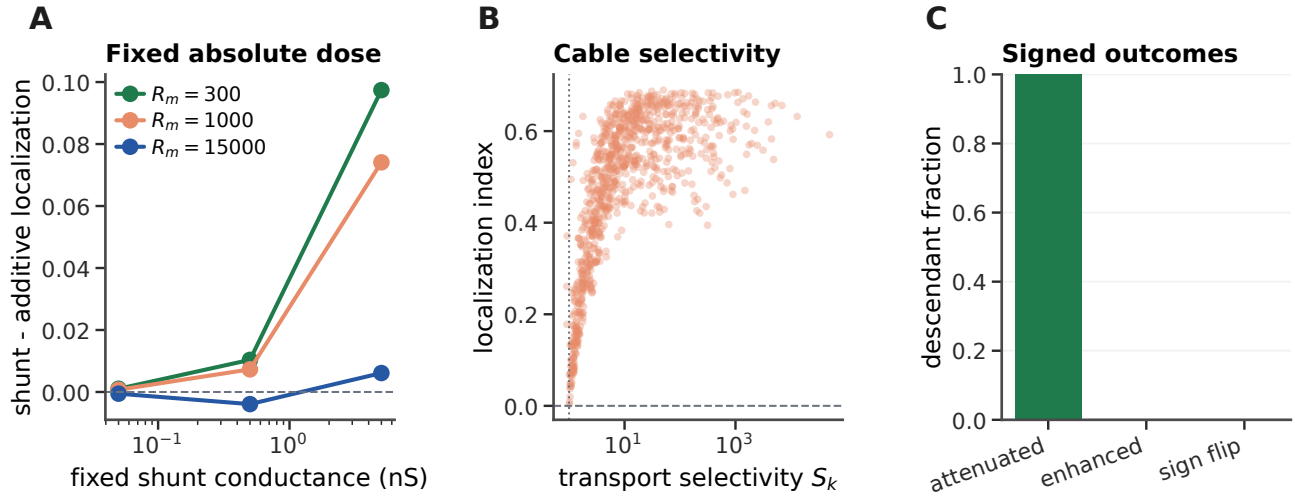


Figure S11: **Fixed-absolute dose and background conductance bound passive focal-shunt selectivity.** **A**, Shunt-minus-additive localization across fixed absolute conductance doses at three membrane-resistance settings with no added background conductance. Each curve averages focal sites within each of eight cells before the cell mean. **B**, Transport selectivity S_k versus full synaptic-gradient localization at unit input-conductance-normalized dose across 101 sites, three membrane resistances and three background levels. Points are site-regime values and are descriptive because sites and regimes are repeated within cell. The dotted line marks $S_k = 1$. **C**, Fraction of descendant gradients attenuated, enhanced or sign reversed at $R_m = 1,000 \Omega \text{cm}^2$, background multiplier one and unit input-conductance dose. All tested passive focal-shunt conditions had the same qualitative signed outcome: attenuation without enhancement or sign reversal. Specific resistances and background levels are sensitivity parameters, not cell-specific fits.

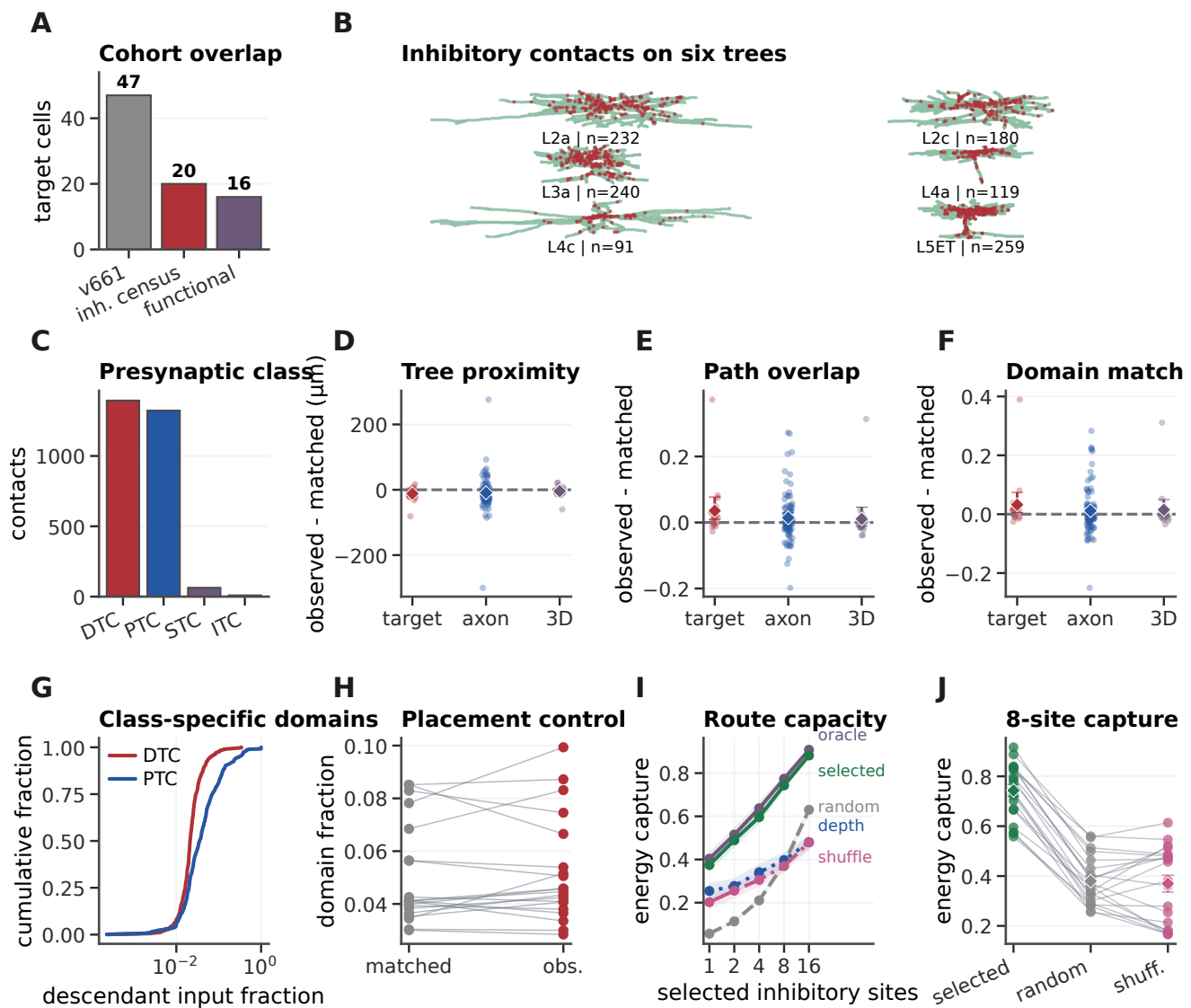


Figure S12: Synapse-resolved inhibition reveals class-dependent dendritic scales. **A**, Overlap between the 47-cell v661 sensitivity cohort, the v795 inhibitory census and available functional summaries. **B**, Six layer-diverse target reconstructions; red points are mapped known inhibitory contacts. **C**, Contacts in the overlap by the published inhibitory targeting class. DTC, distal dendrite-targeting; PTC, perisomatic-targeting; STC, sparsely targeting; ITC, inhibitory-targeting. **D–F**, Observed-minus-matched contrasts in tree distance, shared path and weighted descendant-domain overlap. Target denotes the primary 20-target analysis; axon first averages within each of 92 presynaptic axons; 3D jointly matches path distance and Euclidean proximity within target. Diamonds and intervals show unit means and bootstrap 95% intervals. Only tree distance persisted after axon aggregation, and no endpoint persisted after joint 3D matching. **G**, Descendant-domain size for DTC and PTC contacts, retaining one contact per axonal clump. **H**, Overall inhibitory-contact domain size versus matched locations; no reliable global localization advantage was detected. **I**, Weighted capture of the actual inhibitory-route dictionary as a function of site budget. **J**, Eight-site capture within each of 20 targets. Sites in **I,J** are selected by anatomical leverage; this is a structural capacity analysis, not evidence that the routes carried teaching signals in vivo. Target cells are nested within one mouse.

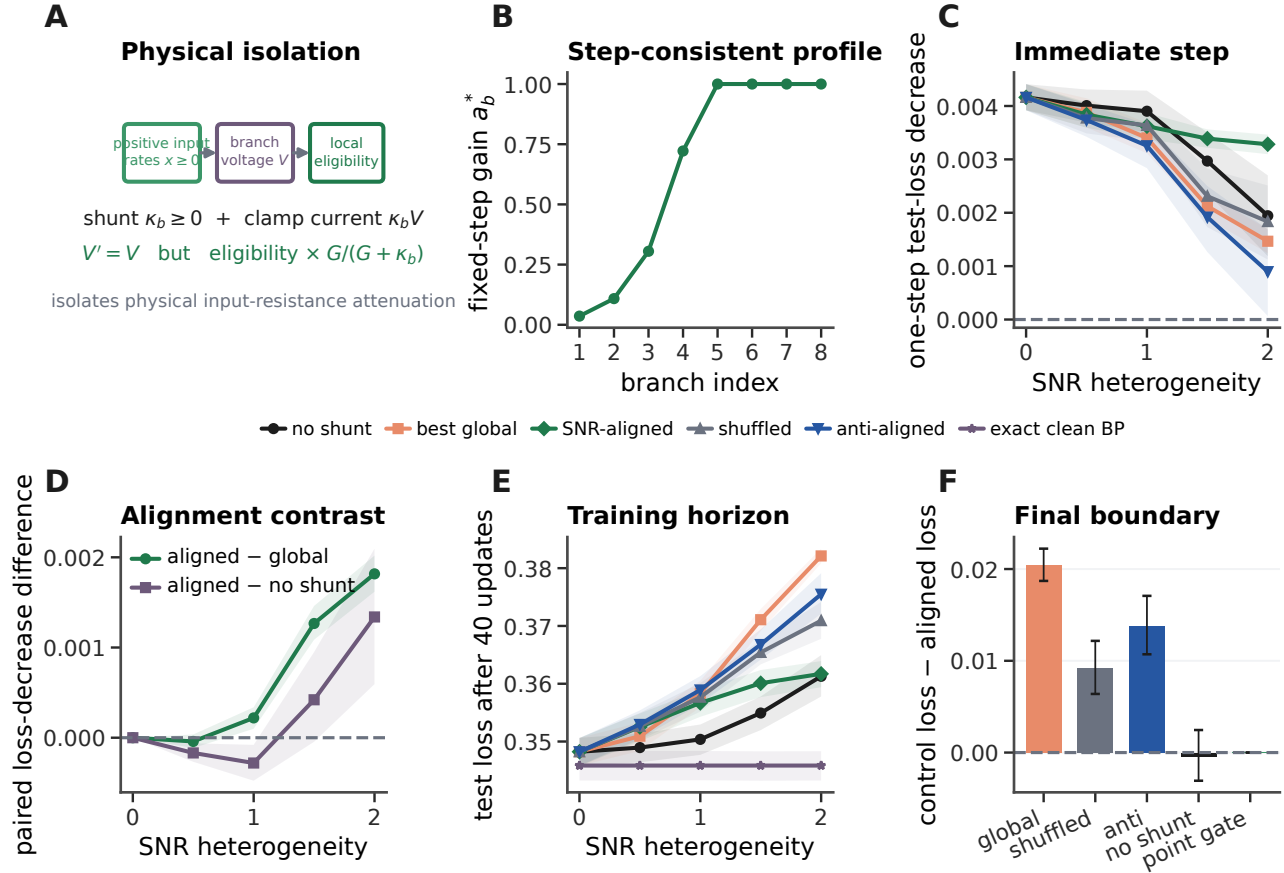


Figure S13: **A state-matched positive conductance approximates the predicted fixed-step branch-reliability factor, with a bounded training benefit.** **A**, A nonnegative shunt κ_b is paired with compensating current $\kappa_b V_b$, preserving branch voltage while reducing physical input resistance and local eligibility. **B**, Prescribed reliability profile $\min\{1, S_b/[c(S_b + N_b)]\}$ at maximum heterogeneity with $c = 0.5$. **C**, Mean one-step held-out loss decrease across 50 paired seeds. Lines and shading show seed means and 95% paired-seed bootstrap intervals. **D**, Aligned-minus-global and aligned-minus-no-shunt one-step contrasts. At heterogeneity two, the former is positive in 49/50 seeds; the latter has a positive mean interval, is positive in 33/50 seeds and has two-sided Wilcoxon $P = 0.0030$. **E**, Held-out loss after 40 updates. Exact clean backpropagation is the unattenuated optimization reference. **F**, Paired final-loss contrasts at maximum heterogeneity, displayed as control minus aligned so positive values favor aligned shunting. Aligned shunting beats the global, shuffled and anti-aligned shunts but has no reliable final difference from the unshunted noisy rule. The independently computed explicit point gate receiving the same sample-dependent factors agrees to numerical precision. All input rates and excitatory and shunt conductances are nonnegative. The oracle state clamp and fixed reliabilities isolate mechanism; they do not constitute autonomous shunt learning.

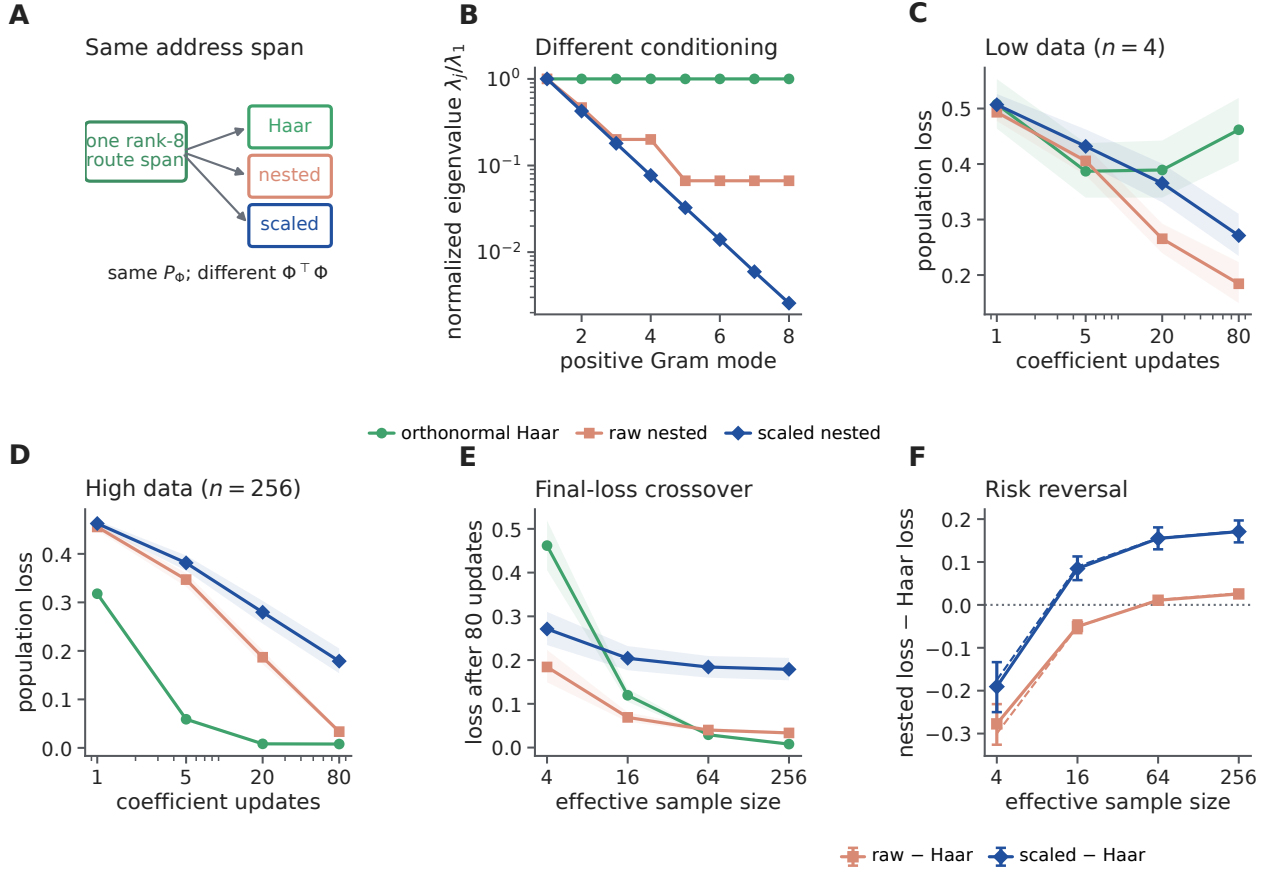


Figure S14: Same address span produces a finite-data coefficient-learning crossover. **A**, Orthonormal Haar, redundant nested and statically scaled nested dictionaries parameterize the identical rank-eight route span. **B**, Normalized positive eigenvalues of their field Gram operators; condition numbers are 1, 15 and 388.52. **C,D**, Mean population-loss trajectories under common noisy local target estimates at effective sample sizes 4 and 256. Shading shows 95% paired-seed bootstrap intervals across 50 seeds. At low data, slow nested modes filter variance; at high data, their retained bias dominates. **E**, Final loss across the sample-size series. **F**, Paired nested-minus-Haar contrasts. Solid points and bars are observed means and intervals; dashed curves are the exact finite-time risk from Eq. 53. The reversal is a bias–variance effect within one span. Gram-preconditioned controls (not shown) make all parameterizations agree to 1.16×10^{-15} and do not assert a biological implementation of the preconditioner.

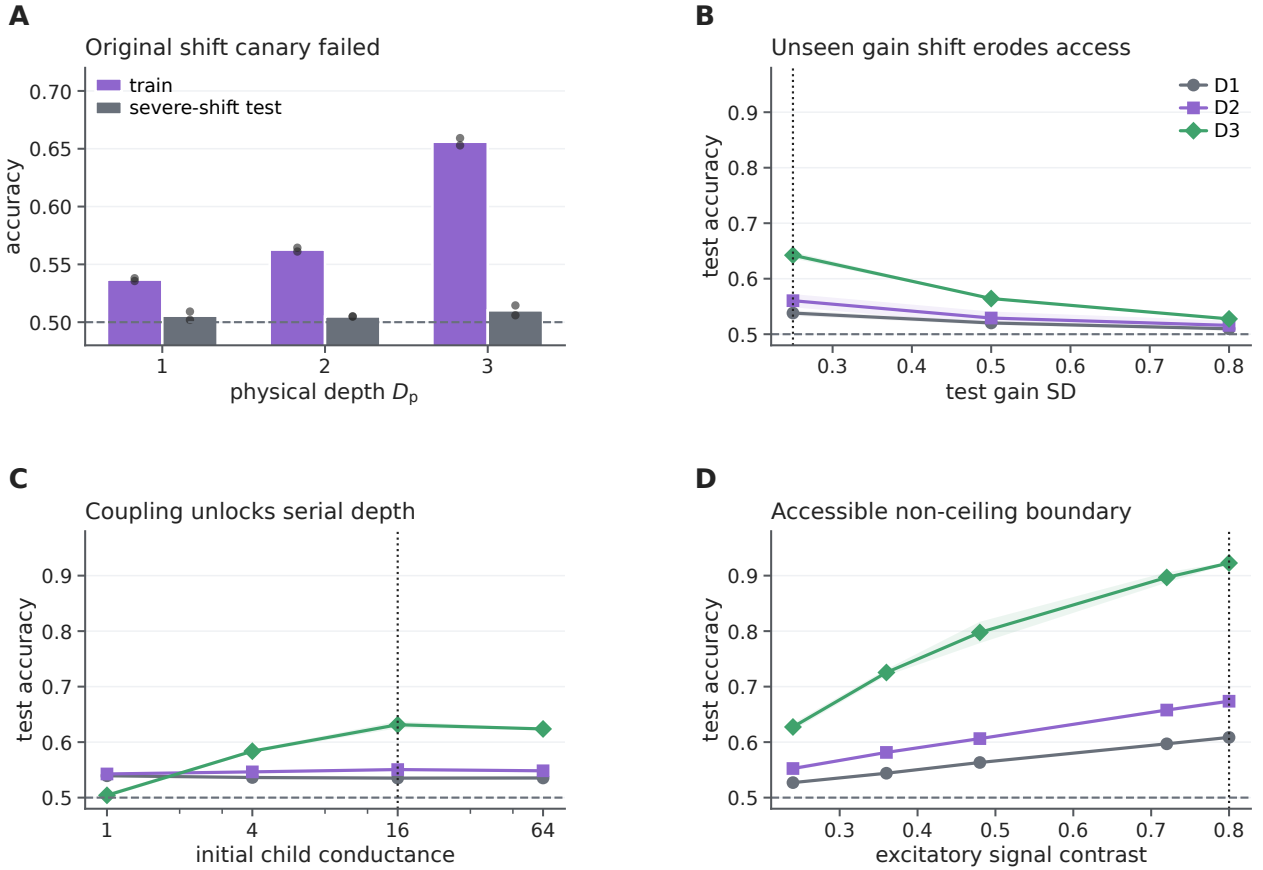


Figure S15: **Transparent calibration of the nonlinear physical-depth operating point.** **A**, The original aligned shunting-BP canary at signal contrast 0.24 fit the training distribution increasingly with depth but remained at chance under the prespecified severe gain shift (test gain SD 1.2). Bars are two-seed means and dots are the individual exploratory seeds. **B**, With training gain SD fixed at 0.25, increasing unseen test gain variation eroded accessibility at every depth. The dotted line marks the matched train/test condition. **C**, At matched gain and signal contrast 0.24, stronger child coupling selectively made serial D3 computation accessible; the dotted line marks child conductance 16 used subsequently. **D**, At matched gain and child conductance 16, increasing excitatory signal contrast moved D1–D3 away from chance without saturating D1 or D2. The separately run 0.80 boundary (dotted line) passed the frozen accessibility gate and selected the fresh-seed operating point. Lines show two-seed means; shading spans the two exploratory values and is not a confidence interval. All pilot, ladder and boundary seeds were excluded from the confirmatory contrasts.

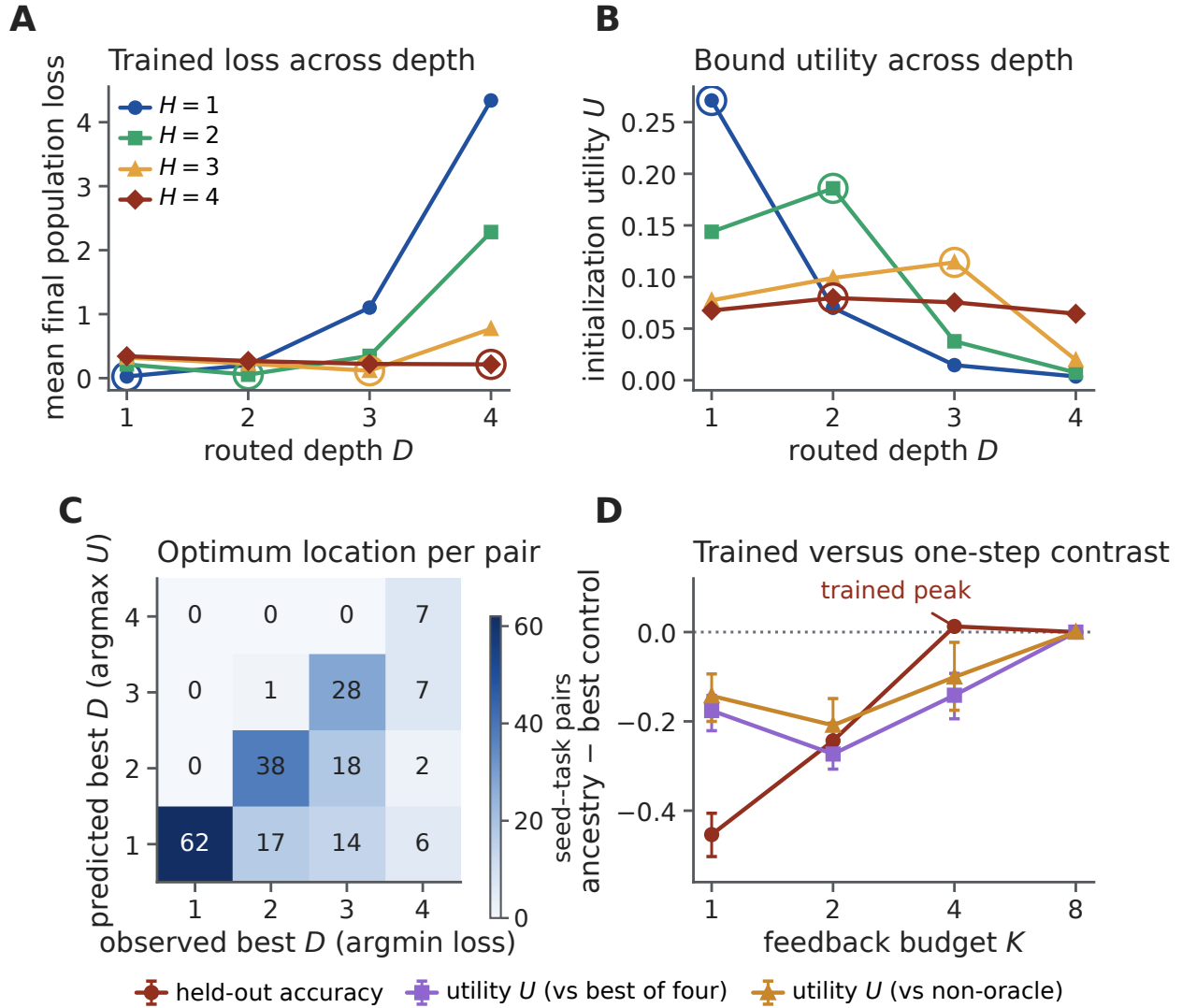


Figure S16: **The credit-operator bound locates the trained depth optimum but not the trained ancestry advantage.** **A**, Mean final population loss across routed depth D_r for task depths $H = 1\text{--}4$ in the frozen hierarchy-depth phase experiment (aligned tree, $n = 50$ seeds); open circles mark the loss minimum. **B**, Mean initialization utility (Supplementary Eq. 44) for the same conditions; open circles mark the utility maximum. The group-level argmax matches the trained optimum exactly at $H = 1, 2, 3$; at the full-rank boundary $H = 4$ the one-step bound is conservative and favours $D_r = 2$. **C**, Per-pair predicted (argmax utility) versus observed (argmin loss) best depth over all 200 seed-task pairs; 135 pairs lie on the diagonal. **D**, Ancestry-minus-best-control contrast across feedback budget K in the 2,700-fit factorial, for trained held-out accuracy and for initialization utility against the published four-control set and its three implementable (non-oracle) members. Points and bars are means and 95% seed-bootstrap intervals ($n = 20$). The utility reproduces the exact ties at $K = 8$, where all spans coincide, but not the trained $K = 4$ peak: the trained anatomical advantage accrues along the training trajectory rather than from one-step initialization geometry. Analysis: `scripts/analyze_interior_optimum.py`.

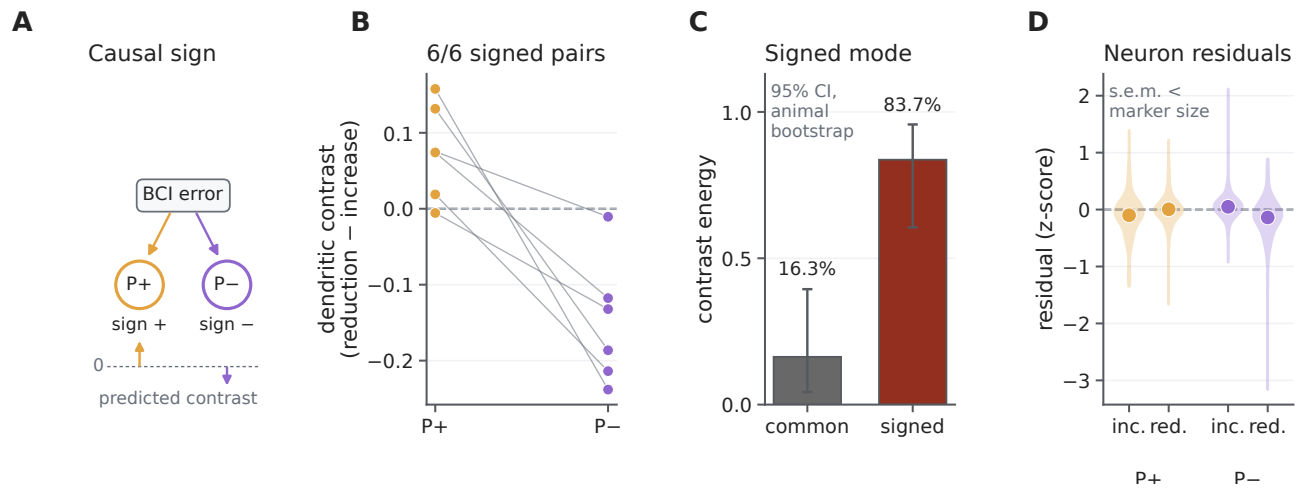


Figure S17: **Published animal data support signed neuron-specific teaching coordinates.** **A**, In the source brain-computer-interface experiment, the causal mapping assigned opposite signs to P+ and P- populations, predicting opposite dendritic contrasts. **B**, Error-reduction minus error-increase standardized dendritic residual for P+ and P- in six animals; lines pair the two populations within animal. **C**, Fraction of the two-coordinate contrast energy in a common scalar mode and the signed P+/P- mode; intervals are 95% animal-bootstrap intervals. **D**, Descriptive neuron-level residual distributions from the published source table; dots show means. Animal, not neuron, is the inferential unit. This external consistency test supports signed neuron identity but does not resolve within-tree transport or uniquely identify a conductance mechanism.

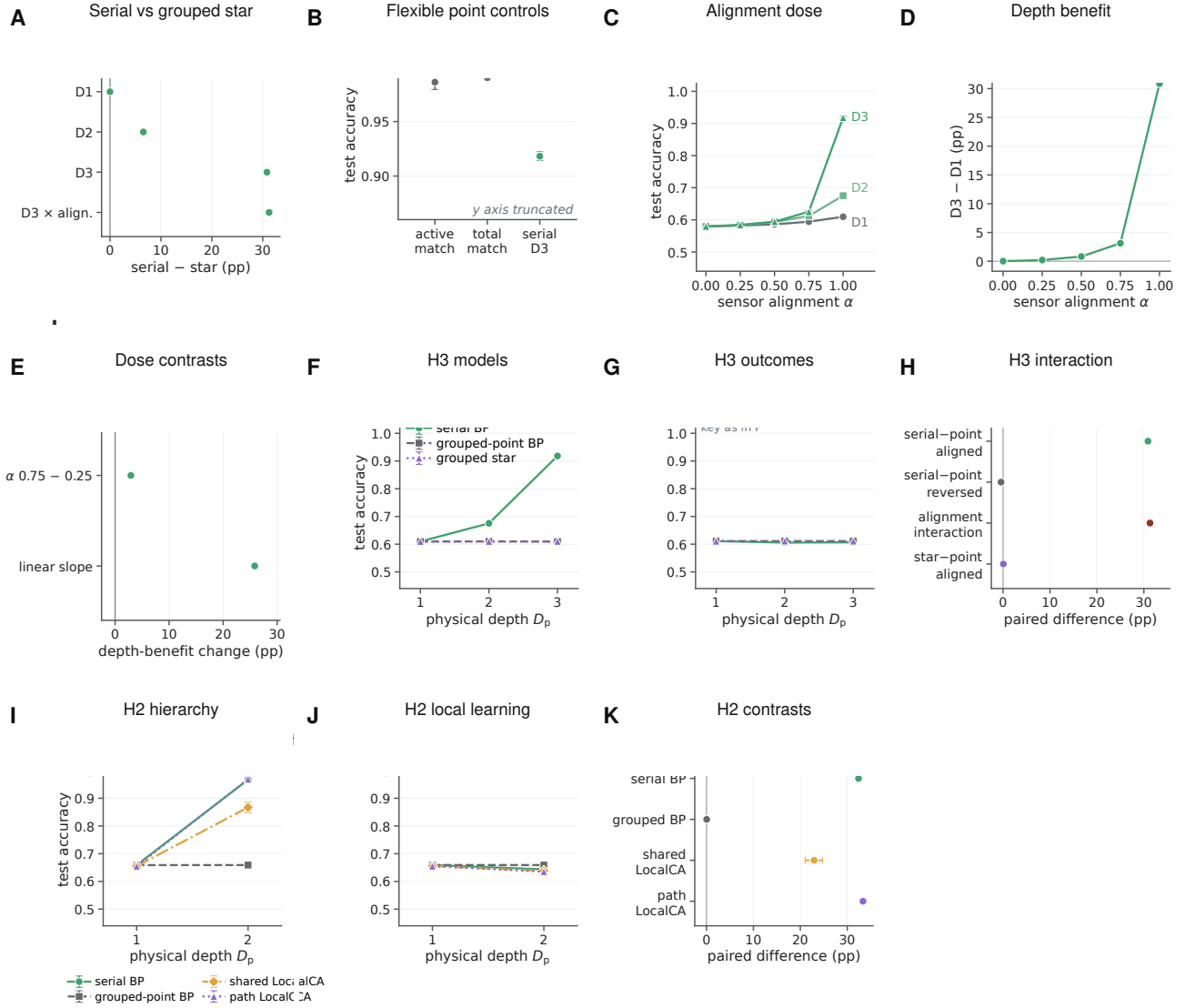


Figure S18: **Physical-depth controls, alignment interpolation and hierarchy replication.** **A**, Paired serial-minus-grouped-star effects and the D3 alignment interaction. **B**, Flexible point networks matched to active or total parameter count versus the serial D3 tree. **C–E**, Accuracy, D3-minus-D1 benefit and paired dose contrasts across sensor–task alignment α . Intermediate levels were specified after the endpoints and are an interpolation, not an independent replication. **F–H**, H3 serial tree, literal grouped-point and grouped-star controls under aligned and reversed placement. **I–K**, Independent H2 hierarchy with serial BP, grouped-point BP and shared- or path-transport LocalCA. Lines are paired-seed means and intervals are 95% seed-bootstrap intervals ($n = 10$). The two hierarchies belong to one calibrated synthetic task family.

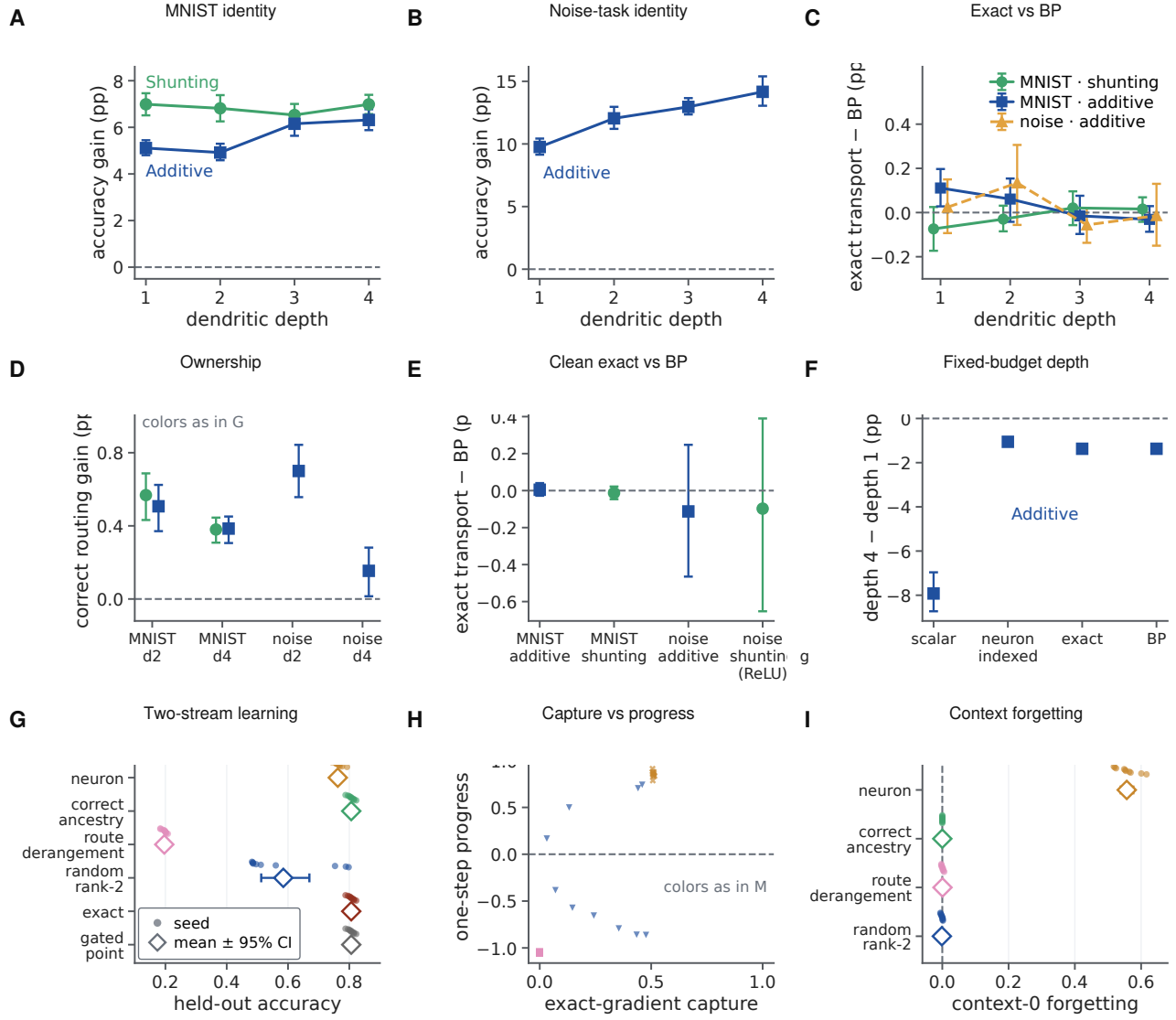


Figure S19: **Prospective identity, ownership and route diagnostics.** **A,B**, Neuron-indexed-minus-scalar accuracy across depth on MNIST and the input-valid additive noise task. **C**, Exact-transport-minus-BP accuracy in the retained historical conditions. **D**, Correct-minus-deranged neuron-to-tree ownership at matched bandwidth. **E**, Depth-averaged exact-minus-BP accuracy in the independent clean-source audit. **F**, Depth-four-minus-depth-one accuracy at fixed terminal branches and contacts. **G**, Held-out two-stream credit-reversal accuracy. **H**, Exact-gradient capture versus norm-matched one-step progress. **I**, Forgetting of the first context after adaptation to the second. Small points are paired seeds and larger symbols and bars are means and 95% seed-bootstrap intervals. Exact transport is an information ceiling.

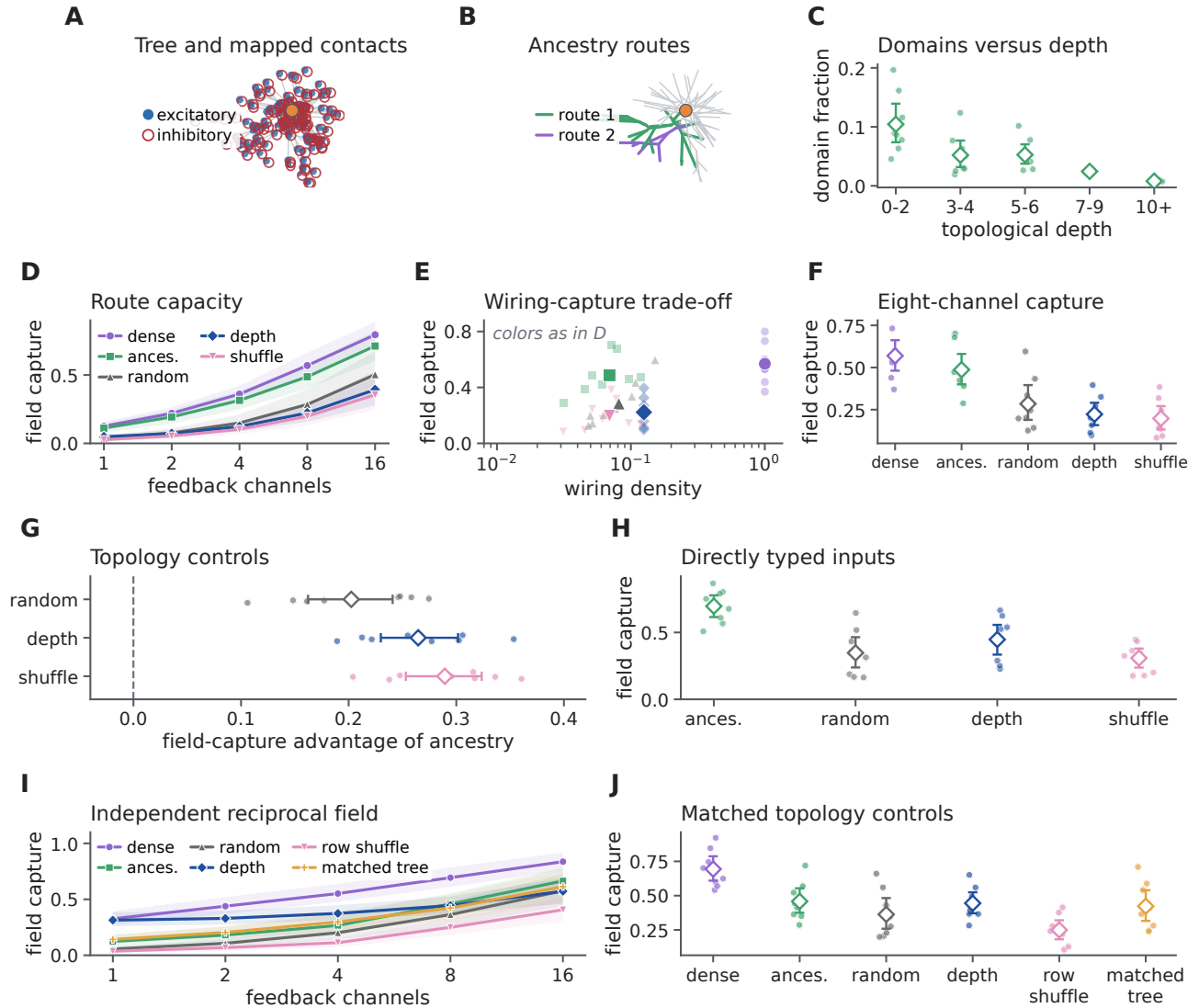


Figure S20: Detailed morphology-route capacity and control analyses. **A**, Reconstructed arbor with mapped excitatory and inhibitory contacts. **B**, Example ancestry-defined route domains. **C,D**, Route support and participation rank. **E,F**, Model-derived field capture across channel budget and the eight-channel comparison. **G**, Paired ancestry-minus-control contrasts. **H**, Eight-channel capture after restricting the analysis to directly typed presynaptic inputs. **I,J**, Independent reciprocal-cable field capture and its matched topology controls. Points are reconstructed-cell summaries; intervals are 95% cell-bootstrap intervals. This figure preserves the complete diagnostic display underlying the focused main Figure 6.

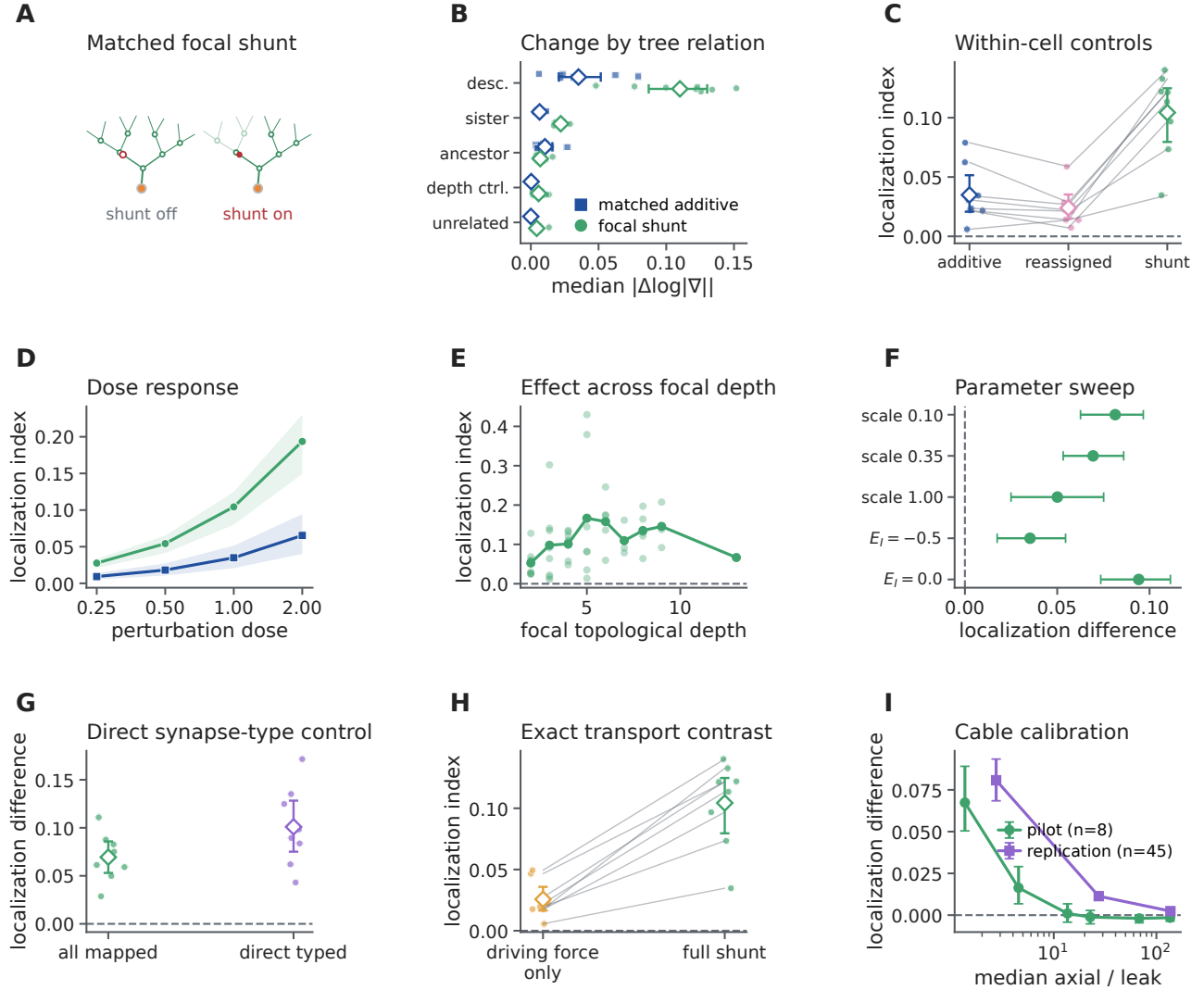


Figure S21: **Detailed focal-shunting controls and electrotonic boundary.** **A**, State-matched focal intervention. **B**, Exact-gradient change by topological relation. **C**, Within-cell shunt, additive and relation-reassignment controls. **D**, Perturbation-dose response. **E**, Effect across focal-site depth. **F**, Conductance-scale and inhibitory-reversal sensitivity. **G**, Direct presynaptic-type control. **H**, Exact factor-freeze contrast isolating transported error. **I**, Physical cable calibration across axial-to-leak regimes. Sites are nested within eight reconstructed cells; intervals are 95% cell-bootstrap intervals.

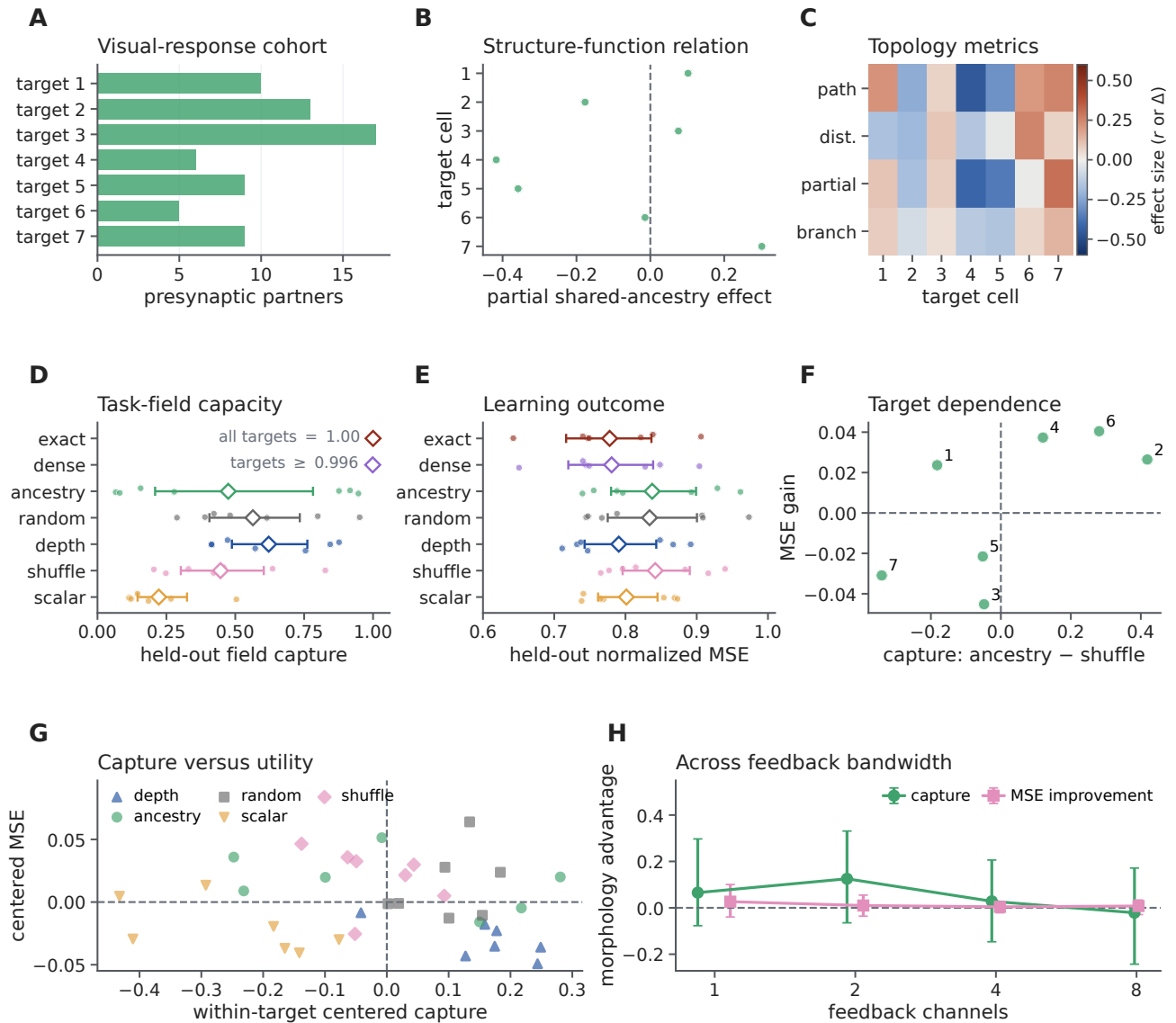


Figure S22: **Detailed measured-response topology and learning boundary.** **A**, Seven-target response cohort. **B**, Geometry-controlled structure–function relation. **C**, Target topology metrics. **D,E**, Four-channel task-field capture and held-out learning across exact, dense, ancestry and matched control dictionaries. **F**, Target-level dependence of morphology-minus-shuffle effects. **G**, Capture–utility association after accounting for target identity. **H**, Morphology-minus-shuffle contrasts across feedback bandwidth. Stimulus splits and scans are nested within seven target cells.

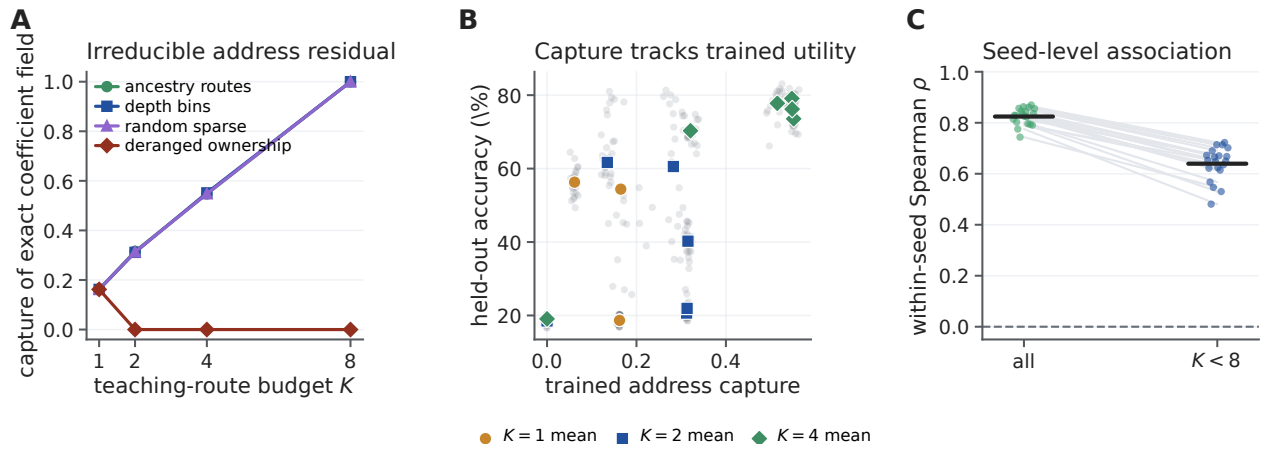


Figure S23: The trained partition residual explains bandwidth and ownership failures but not fine support-specific differences. **A**, Capture of the exact per-example coefficient field by the best coefficient on each available route after training. Curves and intervals are means and 95% seed-bootstrap intervals over 20 independent seeds. Ownership derangement excludes the target leaf and has zero capture for $K > 1$; three partitions that retain the target have similar capacity at matched K . **B**, Trained address capture versus held-out accuracy for the six restricted route families at $K < 8$. Small grey points are seed-condition means across the four implementation-equivalent representations; large symbols are condition means. **C**, Spearman correlations computed separately within each independent seed for all route conditions and after excluding $K = 8$; lines pair the same seed and horizontal bars show means. This deterministic reconstruction is a post-hoc diagnostic, not an independent prediction test.

Table S2: Confirmatory subtree-address factorial. Accuracy contrasts are paired across 20 seeds. The best matched non-anatomical control is selected within seed from route derangement, depth-interleaved, random-sparse and random-rank fields. Positive contrasts favor correct ancestry.

K	Correct accuracy	Versus deranged	Versus best control	Versus rewired tree
1	0.1866	0.0000	−0.3583	0.0000
2	0.4486	0.2639	−0.1540	0.2315
4	0.8010	0.6106	0.0127	0.0502
8	0.8100	0.5587	0.0000	0.0000

For the best-control contrast, 0/20, 5/20, 15/20 and 0/20 paired seeds favored ancestry at $K = 1, 2, 4, 8$, respectively; the full-rank conditions are exactly tied by construction. At $K = 4$, the paired bootstrap 95% interval was 0.0059–0.0197 and the two-sided Wilcoxon value was $P = 0.0048$ (Benjamini–Hochberg-adjusted $P = 0.0064$ across $K = 1, 2, 4, 8$). Correct-versus-rewired intervals were 0.210–0.254 at $K = 2$ and 0.042–0.060 at $K = 4$, with 20/20 seeds positive in both comparisons.

Table S3: Credit-operator phase experiment and frozen-factorial reanalysis. Phase contrasts use 50 paired confirmatory seeds. The depth contrast combines four task depths within seed; its positive reported magnitude favors matched depth. Utility correlations use 20 seed blocks from the existing factorial.

Endpoint	Effect	95% interval	Positive	Control / interpretation
Ancestry–random capture, $\rho = 1$, $K = 4$	0.5520	0.5353–0.5686	50/50	Dense rank- K upper bound; full rank tied
Matched versus best mismatched depth	0.0957	0.0782–0.1134	48/50 seeds	Four task-depth contrasts averaged within seed; $D_r = 4$ equals full stochastic BP
Aligned versus best global reliability gain	1.2415	1.1160–1.3728	50/50	Explicit point gate matched exactly
Utility versus observed one-step progress, ρ	0.9374	0.9336–0.9412	—	Seed-block Spearman interval
Utility versus final accuracy, ρ	0.9157	0.9047–0.9292	—	Secondary initialization diagnostic

Two-sided Wilcoxon values for the spectral, depth and reliability contrasts were 1.78×10^{-15} , 3.00×10^{-13} and 1.78×10^{-15} , respectively.

All 25 signal–noise condition means followed the analytical projection boundary. Static route-wise gains preserved projectors to 1.33×10^{-15} . The phase experiment is a fixed-state quadratic validation, not a trained positive-conductance network.

Table S4: Step-consistent state-matched positive-conductance reliability experiment at maximum branch-SNR heterogeneity. Effects are paired across 50 confirmatory seeds. Positive one-step effects favor aligned shunting; negative final-loss effects favor the left-hand aligned condition.

Endpoint	Effect	95% interval	Positive	Two-sided Wilcoxon P
Aligned-global, one-step loss decrease	0.001817	0.001618–0.002020	49/50	3.55×10^{-15}
Aligned-no shunt, one-step loss decrease	0.001339	0.000594–0.002096	33/50	0.0030
Aligned-global, final test loss	–0.020404	–0.022220––0.018705	0/50	1.78×10^{-15}
Aligned-shuffled, final test loss	–0.009204	–0.012171––0.006387	7/50	6.92×10^{-9}
Aligned-anti-aligned, final test loss	–0.013781	–0.017077––0.010708	3/50	9.52×10^{-13}
Aligned-no shunt, final test loss	0.000425	–0.002456–0.003071	33/50	0.154
Aligned-explicit point gate, final test loss	0	0–0	0/50; 50 ties	1

The independently computed state-matched unattenuated current control agreed with the unshunted noisy rule to numerical precision. Exact clean backpropagation had lower mean final loss than every attenuated condition. Shunts were fixed from initial oracle reliabilities and forward voltage was clamped exactly; excitatory conductance parameters, not shunts, were trained.

Table S5: Same-span noisy coefficient-learning contrasts after 80 updates. Effects are paired across 50 confirmatory target-and-noise seeds. Positive values mean greater loss than the orthonormal Haar coordinates.

Coordinate contrast	n_{eff}	Effect	95% interval	Positive	Two-sided Wilcoxon P
Raw nested-Haar	4	–0.2774	–0.3260––0.2313	0/50	1.78×10^{-15}
Scaled nested-Haar	4	–0.1906	–0.2500––0.1335	10/50	1.05×10^{-7}
Raw nested-Haar	256	0.02555	0.02171–0.02952	50/50	1.78×10^{-15}
Scaled nested-Haar	256	0.17091	0.14576–0.19686	50/50	1.78×10^{-15}

All dictionaries had the same rank-eight projector. Their Gram condition numbers were 1, 15 and 388.52. Median exact-risk crossover sample sizes were 38.53 for raw nested and 8.25 for scaled nested coordinates. Oracle Gram-preconditioned field trajectories agreed to 1.16×10^{-15} .

Table S6: Active steady-state focal-conductance ensemble. Values first average 64 accepted channel draws and focal sites within each of eight reconstructed cells.

Dose	Shunt localization	Additive localization	Shunt-additive	Shunt energy ratio
0.25	0.1587	0.0398	0.1189	0.7154
1	0.5287	0.1470	0.3817	0.3302
4	1.3387	0.4657	0.8730	0.0625

Dose is relative to local input conductance. Shunt-localization cell-bootstrap intervals were 0.1538–0.1628, 0.5130–0.5414 and 1.3066–1.3642. At every dose and in every cell, all modeled descendant gradients were attenuated without enhancement or sign reversal. Draws are nested sensitivity samples, not independent biological replicates.

Table S7: Functional topology across all eligible scans. Statistics were computed within 13 scans, averaged within each of seven target cells, and then summarized across targets.

Endpoint	Mean	95% target-bootstrap interval	Positive targets
Shared-path Spearman ρ	–0.0245	[–0.2142, 0.1466]	4/7
Negative tree-distance Spearman ρ	–0.0175	[–0.1297, 0.1016]	3/7
Partial shared-path ρ	–0.0485	[–0.2237, 0.1183]	3/7
Same-major-branch contrast	–0.0027	[–0.0833, 0.0699]	4/7

Two-sided exact Wilcoxon values were 1.0, 0.8125, 0.8125 and 1.0, respectively. Pairwise partner observations and multiple scans are nested within targets; all targets are from one mouse.

Table S8: Complete reconstructed-tree learning across all eligible scans. Values average ten held-out-stimulus splits within scan, then scans within each of seven target cells.

Teaching field	Normalized held-out MSE	Common-checkpoint update capture	Fits
Exact compartment error	0.7877	1.0000	130
Topology routes	0.8321	0.2322	130
Random routes	0.8333	0.3584	130
Site-shuffled routes	0.8391	0.4392	130

Positive MSE improvement favors topology. Topology improved over site shuffling by 0.0070 (95% interval -0.0207 to 0.0321) and over random routes by 0.0013 (-0.0090 to 0.0113). All restricted dictionaries had identical effective rank within run. None of 520 fits was excluded.

Table S9: Frozen architecture and data specification for the regular-tree MNIST experiments.

Component	Specification
Data	MNIST tensors; 784 flattened pixels; no normalization or augmentation; seed-controlled 48,000/12,000 train/validation split; official 10,000-image test set.
Modeled population	One layer of 128 excitatory modeled neurons; no separate inhibitory-neuron population.
Tree	[3,3]: nine distal, three proximal and one somatic compartment per neuron; no somatic input synapses.
Input contacts	Per dendritic compartment, separate TopK arrays selected 40 excitatory and 20 inhibitory inputs from 784 candidates per sign; 92,160 active contacts among 2,408,448 candidate synaptic parameters.
Parameterization	Nonnegative synaptic and child-coupling conductances obtained by softplus transformation; inactive local weights were not updated.
Reactivation	Learned $[\tanh(m(V - b)) + 1]/2$ at distal, proximal and somatic stages. Shunting used analytical initialization; raw additive used 0.1/0.9 occupancy-quantile calibration.
Readout	Linear map from 128 soma activities to 10 class scores.
Model size	2,414,602 trainable raw parameters in the archived feedback runs, of which 2,408,448 were candidate synaptic parameters.

Table S10: Training protocols for the regular-tree results. Each protocol used mini-batches of 256, categorical negative log likelihood, no weight decay, no mixed precision and restoration of the best validation checkpoint.

Protocol	Seeds; epochs	Core output	Rule and feedback	Adam learning rates
Feedback definition	42–56; 180	Identity	3F, local decoder; matched-width/scalar fallback or neuron-indexed sharing	TopK 1.5×10^{-3} ; child coupling 7×10^{-4} ; reactivation 5×10^{-4} ; decoder 1.5×10^{-3}
Exact-transport factorial	42–46; 200	ReLU	3F or 5F crossed with local or backpropagated decoder; exact path-transported compartment error	TopK 1.5×10^{-3} ; child coupling 7×10^{-4} ; reactivation 5×10^{-4} ; decoder 1.5×10^{-3}
Backpropagation reference	42–46; 200	ReLU	Standard backpropagation through the matched shunting tree and decoder	TopK and decoder 10^{-3} ; child coupling and reactivation 5×10^{-4}

Adam used $(\beta_1, \beta_2) = (0.9, 0.999)$ and $\epsilon = 10^{-8}$. The 3F local-gradient elementwise clip was 5. The exact-transport and backpropagation protocols shared architecture, seeds, epoch budget, and checkpoint-selection procedure, but used the method-specific learning rates listed above. Both were separate from the 180-epoch feedback-definition protocol.

Table S11: Archived provenance and execution records for the regular-tree results. “Not archived” means that the value was not recoverable and was not inferred from the current environment.

Cohort	Surviving run artifacts	Hardware record	Software record
Clean feedback cohort, 60 runs	All resolved configurations, checkpoints, logs, final metrics, frozen sweep manifests and SHA-256 hashes	NVIDIA H200; 4.006 summed GPU-hours across both arrays	Python 3.10.13; Git commit 4ba38bde; tracked worktree recorded as dirty; hashes of generator and critical model sources frozen
Prospective depth-by-feedback cohort, 640 executions (480 retained)	All resolved configurations, checkpoints, learning curves, held-out metrics, resource records, frozen manifests, source hashes and the outcome-independent 1,840-run validity ledger	73.29 summed GPU-hours over heterogeneous NVIDIA accelerators	Frozen source identity plus a verified equivalence audit for two later changes that are inactive in every resolved configuration; signed-noise shunting cells are excluded from inference
Prospective checkpoint diagnostic, 3,200 historical rows (2,400 retained)	Five feedback-field labels and four step sizes at each of 160 checkpoints; publication analysis uses 120 input-valid independently trained checkpoints	Fixed-checkpoint analysis; no additional model training	Complete balance and finite-value checks; checkpoint-clustered bootstrap analysis and exclusion ledger frozen with the source tables
Clean exact-transport/backpropagation audit, 320 runs	Resolved configurations, seed records, checkpoints, learning summaries, final metrics, scheduler and training logs, frozen sweep manifests and SHA-256 hashes	NVIDIA H200 for every resource-corrected rerun; initial incomplete attempts are excluded and archived	Detached tracked-clean root commit 74792ca; nested synthetic-data commit 4f3612a5; signed transfer retained for additive noise, explicit ReLU transfer used for positive-conductance shunting noise
Exact-transport factorial, 20 runs	Resolved configurations, checkpoints, logs and grouped result table	NVIDIA A100-SXM4-40GB; 2.953 GPU-hours total	Exact commit, Python and package environment not archived
Backpropagation reference, 5 runs	Resolved configurations, checkpoints, logs and grouped result table	NVIDIA A100-SXM4-40GB; 0.474 GPU-hours total	Exact commit, Python and package environment not archived

Portable sweep files, all resolved configurations, source-data tables, source hashes and validation records for the clean 15-seed feedback cohort and the 320-run detached audit are supplied with the article. Hardware times are sums of Slurm elapsed time over GPU tasks, not wall-clock latency.

Table S12: Notation used in the theory and analyses.

Symbol	Definition
u, n, i	Indices for a neuron, compartment and synapse, respectively.
V_n, a_n	Conductance-stage voltage and transmitted activity of compartment n ; $a_n = f_n(V_n)$.
$g_i, g_{c \rightarrow n}^{\text{den}}$	Scalar synaptic conductance and child-to-parent coupling conductance. These are distinct from bold gradient symbols.
E_i, x_i	Synaptic reversal potential and presynaptic activity.
$g_n^{\text{tot}}, R_n^{\text{tot}}$	Total compartment conductance and its reciprocal input resistance.
$\delta_u, \delta_{u,k}$	Neuron-level teaching coordinate and its route- k refinement within neuron u .
$\delta_{0,u}$ (or δ_0^V), $\partial \mathcal{L} / \partial V_n$	Exact somatic error for neuron u (single-tree voltage-space notation in parentheses) and exact compartment voltage-space error.
$\alpha_n^{\text{cond}}, \tilde{\alpha}_n$	Conductance-only and effective path gains; the latter includes reactivation derivatives.
α, ρ	Sensor alignment in the physical-depth task and covariance alignment in the spectral phase experiment.
$\mathbf{w}, \mathbf{g}, \hat{\mathbf{g}}$	Parameter vector, exact population gradient, and its unbiased stochastic estimate.
$\mathbf{q}$	Adjoint field; q_n is the loss sensitivity to a small current injected at compartment n .
$M, \Sigma, U(M)$	Credit operator, stochastic-gradient covariance, and the signal-noise phase utility.
K	Number of communicated feedback channels or route fields.
H, D_r	Task-hierarchy depth and routed address depth in the fixed-state phase experiment.
D_p ; D1–D3	Physical dendritic stage count; D1, D2 and D3 mean $D_p = 1, 2, 3$. In $[b_1, \dots, b_{D_p}]$, branching factors are ordered soma-to-distal.
A_T	Binary synapse-by-route ancestry matrix of tree T .
Γ_{ik}	Direct ancestry-route sensitivity from candidate route k to excitatory site i .
$\Phi_{T,K}$	K -channel feedback dictionary implemented by topology, route gains, channel allocation and weights.
W	Positive diagonal coordinate-weight matrix; summed excitatory synaptic area in the structural MICrONS analysis.
$\bar{P}_\Phi$	Ordinary Euclidean orthogonal projector onto the columns of $W^{1/2} \Phi_{T,K}$.
$\mathcal{C}_K$	Field-energy capture at feedback-channel budget K in the stated coordinates.
G, R_a, R_m	Passive conductance matrix, axial resistivity, and specific membrane resistance.
B	Mini-batch size when used in $\langle \cdot \rangle_B$.
S_K, C_g	Feedback-channel assignment matrix in Eq. 26 and gradient covariance, respectively.

Table S13: Feedback objects and the information each supplies.

Feedback field	Communicated values	Spatial assignment	Interpretation
Global scalar	Robust mean-or-signed-maximum reduction in Eq. 23	Same value at every receiving unit	Deliberately removes neuron and branch identity.
Matched-width/scalar fallback	Somatic vector at a same-width stage; otherwise the same robust scalar	Coordinate-wise only at the matched stage	Historical implementation; not ancestry sharing.
Neuron-indexed ancestry sharing	One coordinate per modeled neuron	Each coordinate repeated over all compartments descended from that soma	Tests neuron identity and correct neuron-to-tree ownership, not distinct within-tree addresses.
Correct sibling-subtree routing	K coefficients within one modeled neuron	Each coefficient assigned to its context-relevant subtree	Phase-1 trained test of within-neuron address support at $K = 2$.
Within-neuron derangement	The same K coefficients as correct routing	Fixed permutation among sibling routes inside the neuron	Alignment control at matched coordinate count and sparse nonzeros.
Rank- K random	K projected coordinates	Fixed random compartment mixing	Bandwidth control without anatomical routes.
Selected paths	K channel coefficients	Sparse real ancestry domains	Tests topology-matched route capacity.
Exact transport	Somatic error and exact sample-specific path derivative	Distinct exact value at every compartment	Information oracle; not proposed as an available biological signal.

Table S14: Principal regular-tree learning and gradient results. Values are mean $\pm$ s.d. over independent seeds. Accuracy uses checkpoints trained with the named feedback field; both gradient-cosine entries within an architecture are fixed-checkpoint diagnostics evaluated at its neuron-indexed-trained checkpoints.

Architecture	Feedback or learning condition	n	Test accuracy (%)	Dendritic-gradient cosine
Shunting	Matched-width/scalar fallback, 3F	15	90.54 ± 0.54	0.012 ± 0.059
Shunting	Neuron-indexed ancestry sharing, 3F	15	97.16 ± 0.11	0.708 ± 0.027
Additive	Matched-width/scalar fallback, 3F	15	91.76 ± 0.78	-0.001 ± 0.047
Additive	Neuron-indexed ancestry sharing, 3F	15	97.14 ± 0.11	0.625 ± 0.035
Shunting	Exact transport, 3F, local decoder	5	97.18 ± 0.12	Exact field
Matched reference	Backpropagation	5	97.03 ± 0.08	1 by definition

Table S15: Input-valid prospective depth-by-feedback results. Accuracy contrasts are ranges of condition means over one to four dendritic stages. MNIST retains both neuron models; the noise task retains the additive model. The exact-transport comparison reports the largest absolute difference between the three-factor local rule and matched backpropagation among the 12 retained task-model-depth conditions.

Task	Contrast	Accuracy difference	Paired-seed ordering
MNIST	Neuron indexed minus scalar	4.92–6.99 points	percentage Positive in 80/80 pairs
Noise resilience, additive	Neuron indexed minus scalar	9.76–14.16 points	percentage Positive in 40/40 pairs
Retained conditions	Exact 3F minus matched backpropagation	At most 0.14 percentage points in absolute mean difference	12 matched task-model-depth conditions

Each condition contains ten independent paired seeds. Exact transport is an information oracle rather than a proposed biological signal. Synthetic-noise shunting rows with a signed transfer output are excluded by an outcome-independent conductance-validity rule.

Table S16: Detached tracked-clean exact-transport/backpropagation audit. Within each task and core, accuracy was first averaged over depths one through four for each paired seed. Values are percentages; intervals are 20,000-draw paired- seed bootstrap intervals and P values are descriptive two-sided Wilcoxon tests.

Task	Core	Exact accuracy	BP accuracy	Exact – BP (95% interval)	Exact > BP	P
MNIST	Additive	97.013	97.008	0.005 (–0.028–0.041)	4/10	1.000
MNIST	Shunting	96.960	96.973	–0.013 (–0.047–0.022)	4/10	0.492
Noise resilience	Additive	33.378	33.490	–0.112 (–0.465–0.247)	4/10	0.625
Noise resilience	Shunting (ReLU)	36.690	36.788	–0.098 (–0.653–0.390)	7/10	1.000

The 320 runs comprise 160 seed-, task-, core- and depth-matched method pairs. One MNIST-additive depth-averaged seed was tied. Exact transport is an information-oracle implementation control; intervals containing zero do not constitute a formal equivalence test.

Table S17: Bandwidth-matched neuron-to-tree routing control. Values are correct minus fixed-deranged held-out accuracy in percentage points. Intervals are paired-seed bootstrap intervals; P values are exact two-sided Wilcoxon tests.

Task	Neuron model	Depth	Difference (95% interval)	Positive pairs	P
MNIST	Additive	2	0.51 (0.37–0.62)	10/10	0.0020
MNIST	Additive	4	0.39 (0.31–0.45)	10/10	0.0020
MNIST	Shunting	2	0.57 (0.43–0.69)	10/10	0.0020
MNIST	Shunting	4	0.38 (0.31–0.44)	10/10	0.0020
Noise task	Additive	2	0.70 (0.56–0.84)	10/10	0.0020
Noise task	Additive	4	0.15 (0.02–0.28)	8/10	0.1055

Five contrasts remained significant after Benjamini–Hochberg correction across the six tests; the additive depth-four noise contrast did not. Across conditions, 58/60 paired seeds favored correct assignment. The modest effect relative to the much larger scalar-to-neuron-indexed contrast separates a dominant coordinate-bandwidth contribution from a smaller assignment-specific routing contribution.

Table S18: Outcome-independent input-validity accounting for the prospective artificial-tree cohorts. “Retained” counts the units used in current figures and inference. The entire inhibitory-dose family is excluded because its prespecified endpoint compares against invalid signed-input shunting cells.

Cohort	Original units	Signed-shunting exclusions	Retained units
Depth by feedback	640 runs	160	480
Matched routing	160 runs	40	120
Inhibitory dose	400 runs	200	0
Spatial topology	320 runs	80	240
Fixed-budget depth	320 runs	160	160
Checkpoint diagnostic	160 checkpoints	40	120

The exclusion rule is task plus core plus resolved transfer activation; no accuracy or gradient outcome enters it. The complete 1,840-run historical ledger and all retained row tables are supplied as Source Data.

Table S19: Audit of fixed random and spatial sparse input maps. Each row summarizes 1,280 neuron-level maps: 128 neurons for each of ten topology seeds. Both maps contain 336 active contacts per neuron.

Connectivity property	Random map	Spatial map
Unique input coordinates, mean $\pm$ s.d.	276.2 $\pm$ 5.9	336.0 $\pm$ 0.0
Input coverage fraction	0.352	0.429
Repeated-contact fraction	0.178	0.000
Mean pairwise branch Jaccard overlap	0.0139	0.0000
Feature-degree coefficient of variation	1.507	1.155

Table S20: Prospective fixed-checkpoint credit-to-step audit at relative step 10^{-5} . Values are means over 120 input-valid independently trained backpropagation checkpoints. One-step progress is the loss decrease from a norm-matched candidate update divided by that from the exact update.

Feedback field	Gradient cosine	Gradient capture	Relative progress	Descent checkpoints
Global scalar	0.027	0.003	0.013	75/120
Neuron indexed	0.586	0.355	0.636	120/120
Exact transport	1.000	1.000	1.000	120/120

Across scalar and neuron-indexed fields, gradient cosine predicts relative one-step progress with Spearman $\rho = 0.893$ (checkpoint-clustered 95% bootstrap interval 0.860–0.923); gradient capture gives $\rho = 0.859$ (0.825–0.890). These are fixed-state optimization diagnostics, not convergence or generalization comparisons.

Table S21: Selected endpoints from the expanded regular-tree regime analyses. Values are test-accuracy means $\pm$ s.d. over five independent seeds.

Analysis	Contrast	Shunting or structured feedback	Additive or restricted feedback
Depth stress	One to four dendritic layers	$63.5 \pm 6.2\%$ to $57.4 \pm 4.2\%$	$54.9 \pm 5.4\%$ to $29.7 \pm 10.5\%$
Broadcast-noise stress	$\sigma = 0$ to $\sigma = 1$	0.617 ± 0.022 to 0.469 ± 0.062	0.465 ± 0.052 to 0.114 ± 0.009
Noise-task feedback	Scalar versus rank eight	0.662 ± 0.016 versus 0.795 ± 0.019	Exact transport: 0.847 ± 0.018
Flattened CIFAR-10	Scalar versus rank four	$29.7 \pm 1.3\%$ versus $35.0 \pm 1.0\%$	Exact transport: $47.2 \pm 0.6\%$; BP: $49.5 \pm 0.4\%$

The first two rows compare complete forward architectures and therefore establish robustness regimes rather than a causal backward effect of shunting. Exact transport is an information oracle. The flattened CIFAR-10 analysis is a harder-data mechanism control, not a state-of-the-art image classification comparison.

Table S22: Reconstructed-cell inputs to the structural analysis. Synapse counts are mapped and classified excitatory (E) or inhibitory (I) contacts.

Root ID	Type	Depth	Segments	E synapses	I synapses	Kernel rank	Map pass
864691135060651419	L2IT	11	146	4661	1378	38.36	0.983
864691135277186789	L3IT	10	104	3780	983	14.79	0.981
864691135324623260	L4IT	6	40	1534	506	16.66	0.967
864691135196576298	L5IT	5	22	1254	394	11.00	0.974
864691135409937097	L2IT	9	78	2870	744	21.15	0.984
864691135739661169	L5IT	6	34	970	446	11.96	0.982
864691135810666525	L2IT	14	128	4725	1934	31.29	0.986
864691136043380566	L2IT	9	64	1987	503	21.25	0.986
Total or mean		8.75	77.0	21,781	6,888	20.81	0.980

Table S23: Eight-channel routing capacity on model-derived perturbation fields in eight reconstructed cells. Intervals resample cells and Monte Carlo streams hierarchically.

Dictionary	Mean capture	95% interval	Wiring density	Ancestry advantage
Dense PCA oracle	0.569	0.481–0.665	1.000	–
Ancestry-selected paths	0.487	0.400–0.581	0.069	–
Random nonempty paths	0.285	0.191–0.398	0.081	0.202 (0.161–0.242)
Depth-only bins	0.222	0.157–0.292	0.125	0.265 (0.229–0.302)
Ancestry shuffle	0.198	0.132–0.272	0.069	0.289 (0.253–0.324)

The morphology advantage was positive in 8/8 cell means for each structural control; the two-sided exact paired Wilcoxon value was $P = 0.0078125$ in each exploratory comparison. The cohort was selected before this journal synthesis but was not a population-random sample.

Table S24: Eight-channel ancestry dictionaries evaluated on an independently generated reciprocal-cable shunt-response operator. Capture is the expected weighted response energy in the stated subspace.

Dictionary	Mean capture	Ancestry difference	Ordered cells
Dense SVD oracle	0.694	–	–
Ancestry-selected paths	0.458	–	–
Random real paths	0.363	0.095	8/8
Depth bins	0.445	0.014	5/8
Row-shuffled selected paths	0.249	0.209	8/8
Degree- and depth-matched surrogate tree	0.423	0.035	6/8

The target was the finite-difference response of the exact soma-state-matched excitatory log-gradient field to every focal shunt, rather than a field sampled from the tested ancestry kernel. Ancestry-minus-control cell-bootstrap intervals were 0.064–0.124 for random paths, –0.037–0.068 for depth bins, 0.179–0.244 for row shuffling, and 0.0069–0.0658 for matched surrogate trees. The paired two-sided Wilcoxon values were 0.0078, 0.844, 0.0078, and 0.078, respectively. The cell-bootstrap interval for the matched surrogate excluded zero whereas the prespecified exact paired test did not; we therefore treat that comparison as inconclusive. This remains a passive-model response on one mouse’s reconstructed anatomy.

Table S25: Focal-perturbation sensitivity at unit dose. The primary condition is the first row. Scale multiplies normalized excitatory and inhibitory synaptic conductance; E_I is the inhibitory reversal potential in normalized units.

Scale	E_I	Shunt localization	Additive localization	Difference	Positive cells	P
0.35	-0.2	0.104	0.035	0.069	8/8	0.0078
0.10	-0.2	0.101	0.019	0.081	8/8	0.0078
1.00	-0.2	0.114	0.064	0.050	7/8	0.0156
0.35	0.0	0.115	0.021	0.094	8/8	0.0078
0.35	-0.5	0.089	0.054	0.035	7/8	0.0156

Values are means of site-averaged cell effects. P is the exact two-sided paired Wilcoxon value over reconstructed cells. Sensitivity rows reuse the same eight anatomical cells and are not independent replications.

Table S26: Physical passive-cable sensitivity of the focal shunt contrast. R_a is axial resistivity and R_m is specific membrane resistance.

Cohort	(R_a, R_m)	Median axial/leak	Difference	Positive cells	95% interval
Original eight	(150, 300)	1.36	0.0674	8/8	0.0502–0.0898
Original eight	(150, 1000)	4.53	0.0164	8/8	0.0068–0.0290
Original eight	(150, 3000)	13.60	0.0010	5/8	-0.0043–0.0068
Original eight	(150, 15000)	68.02	-0.0019	2/8	-0.0046–0.0000
v661 sensitivity	(150, 300)	2.74	0.0808	45/45	0.0687–0.0937
v661 sensitivity	(150, 3000)	27.41	0.0113	45/45	0.0090–0.0138
v661 sensitivity	(150, 15000)	137.07	0.0024	45/45	0.0018–0.0030

R_a is in Ωcm and R_m in Ωcm^2 . Synaptic area was calibrated relative to the median physical leak and is not a measured conductance. Specific resistances are sensitivity settings rather than fitted cell-specific values. Complete R_a/R_m combinations are supplied as source data.

Table S27: Target-level inputs for the measured-response branch-model analysis. Every target contributed 464 trials, 280 unique stimulus identities and 136 repeated identities; all extracted response values were finite.

Nucleus ID	Root ID	Type	Session	Scan	DANDI asset prefix	Sites	Branches
256456	864691135810666525	L2IT	6	6	06c05eec	10	5
256550	864691135060651419	L2IT	9	4	ad27b3c6	13	5
258209	864691135277186789	L3IT	8	5	436f762f	17	8
264871	864691135739661169	L5IT	7	3	67384ebb	6	4
264888	864691135196576298	L5IT	7	3	67384ebb	9	5
264889	864691135324623260	L4IT	6	6	06c05eec	5	3
291111	864691135409937097	L2IT	6	4	dc68c26d	9	6

Full DANDI asset identifiers and NWB paths are included in the machine-readable target manifest. “Sites” are connected, functionally imaged presynaptic partners after automatic conservative matching; “Branches” are occupied first-level dendritic branches in the fitted model.

Table S28: Four-channel measured-response task analysis. Lower normalized MSE is better. Values average ten fixed stimulus splits within each of seven target cells before averaging targets.

Feedback field	Held-out field capture	Held-out normalized MSE
Exact backpropagation	1.000	0.778
Dense PCA oracle	0.999	0.781
Ancestry-selected paths	0.475	0.837
Random nonempty paths	0.564	0.834
Depth-only bins	0.621	0.791
Ancestry shuffle	0.446	0.842
Scalar broadcast	0.222	0.801

Ancestry routes minus ancestry-shuffle capture was 0.028 (target bootstrap 95% interval -0.146 to 0.206; 3/7 targets positive). The reduction in error from morphology relative to ancestry shuffle was 0.004 (interval -0.021 to 0.027; 4/7 targets improved). Neither contrast was reliable.

Table S29: Synapse-resolved inhibitory-census endpoints. Differences are first averaged within each of 20 postsynaptic targets. Confidence intervals bootstrap target cells. Directional one-sided Wilcoxon tests were frozen before the join for the first four endpoints and the three selected-site capacity comparisons; overall placement is two sided.

Endpoint	Mean difference	95% interval	Ordered targets	<i>P</i>
Same-axon tree distance (μm)	-12.003	[-21.460, -4.151]	15/20 lower	0.0032
Same-axon shared path	0.0357	[0.0091, 0.0766]	15/20 higher	0.0028
Same-axon domain overlap	0.0322	[0.0060, 0.0743]	14/20 higher	0.0053
DTC minus PTC median domain fraction	-0.0162	[-0.0231, -0.0105]	20/20 lower	9.5×10^{-7}
All contacts minus matched locations	0.0016	[-0.0022, 0.0056]	10/20 higher	0.546
Eight selected sites minus random sites	0.3638	[0.3326, 0.3948]	20/20 higher	9.5×10^{-7}
Eight selected sites minus depth bins	0.3462	[0.3126, 0.3793]	20/20 higher	9.5×10^{-7}
Eight selected sites minus ancestry shuffle	0.3738	[0.3437, 0.4037]	20/20 higher	9.5×10^{-7}

Table S30: Dependence and spatial sensitivity of the repeated inhibitory-contact analysis. Contrasts use the same endpoint sign as Table S29.

Control	Endpoint	Mean difference	Ordered units	<i>P</i>
Presynaptic-axon aggregation	Tree distance (μm)	-9.368	62/92 lower	0.0008
Presynaptic-axon aggregation	Shared path	0.0143	45/92 higher	0.104
Presynaptic-axon aggregation	Domain overlap	0.0117	41/92 higher	0.393
Joint 3D and path matching	Tree distance (μm)	-4.272	13/20 lower	0.108
Joint 3D and path matching	Shared path	0.0107	8/20 higher	0.826
Joint 3D and path matching	Domain overlap	0.0151	9/20 higher	0.54

The first control averages connection contrasts within each of 92 presynaptic inhibitory axons before testing. The second retains the target as the unit and draws matched sites by joint percentile rank of log path distance and three-dimensional Euclidean proximity. All units remain nested within one mouse. Only the coarse tree-distance contrast persisted after presynaptic-axon aggregation; no endpoint persisted after joint spatial matching.

Table S31: External six-animal signed-credit test from Francioni et al. source data. The contrast is error-reduction minus error-increase soma-dendrite residual. Directions follow the known causal mapping.

Contrast	Mean	95% interval	Ordered animals
P+	0.0753	[0.0295, 0.1214]	5/6 positive
P-	-0.1498	[-0.2054, -0.0866]	6/6 negative
P+ minus P-	0.2251	[0.1416, 0.3108]	6/6 positive

Exact one-sided random-sign permutation *P* values were 0.0313, 0.0156, and 0.0156. The signed and common modes contained 83.7% and 16.3% of pooled squared projection energy across the six animal contrast vectors, respectively. The signed fraction was 45.8–97.1% across individual animals, with an animal-bootstrap 95% interval of 60.6–95.7% for the pooled fraction and a leave-one-animal-out range of 76.5–90.2%. Animal-level values are provided in Source Data.

Table S32: Interpretive boundary for each evidence class.

Evidence	Supported conclusion	Not established
Exact factorization	Conductance-tree gradients separate local eligibility from transported compartment error.	A biological circuit computes exact gradients or has exact path transport.
Regular-tree feedback	Preserving neuron identity closes most of the scalar-feedback gap in both tested architectures; exact transport makes 3F sufficient; feedback quality orders norm-matched one-step progress.	Universal advantage of shunting, or superior accuracy, memory or latency relative to backpropagation.
Subtree-address factorial	Task-aligned nested ancestry has a small advantage over matched non-anatomical routes at intermediate bandwidth and over a rewired tree at intermediate budgets.	A uniquely dendritic advantage; matched gated point and flat implementations reproduce the same result, and unrestricted low-rank routes win at the smallest budgets.
Credit-operator phase	Retained signal, admitted noise and task hierarchy predict when a restricted route or additional address depth improves a stochastic update.	Superiority to exact full-batch backpropagation, a positive-conductance forward mechanism or learned branch reliability.
Physical-depth controls	Across $H = 2$ and $H = 3$ variants of one mechanism-matched task family, serial divisive composition beats resource-identical star and literal grouped-point controls only when task factors match physical order; shared and path LocalCA retain the aligned H2 effect. Flexible parameter-matched point MLPs beat the serial tree, and soma-broadcast controls localize the BP-local separation.	A dendrite-exclusive expressivity advantage, natural-task prevalence, biological availability of exact path transport or optimizer-independent equivalence of BP and LocalCA.
State-matched reliability conductance	Fixed positive shunts approximate the step-consistent oracle reliability ordering and outperform global, shuffled and anti-aligned shunts under heterogeneous branch noise.	Exact block-scalar attenuation, autonomous shunt learning, a dendrite-exclusive operation or sustained final-loss superiority over unshunted noisy learning.
Same-span coefficient learning	Coordinate conditioning creates the finite-data bias-variance crossover predicted by exact linear risk, despite identical route capacity.	A biological Gram preconditioner, unconditional benefit of orthogonal or ill-conditioned coordinates, or new address span.
Prospective input-validity audit	Identity, ownership, spatial and fixed-budget conclusions persist in the 1,000 retained artificial-tree runs; the inhibitory-dose family is excluded.	Any claim derived from signed synthetic inputs driving positive-conductance shunting cells.
Fixed spatial topology	Spatial partitioning improves forward learning and input coverage at fixed contact count.	Task-aligned dendritic credit routing; the effect persists under backpropagation and a randomly projected task.
MICrONS capacity	Real ancestry compresses model-matched fields and captures independently generated reciprocal-cable responses better than random or row-shuffled routes.	An advantage of fine topology over coarse depth and degree, task-gradient alignment, endogenous credit routing, or learning in the recorded mouse.
Inhibitory census	Inhibitory classes occupy different dendritic scales and repeated contacts are spatially co-targeted; actual-route dictionaries are internally compressible.	Geometry-independent descendant-route organization, inhibitory activity, a teaching signal, or biological selection of leverage-ranked routes; all targets come from one mouse.
Focal perturbation	In the normalized proxy, an exact factor-freeze control attributes descendant localization to changed adjoint transport; physical calibration specifies where the effect is large or small.	That the effect is universal across electrotonic regimes or that inhibitory synapses conveyed teaching signals during the animal's experience.
Active focal ensemble	Descendant-enriched attenuation survives exact local linearization at 512 accepted active-conductance steady states.	Channel kinetics, spikes, cell-specific channel fits or independent biological replication by the nested channel draws.
Measured responses	All 13 eligible scans and the 520-fit complete-tree model do not show reliable alignment between the tested anatomical routes and this visual-response learning objective.	Absence of topology-function organization in cortex or under other response windows, coverage and biophysical calibration.
Projection coefficients	Each dictionary's best attainable capacity can be compared under a fixed channel budget.	A locally realizable circuit for learning the oracle coefficients.
Alignment-controlled optimization	Rotating exact credit into a fixed reconstructed-tree basis changes both capture and loss reduction under matched quadratic objectives.	That the reconstructed cells acquired or used the imposed gradients in vivo.
Signed animal contrast	Known P+/P- causal sign predicts the dominant dendritic population contrast across six animals.	Within-tree routing or a unique conductance/shunting mechanism.
Interference identity	Route overlap and off-route leakage are sufficient to produce exact one-step forgetting in a static quadratic task.	Temporal neural dynamics or a causal explanation of the published motor-learning experiment.

Table S33: Alignment-controlled capture and one-step optimization. Values are means across eight reconstructed cells after averaging 40 nested Monte Carlo streams per cell. One-step progress is normalized by a norm-matched full-gradient update.

Aligned energy	Dictionary	Field capture	One-step progress
0.0	Ancestry-selected paths	0.000	0.000
0.0	Random paths	0.135	0.307
0.0	Depth bins	0.153	0.331
0.0	Ancestry shuffle	0.152	0.331
0.4	Ancestry-selected paths	0.400	0.615
0.4	Random paths	0.197	0.390
0.4	Depth bins	0.191	0.385
0.4	Ancestry shuffle	0.172	0.359
1.0	Ancestry-selected paths	1.000	1.000
1.0	Random paths	0.290	0.460
1.0	Depth bins	0.247	0.444
1.0	Ancestry shuffle	0.201	0.373

At 40% alignment, morphology capture exceeded the three controls by 0.203–0.228 and the difference was positive in 8/8 cell means. At zero alignment morphology lost to all controls in 8/8 cells; at complete alignment it exceeded all controls in 8/8 cells. Monte Carlo streams and gradient fields are nested sensitivity samples, not biological replicates.

1180 **References**

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